

PERFORMANCE WORK STATEMENT (PWS)

Fort Carson Directorate of Public Works (DPW) Base Operations and Maintenance Contract

SECTION C.1 GENERAL INFORMATION

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C.1.1. GENERAL

This is a non-personal services contract to provide Base Operations and Maintenance Services. The Government shall not exercise any supervision or control over the Contractor. The prime Contractor is responsible for ensuring Subcontractors' compliance with all Performance Work Statement (PWS) requirements. The Contractor shall not perform any inherently governmental actions under this contract. The Performance Work Statement (PWS) provides guidelines on how the Contractor's personnel shall identify themselves. In all communications with the Government or other contractors in connection with this contract, the Contractor's employees shall state that they have no authority in any way to change the other contract and that if the other contractor believes this communication to be a direction to change its contract, it should notify the Contracting Officer for that contract and not carry out the direction until a clarification has been issued by that contract's Contracting Officer.

C.1.1.1. Scope

The Contractor shall plan, program, administer, manage, document, and execute the work necessary to provide Directorate of Public Works (DPW) functions and services specified in this contract. The Contractor shall ensure that all work meets standards specified in the contract, or documents referenced in individual tasks of Section C.5. The Contractor shall comply with applicable Federal, State, and local laws, regulations, and directives to include applicable publications in Section C.6. The Contractor shall perform all related Contractor administrative services required to perform work such as, but not limited to, material requisitioning, quality control (QC), financial control, and correspondence. The Contractor shall maintain accurate and complete records, files, and libraries of documents. The Contractor shall provide all management, supervision, materials, supplies, labor, and equipment, except that specified herein as Government-furnished, necessary to accomplish the services outlined in this contract. This work includes required maintenance and repair to facilities, structures, buildings, and utilities that are classified as real property to include installed building equipment as defined in Army Regulation (AR) 420-1, Army Facilities Management. The work to be performed under this contract will be performed at Fort Carson, Fort Carson water right areas located off the base as outlined in C.5.4.10, and Pinon Canyon Maneuver Site (PCMS) located in Model, Colorado, approximately 120 miles south of Fort Carson. The scope of this PWS is for real property facilities and systems only, unless specifically stated otherwise.

C.1.1.2. Period of Performance (POP)

It is anticipated that the POP for the effort in this PWS will consist of 6 Months. The total contract performance is expected to cover six months from contract award as indicated below:

C.1.1.3. Background

Fort Carson has contracted the DPW Base Operations and Maintenance services since the 1980's. The contracted effort has changed over the years to meet Division, Garrisons, and DPW

missions for our customers. The work described in this contract involves a broad range of non-personal base operations and maintenance services in support of the Directorate of Public Works (DPW) headquartered at Fort Carson, Colorado.

C.1.1.4. Objectives

- Provide healthy, safe spaces, and reliable essential city services to Fort Carson Soldiers, families and civilians;
- Operate as an effective team providing services as an extension of DPW;
- Provide effective stewardship of Installation Resources;
- Be the expert on Fort Carson utility systems for proper operations;
- Provide world-class maintenance and repair services while performing the contract; and
- Perform maintenance and repair requirements effectively and efficiently.

C.1.2. PERFORMANCE WORK STATEMENT (PWS) DOCUMENT INFORMATION

C.1.2.1. Technical Exhibits (TEs)

TEs are used to provide supplementary information to the contract, and may be presented in the form of photographs, tables, graphs, or maps. TEs for Section C will be referenced by the paragraph number in which the requirement is identified. All TEs are listed in Section C.7 of this PWS. Although the Government has made best efforts to provide complete and accurate information in the TEs, there may be errors or omissions. If the Contractor finds an error or omission during the course of contract performance to any TE(s), the Contractor shall submit the corrected TE to the COR and KO within five business days. The COR will validate the information and provide the KO an updated TE for inclusion into the contract documents.

C.1.2.2. Contract Reports

The Contractor shall compile data, prepare required forms, reports, and documents, and submit information as specified by the Report List in this contract. Reports and plans for Section C of the PWS will be referenced by the paragraph number in which the requirement is identified. There are two reports required of the Contractor as part of the contract documents not specifically listed elsewhere in the PWS: Motor Vehicle Licensing Requirements per DoD Directive 4500.36-R (Report C.1.2.2-B) and Affirmative Procurement of Bio-based Products per FAR 52.223-2 Report C.1.2.2-A. All reports and plans are listed in Section C.8 of this PWS.

C.1.2.3. Publications and Forms

The publications and forms listed in this Section shall be considered mandatory. The Contractor is obligated to comply with these publications and forms in their entirety.

C.1.2.4. Pagination

Pagination for all parts of Section C begins with the letter “C,” followed by the section number and the page number (e.g., page 12 of Section C.1 is C.1-12).

C.1.3. TRANSITION (PHASE-IN AND PHASE-OUT) PLAN

The Contractor shall develop comprehensive plans for phasing out Contractor performance to the level prescribed and within the time allowed under the terms of this contract.

C.1.3.1. Phase-In Period

During the Phase-In period, the Contractor shall prepare to assume full responsibility for all areas of operation IAW the terms and conditions of this contract no later than (NLT) 45 days from the start date for the base period. The Government will attempt to make all Government-furnished property (GFP), including facilities and equipment, accessible to the Contractor at the start of the Phase-In period. The transfer of all GFP will be in accordance with the Contractor’s Phase-In plan and the incumbent Contractor’s phase-out plan. The Contractor shall take all actions necessary for a smooth transition of the contracted operations. The Contractor shall submit the Phase-In plan to the KO and COR NLT 15 days after award IAW C.8, report C.1.3.1. During the Phase-In period, the Contractor shall perform, at a minimum, all of the following tasks:

- establish the Project Management Office;
- recruit and hire necessary personnel;
- obtain all required certifications;
- obtain minimum required security clearances;
- participate in joint inventories of the GFP;
- develop and submit any required deliverables;
- obtain necessary access to Government computer network system;
- attend post-award meetings as required; and
- complete all necessary training to support the services listed in each functional area.

C.1.3.2. Phase-out Period

The Contractor shall develop and implement a plan for the smooth and orderly transfer of performance responsibility to a successor. The plan shall contain details that include time lines for events, personnel involved, and interface with the Government and follow-on Contractor. The plan shall detail provisions for training the successor’s personnel on Government-furnished Automated Information Systems (AIS) used in performance of this PWS, specialized equipment, utilities systems, and ongoing work (work-in-progress) that the successor would be required to complete. The plan shall also include a joint inventory among the incumbent, the Government and the successor Contractor. Reconciliation of all property accounts and requisitions, turn-in of excess property, and cleanup of Contractor work areas is required. The Contractor shall correct any damage to Government-owned property caused by employee negligence or mistake, intentional or otherwise.

C.1.3.3. Current Operation Observation (End of Contract)

During the phase-out period, the incumbent Contractor shall permit the successor Contractor (and the successor Contractor's employees) to observe and become familiar with all operations specified in this PWS without disruption to current services. The Contractor shall not defer any specified work for the purpose of transferring responsibility to the successor Contractor. The Contractor shall fully cooperate with the successor Contractor and the Government while continuing to perform contractually during this observation period.

C.1.4. CONTRACTOR ADMINISTRATION

C.1.4.1. Project Manager, Alternate Project Manager

The Contractor shall perform continual project management and shall provide a Project Manager on-site during working hours from the first day of the Phase-In period through the duration of the contract. The Project Manager shall be a full-time position dedicated solely to this contract, with education and/or experience identified in Section C.1.4.11.1.2. The Project Manager shall coordinate overall management and shall be the central point of contact with the Government for performance of all work under the contract. The Project Manager shall have full authority to obligate Contractor funding and resources to complete the work in this contract. The Government reserves the right to approve replacement of key personnel IAW Section I, Contract Clauses. Any limitations on the Project Manager set forth by the Contractor shall be included in the letter of designation. An Alternate Project Manager shall have the same authority as the Project Manager and act for the Project Manager during the Project Manager's absence.

During off duty hours the Project Manager or designated Alternate Project Manager shall return all calls from the COR within two hours of receipt. The Project Manager and Alternate Project Manager(s) shall be proposed, in writing, to the KO and the COR 5 days prior to Phase-In period and within 5 business days of change in the Project Manager or Alternate Project Manager(s) occurs IAW C.8, report C.1.4.1.

C.1.4.1.1. Meetings, Conferences, and Briefings

The Contractor shall provide effective communication and necessary information that relates to the services required by the Government under this contract by conducting, attending, and participating in scheduled and unscheduled meetings, conferences, and briefings as requested by the Government. Meetings, conferences, and briefings may occur on-site or off-site and may begin or end outside of normal work hours. The Project Manager or designated Alternate Project Manager, contract manager, supervisory, or other personnel knowledgeable of the subject matter pertinent to this contract shall participate in meetings with Fort Carson customers; Federal, State, and local agencies and their representatives; and other Contractors as required by the Government.

The Contractor shall notify the COR prior to holding any meetings called by the Contractor that include Fort Carson customers; or Federal, State, and local agencies and their representatives.

Minutes for all meetings between the Contractor and Government shall be prepared and submitted to the KO and COR NLT three days after meeting IAW C.8, report C.1.4.1.1.

C.1.4.1.2. The Contractor shall provide performance update meetings on a weekly basis, unless cancelled by the Government. These update meetings shall provide metrics on meeting contract performance requirements to the Government including status of work being performed and any issues completing work or meeting contract performance. There shall be data indicating performance in each shop, facility maintenance, barracks repairs, service order completion, service order backlog, and other information. Specific meeting information will be discussed with the Contractor.

C.1.4.2. Work Schedule and Hours of Operation

Contractor work schedules shall vary to accommodate requirements as specified in the functional areas. Variances in working hours, other than those specified in the functional areas, must be approved by the KO and submitted 30 days in advance of the proposed change. Core operational hours for Fort Carson are 7:00 a.m. to 5:00 p.m. Monday through Friday excluding Federal Holidays.

C.1.4.2.1. Federal Observed Holidays

During these Holidays, the Contractor is expected to only perform services that are identified to be 24/7 operations such as utility plants and customer service operations. If emergencies are reported, the Contractor is required to respond during holidays.

New Year's Day	Labor Day
Martin Luther King Jr.'s Birthday	Columbus Day
President's Day	Veteran's Day
Memorial Day	Thanksgiving Day
Independence Day	Christmas Day

When a holiday falls on a Sunday, the following Monday will be observed as a holiday, and, when a holiday falls on a Saturday, the preceding Friday is observed as a holiday by U.S. Government agencies. Days of Non-Scheduled Activities (DONSA) are not considered Federal holidays.

C.1.4.3. Curtailment of Operations

When Fort Carson's operations are curtailed due to manmade or natural conditions on a working day, it may be necessary for the Government to interrupt that day's work under this contract. The Contractor shall reschedule the work on any day acceptable to the Government. If the KO agrees, the Contractor may continue the work as scheduled. The KO will provide verbal announcements of immediate curtailment of operations to the Contractor.

C.1.4.3.1. Essential Services

The following services are identified as essential during the performance of this contract IAW DFARS Clause 252.237-7023. Continued performance of these services without disruption is required during curtailment of operations. The Contractor is required to provide a written plan to the KO identifying how they will perform these services when notified of curtailment of operations by the KO. The written plan will be incorporated into the contract

C.5.1.7.1.2 Priority 1 DMO's	C.5.7.14 Microgrid Operation and Maintenance
C.5.2.1 Work Reception and Customer Services	C.5.8.1 Heating Systems
C.5.5.4 Snow Removal and Ice Control	C.5.8.2 Refrigeration, Ventilation and Air Conditioning
C.5.7.1 Primary Electrical Distribution	C.5.9.5 Fire Suppression Systems
C.5.7.3 Airfield Lighting System	C.5.10 Gas Distribution and Regulation System
C.5.7.5 Emergency Generator Systems	C.5.13 Sanitary Wastewater Treatment and Collection System
C.5.7.7 Uninterruptible Power Supply Systems	C.5.15 Potable Water Systems and Distribution

C.1.4.3.2. Fort Carson may experience delayed reporting or base closures due to inclement weather or other instances as directed by the Senior Fort Carson Commander. The Contractor may or may not follow the Senior Commander's reporting instructions for non-essential personnel. Personnel involved in performing the essential services identified in this section shall report as regularly scheduled. The Contractor shall develop a company policy regarding reporting procedures for its personnel. Under a cost reimbursement contract, the Government will only pay for direct labor hours associated with performance of this contract. The Contractor shall not bill the Government for any hours not performed as a result of the delay or closure. Delayed reporting does not release the contractor from meeting priority 1 and 2 service order response requirements.

C.1.4.4. Days of Non-Scheduled Activity (DONSA)

Fort Carson has historically granted military personnel (Enlisted and Officers) DONSA (also referred to as training holidays) in addition to Federal holidays. DONSA are not considered a day of excused absence for the Contractor's employees. DONSA may affect access to Fort Carson facilities. If a DONSA affects scheduled contract work or access to facilities, the work in the affected facilities may be rescheduled to compensate for the DONSA. Examples of Fort Carson DONSA are as follows: Friday after Thanksgiving, Friday before Memorial Day, and Friday before Labor Day.

C.1.4.5 Reserved

C.1.4.6. Non-interference with Government Operations

Contractor operations shall not unduly interfere with Government work in any area where the Contractor is performing any repair or maintenance work. The Contractor shall coordinate with Government operations to minimize interference with Government work. See C.5.1.11 for further information.

C.1.4.7. Interfaces

C.1.4.7.1. Customer Service

The Contractor shall provide timely, courteous, professional customer service in all situations dealing with customers serviced by DPW.

C.1.4.7.2. Other Contractors

The Contractor shall be responsible for any support services to other Contractors within the scope of this contract as required by the Government. There are other Contractors performing services on Fort Carson not associated with this contract but are serviced by the contract based on their assigned facilities or contract requirement. Some examples are Contractors who deliver and pick up material on Fort Carson, Custodial Contractor, Other Directorate Service Contractor's and construction Contractors performing specialized and one-time jobs.

C.1.4.7.3. Disagreements with Customers or Other Government Contractors

The Contractor shall verbally notify the COR of unresolved coordination in receiving support from, or providing support to, customers or other Contractors within two working hours from the time the disagreement occurs. In the event a disagreement between the Contractor and others cannot be resolved by the Contractor, the KO will resolve the disagreement.

C.1.4.8. Contract Data and Information

The Contractor shall make all Contractor generated technical records, reports, files, and other documentation available to the COR or other authorized Government representatives. The Contractor shall obtain KO approval before releasing any information related to this contract to other Government activities or agencies, other Contractors, or private parties. The Contractor shall not use this data for other commercial purposes.

C.1.4.9. Marking Proprietary Information

The Contractor shall clearly mark proprietary records, files, reports, and data as "Proprietary." The Government will make the final determination of the appropriateness of proprietary claims by the Contractor. The Government has the right to verify, challenge, and validate asserted restrictions.

C.1.4.10. Software

DPW corporate automation technology is Microsoft (MS) Windows based, and includes systems listed in C.7, TE-1.4.10. Electronic communications, report submissions, or data required by this contract, shall conform to current Department of the Army (DA) software. Certificate of Networthiness (CoN) is required to add any commercial software to the Government network along with KO approval. The Contractor shall provide all maintenance service agreements and software updates required for this contract.

C.1.4.11. Personnel

The Contractor shall provide a work force possessing skills, knowledge, and training to satisfactorily perform services required by this contract. The Contractor shall insure all employees have and maintain a valid state driver's license to operate a motor vehicle on Fort Carson. The Contractor shall provide its personnel with specific training, licenses, certifications, and clearances as required in this contract. Personnel absence at any time for any reason shall not constitute an excuse for non-performance under this contract. Trades workers shall be journeyman level, or higher. However, the Contractor's classification and employment policies and practices shall avoid requirements for personnel with greater qualifications or skills than necessary to perform the PWS tasks.

C.1.4.11.1. Organization Chart and Employee Roster

The Contractor shall provide the KO and COR an organization chart and employee roster five days prior to contract start, again annually by 1 October and when requested IAW C.8, report C.1.4.11.1.

C.1.4.11.1.1. Key Personnel

At a minimum, the positions listed in C.1.4.11.1.2 shall be identified as key personnel as incorporated into Section I, Contract Clauses, Local Clause 5152.204-4001 Key Personnel Requirements. Key personnel shall be available during business hours Monday through Friday except federal holidays or when the Government facility is closed for administrative reasons.

C.1.4.11.1.2. Qualifications

The Contractor shall ensure their personnel maintain the proper qualification, licenses, certification, and clearances necessary to perform the requirements identified in this PWS. Qualifications for identified key personnel are:

Project Manager	Minimum of 10 years operations experience with minimum of two years as the PM on Operation and Maintenance (O&M) contract greater than \$20 million per year. Minimum education: Bachelor's degree
Alternate Project Manager	Minimum of five years operations experience with minimum of one year as the APM on Operation and Maintenance (O&M) contract

	greater than \$10 million per year. Minimum education: Bachelor's degree
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C.1.4.11.2. Automated Systems Access

Selected contract personnel will be required to access the Fort Carson and DPW network. Permissions will be provided based on access requirements. The Contractor shall provide a list of personnel requiring access to the Government provided databases and systems to the COR NLT five days after phase-in start and as necessary within five days of new hire IAW C.8, report C.1.4.11.2. The Contractor shall be responsible for requesting the investigation or clearance that is required through the G2 Installation Security Personal security prior to any access to Government databases per AR 380-5 Department of Army Information Security Program and AR 25-1 Army Information Management. The Contractor shall be aware that the process to gain automated system access is a lengthy process, and may not be in place prior to contract start.

Before Common Access Card (CAC) issuance, the contractor employee requires, at a minimum, a favorably adjudicated National Agency Check with Inquiries (NACI) or an equivalent or higher investigation in accordance with Army Directive 2014-05. The contractor employee will be issued a CAC only if duties involve one of the following: (1) Both physical access to a DoD facility and access, via logon, to DoD networks on-site or remotely; (2) Remote access, via logon, to a DoD network using DoD-approved remote access procedures; or (3) Physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. At the discretion of the sponsoring activity, an initial CAC may be issued based on a favorable review of the FBI fingerprint check and a successfully scheduled NACI at the Office of Personnel Management.

All contractor employees with access to a government info system must be registered in the ATCTS (Army Training Certification Tracking System) at commencement of services, and must successfully complete the DOD Information Assurance Awareness prior to access to the IS and then annually thereafter.

C.1.4.11.2.1. Data Rights

The Government has unlimited rights to all documents and materials produced under this contract. All documents and materials produced under this contract shall be Government owned and are the property of the Government with all rights and privileges of ownership and copyright belonging exclusively to the Government. These documents and materials may not be used or sold by the Contractor without written permission from the KO. All materials supplied to the Government shall be the sole property of the Government and may not be used for any other purpose. These rights do not abrogate any other Government rights. Any software or data installed on a Government computer that was created by the Contractor shall become property of the Government and the Government will have the right to continue to use the software or data free of charge even after the end of this contract.

C.1.4.11.3. Employment Limitations

C.1.4.11.3.1. Employment of Aliens

The Contractor shall not employ any alien who is not lawfully employable in the US.

C.1.4.11.3.2. Language Requirement

The Contractor shall employ persons able to speak, read, write, and understand English for those positions interacting with Government personnel and customers in the performance of this contract (i.e., where English is used or essential to provide a service in this contract).

C.1.4.11.3.3. Employment of Military or Government Personnel

Employment of off-duty military personnel or Government civilian personnel is generally permissible provided such employment does not violate DoD Regulation 5500.7-R Joint Ethics Regulation (JER). Further, such employment cannot involve the holding of a concurrent Federal civilian position nor result in the direct receipt of Federal compensation for work performed for a Government Contractor.

C.1.4.11.3.4. Installation Access

Access to the Installation requires Contractor employees to present a Government issued Common Access Card (CAC), military retiree identification card, dependent identification card or an issued Installation Access Badge/Pass. The Contractor-furnished identification badge is not considered a valid identification for access. It is used solely for ease of identification of employees on job sites.

The Contractor and all associated sub-contractors employees shall provide all information required for background checks to meet installation access requirements to be accomplished by installation Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must comply with all personal identity verification requirements (FAR clause 52.204-9, Personal Identity Verification of Contractor Personnel) as directed by DOD, HQDA and/or local policy.

If a CAC is not required: Commanders and directors will not grant visitors unescorted installation access without the required identity proofing, vetting against the National Crime Information Center Interstate Identification Index (NCIC-III), the Terrorism Screening Data Base (TSDB) and determination of a valid purpose for entry for all personnel who do not possess a CAC, another Federal personal identity verification card, or other DOD ID card. Non CAC-holding visitors, contractors, vendors, and other personnel as described in paragraph 8-4 of AR 190-13, must have a need validated by a DOD component for one-time, intermittent, or routine physical access to an Army installation. The Contractor and all associated sub-contractors employees shall comply with adjudication standards and procedures using NCIC-III and TSDB (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative).

Contractor employees will obtain an Installation Access Badge/Pass in accordance with local installations procedures.

All personnel are subject to inspection at any time while entering the installation. All vehicles are required to have current vehicle registration and proof of insurance in the vehicle.

An employee's inability to obtain entrance to a Government installation because he/she does not have the required documents shall not excuse untimely performance of the requirements of this contract. Contractor employees may be denied entry to Fort Carson by Military Police (MP) if it is determined that such entry may be contrary to good order, discipline, or the security of Fort Carson.

In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

All Contractor employees requiring access to Army installations, facilities, and controlled access areas shall complete Anti-terrorism (AT) Level I awareness training within 60 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The Contractor shall complete required refresher training. The Contractor shall maintain certificates of completion, for government review, for each affected Contractor, and subcontractor, employee for the duration of the contract. The Level I AT awareness training course (JS-US007-14) is hosted on the Joint Knowledge Online Learning Management System for CAC holders at <https://Jkodirect.jten.mil>. Login and Search for the course on the Course Catalog tab via the number or key word, enroll, and Launch. A standalone version of the course for Non-CAC holder is also available at this link <http://jko.jten.mil/courses/at11/launch.html>. An access link to this course is also located on the Joint Knowledge Online login page. The standalone version is intended for Non-CAC users without a Joint Knowledge Online account ONLY! The standalone course MUST be completed on the computer in which it is started. Course progress is not transferable between computers. The standalone course does provide a completion certificate but will not be tracked on Joint Knowledge Online and no record will be maintained.

The Contractor shall brief all its employees on the local iWATCH program (training standards provided by the requiring activity Anti-Terrorism Officer (ATO)). This local developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within 60 calendar days after contract performance start and within 30 calendar days of new employees commencing performance with the results reported to the COR NLT 75 calendar days after contract award.

C.1.4.11.3.5. Employee Identification Badges and Uniforms

Contractor employees shall be easily identifiable as Contractor employees by wearing the Contractor-furnished identification (ID) badge on the upper front outer garment. Any Contractor-furnished ID badge shall include the Contractor's name and the employee's name, photo, and duty position. The Contractor shall ensure that each Contractor employee obtain a Contractor-furnished ID badge prior to beginning performance.

The Contractor shall establish a dress code for employees ensuring employees shall not wear clothing, in whole or in part, that resembles military uniforms or clothing that would be viewed as unsafe, obscene or inappropriate while performing work under this contract. Uniforms will not be provided by the Government or reimbursed to the Contractor for employees other than for Weed Control personnel and Arc Flash (NFPA 70E) rated uniforms for electricians or as approved by the KO

Contractor Weed Control personnel shall wear uniforms in the performance of their work in accordance with Environmental Protection Agency (EPA) standards. The Weed Control duty uniform shall be worn at all times when handling or applying chemicals. The uniform shall be removed at the end of the workday (or more frequently, if directly contaminated by pesticides) and placed in the laundry basket. Weed Control uniforms shall be laundered in accordance with the Installation Pest Management Plan. Laundry facilities are available in the weed control building and shall be used by the Contractor for this purpose. Weed Control uniforms shall not be worn or taken off the Installation.

C.1.4.11.3.6. Specific Personnel Requirements

All Contractor personnel shall be fully qualified to perform the work in which they are assigned under this contract. Typical license and certification requirements are listed at C.7, TE-1.4.11.3.6)

C.1.4.11.3.7. Conduct of Personnel

The Contractor shall be responsible for the performance and conduct of Contractor and Subcontractor employees at all times. Personnel employed by the Contractor in the performance of this contract or any representative of the Contractor entering Fort Carson shall abide by the security regulations listed in the contract and shall be subject to such checks by the Government as deemed necessary. Contractor personnel, their vehicles, and their property shall be subject to search and seizure provisions of AR 190-13, The Army Physical Security Program upon entering the confines of Fort Carson or PCMS, while on Fort Carson or PCMS and upon leaving the confines of Fort Carson or PCMS.

C.1.4.11.3.8. Personnel Removal

Government rules, regulations, laws, directives, and requirements which are in place or issued during the contract term relating to Fort Carson law and order, administration, and security shall be applicable to all Contractor employees or representatives who enter Fort Carson or who travel using Government transportation. Violation of such rules, regulations, laws, directives, or requirements shall be grounds for removal from the work site. The Government reserves the right to require removal from the job site of any Contractor personnel who endanger persons or property, or whose continued employment is inconsistent with the interest of military security. The Government also reserves the right to require the Contractor to remove from the job site any employee found to be under the influence of alcohol, drugs or any other incapacitating agent or

any employee involved in the theft of Government property. Such removal of employees shall not relieve the Contractor of the requirement for timely performance of contract tasks.

C.1.4.11.3.9. Removal by Fort Carson Commander

The Fort Carson Commander may, at their discretion, bar an individual from Fort Carson for conduct that is determined contrary to good order, discipline, or Fort Carson security.

C.1.4.11.3.10. Removal by the Contracting Officer

The KO may require the Contractor to remove an employee from the work site for reason of misconduct or security. Contractor employees shall be subject to removal from the premises upon determination by the KO that such action is necessary in the interest of the Government.

C.1.4.11.3.11. Entrance Denial by Military Police (MP)

Contractor employees may be denied entry to Fort Carson by the MP if it is determined that such entry may be contrary to good order, discipline, or the security of Fort Carson.

C.1.4.11.3.12. Loitering

Contractor employees shall not loiter in any working or patron area. Upon completion of their assigned duties, Contractor employees shall depart the facility. The Contractor shall allow only authorized personnel to be present in its work areas.

C.1.4.11.3.13. Fraud, Waste, and Abuse

Contractor personnel shall not be party of and be alert to suspected situations of fraud, waste, abuse, and dishonest conduct observed during or in the performance of this contract and report fraud, waste, and abuse to the COR.

C.1.4.11.3.14. Tobacco Use and Drug Free Work Place Act of 1988

The Contractor shall comply with Fort Carson policies for controlling tobacco use in Government buildings, vehicles, and facilities covered in AR 600-63 Army Health Promotion. Although the State of Colorado has legalized the use and possession of marijuana, the Contractor shall comply with the Drug Free Workplace Act of 1988(48 C.F.R. §§ 52.223-6 and 252.223-7004), which requires contractors to develop and maintain drug-free workplace programs. Such programs must include, among other features, disciplinary action or rehabilitation following a criminal drug conviction and a prohibition on the possession or use of controlled substances in the workplace. This includes marijuana, a Schedule I controlled substance under the Controlled Substances Act of 1970. *See* 21 U.S. C. § 812.

C.1.4.11.4. Emergency Preparedness

Pursuant to the Stafford Act; DoD Instruction Nos. 6055.06 and 3025.18, certain DOD Officials have immediate response authority in response to a request for assistance from a civil authority,

under imminently serious conditions. DoD officials may provide an immediate response by temporarily employing the resources under their control, subject to any supplemental direction provided by higher headquarters, to save lives, prevent human suffering, or mitigate great property damage within the United States. Immediate response authority does not permit actions that would subject civilians to the use of military power that is regulatory, prescriptive, proscriptive, or compulsory. These sudden or unusual events could result in a great impact upon Contractor performance and contract requirements.

The Contractor shall establish, maintain, and implement an Event Contingency Plan (ECP) and procedures to meet Government requirements during contingency operations. The plan shall be submitted to the KO and COR 15 days prior to contract start IAW C.8, report C.1.4.11.4. In the event of such contingencies, the Contractor shall continue and expand contract performance, which may include increases in Contractor-furnished property (CFP), extended work hours, and expansion of the contract workforce. The KO or other authorized warranted contracting officer (to be identified after award), will provide a verbal notice to proceed with a not to exceed dollar value. The Contractor shall provide the KO and COR with a rough order of magnitude no later than the next business day after a request to perform these services. The KO will provide a written notice to proceed based upon the submission of the rough order of magnitude.

C.1.4.11.4.1. DPW Emergency Operations Center

If requested by the KO during emergency operations, the Contractor shall have a representative on-site in the DPW Emergency Operations Center to provide required services and coordination.

C.1.4.11.4.3. Emergency, Disaster, and Unusually Severe Weather Plan

The Contractor shall perform requirements when the DPW emergency operations plan is implemented in the event of a national emergency, manmade or natural disasters, utility outages, or unusually severe weather so Fort Carson can continue mission requirements. The Contractor shall perform requirements with their personnel; accomplish all functions in the contract; and respond to any new demands including increased CFP, extended work hours, and expansion of the Contractor work force. Identifiable costs of performing under these abnormal situations may be separately negotiated as determined by the KO.

C.1.4.11.4.4. Strike Contingency Plan

The Contractor shall plan for labor strikes that would affect upon either the Government or Contractor ability to perform work associated with this contract. The Contractor shall submit a Strike Contingency Plan and update this plan as changes occur and shall submit a copy of the proposed plan to the KO and COR NLT start of contract performance IAW C.8, report C.1.4.11.4.4.

C.1.4.11.5. Documentation and Reports

C.1.4.11.5.1. Files and Records

The Contractor shall maintain records made available at contract start date and those generated under this contract. All such records and files shall immediately be made available for review by any agency or individual authorized access by the KO. All files maintained by the Contractor under the provisions of this contract are the property of the Government and shall be returned to the Government upon expiration or termination of this contract.

C.1.5. PERFORMANCE

C.1.5.1. Quality Control Program

The Contractor shall implement an effective Quality Control Program.

C.1.5.1.1. Quality Control Manager (QCM)

The Contractor shall designate a QCM who shall be responsible for implementing and supervising all aspects of the Quality Control Program. The QCM shall be a full time position dedicated solely to this contract. The Contractor QCM shall not be the PM. The duties of the QCM do not have to be limited exclusively to QC tasks. The QCM shall report directly to the Contractor's Corporate Headquarters and not to the PM.

C.1.5.1.2. Program Content

The program shall address all items listed in the PWS. The contract shall be International Standardization Organization (ISO) 9001-2008 Quality Management System compliant NLT the completion date of the base period.

C.1.5.2. Quality Control Plan (QCP)

The Contractor shall develop and submit a QCP to the KO and COR NLT 30 days after contract start and annually 30 days after option period start IAW C.8, report C.1.5.2. The work and any services performed shall be accepted only when in full compliance with the contract specifications. The plan shall explain the manner in which the Contractor shall ensure all contract requirements are being accomplished IAW this contract. The plan shall emphasize deficiency prevention over deficiency detection and subcontract oversight. The plan shall contain a reporting system that will provide information to the KO on Contractor performance. The Contractor shall maintain the QCP for the duration of the contract and submit any changes to the plan for KO review prior to implementation. A sustaining focus throughout the program shall be the attainment of continuous quality improvement.

C.1.5.2.1. Documentation and Enforcement

The QCP shall include methods of documenting and enforcing QC operations of both Contractor and Subcontractor work, including inspection and testing.

C.1.5.2.2. Trend Analyses

The QCP shall include methods of performing trend analysis, metrics, and assessments.

C.1.5.2.3. Customer Complaint Program

The QCP shall include a customer comments, complaint program, and processing system. The primary mechanism for this shall be the Interactive Customer Evaluation (ICE) program. The ICE program allows identification of customer complaints and requires feedback to the

Government and customers on action(s) taken to the complaint or comment. The term “customer” refers to customers internal and external to the organizations and activities identified by this contract. If the Contractor elects to use a separate method, it will be at the Contractor’s sole expense.

C.1.5.3. Quality Control Files

The Contractor shall maintain an auditable electronic file of all inspections or tests conducted by the Contractor, to include any corrective actions taken. This file shall be subject to review by Government personnel designated by the KO.

C.1.5.4. Quality Control Evaluation Meetings

The Contractor QCM shall meet with Government personnel quarterly to review contract performance. Meetings shall include review and analysis of key process indicators, analysis of process deficiencies, financial compliance and problem resolution. At these meetings, Government personnel and the Contractor will discuss Contractor performance as viewed by the Contractor and Government along with any problems arising from Contractor performance. The Contractor and the Government shall make a mutual effort to resolve any problems identified. Provide a copy of the slides and minutes to the KO and COR seven days prior to meeting IAW C.8 report 1.5.4.

C.1.5.4.1. Quality Control Meeting Frequency

The QC evaluation meetings shall be held quarterly or as directed by the KO. The Contractor shall prepare minutes of what transpired in the meeting and submit them to the COR within five working days of the meeting. The COR or KO may require the Contractor to add or change the minutes and both parties shall sign the final iteration prior to the next scheduled meeting as agreement to their accuracy.

C.1.5.5. Government Quality Assurance (QA)

The Government intends to monitor Contractor’s performance under this contract by performing checks contained in a Government’s Quality Assurance Surveillance Plan (QASP) and the performance requirements summary (Attachment –3). Typical procedures include random sampling, planned sampling, scheduled inspections, observations, and validated customer comments. A report of the Government QA results will be provided to the Contractor.

C.1.5.5.1. Unsatisfactory Performance

When contract performance is unsatisfactory or when performance progress is reported to be unsatisfactory, the Contractor will be given a Contract Discrepancy Report (CDR) or Customer Complaint Record. Return completed response to the CDR or complaint as required by the KO.

C.1.6. COMPLIANCE

The Contractor and its employees shall comply with all applicable Federal, State, and local laws while engaged in the performance of this contract. Where laws and regulations are specifically referenced in C.5, they appear in descending order of precedence. Generally, the order of precedence for laws and regulations are Federal, State, local laws, Army regulations, and Fort Carson regulations.

C.1.6.1. Inspection by Government Agencies

The Contractor shall be subject to inspections, audits, and work interruptions by other Government agencies. The Contractor shall provide access to Government-owned facilities and property and cooperate with visiting Government personnel conducting such activities. Inspections, audits, and similar activities may be made by agencies including, but not limited to, the Inspector General (IG), Army Audit Agency (AAA), US Army Corps of Engineers (USACE), Occupational Safety and Health Agency (OSHA), Environmental Protection Agency (EPA), Colorado Department of Labor and Industries, Colorado Department of Public Health and Environment (CDPHE), General Accounting Office (GAO), Internal Review and Audit Compliance Office, Safety Security Integrated Sustainment Maintenance, El Paso County Health Department, US Criminal Investigation Command, Installation Management Agency, and Forces Command (FORSCOM). Such inspections, audits, and work interruptions will not unreasonably delay the work.

C.1.6.1.1. Notification of Inspection, Audit, or Other Activity

C.1.6.1.1.1. Planned Visits

The Contractor shall notify the KO and COR of visits, audits, investigations, or other activities required by Federal, State, local laws, or any non-Fort Carson agencies within four hours of notification IAW C.8, report C.1.6.1.1.1. Notification shall include the name(s), agency(s) of the visitor, reason for visit, and any remarks.

C.1.6.1.1.2. Unplanned Visits

The Contractor shall notify the COR by phone within 15 minutes of unannounced arrival of any agents of any regulatory agency at Government facilities operated by the Contractor. The Contractor shall receive and cooperate with such agents unless otherwise instructed by the KO.

C.1.6.1.1.3. Visit Report

The Contractor shall prepare and submit a detailed visit report to the KO and COR NLT COB the day following the visit IAW C.8, report C.1.6.1.1.3.

C.1.6.2. Environmental Protection

The Contractor and its Subcontractors shall protect environmental resources including, but not limited to, air, water, land, and historical and cultural resources in the performance of work during the entire period of this contract. The Contractor and its Subcontractors shall comply with all applicable Federal, State, and local environmental, natural, and cultural resources, and historic preservation laws and regulations. FC Regulation 200-1 Environmental Management and Protection identifies the environmental protection requirements applicable to Fort Carson. In addition AR 200-1, Chapter 6 outlines applicable requirements for cultural resources.

C.1.6.2.1. Environmental Management System (EMS)

The Contractor shall establish and implement an EMS IAW Executive Order (EO) 13423 Strengthening Federal Environmental, Energy, and Transportation Management. The EMS shall incorporate practices to increase energy efficiency, reduce fuel consumption, conserve water, reduce hazardous material usage, as stated in the EO, Section 2 paragraph (a), (d), and (e). Other efforts identified in the EO the Contractor may support such as Section 2 paragraph (c), (f), (g), (h), and Section 3 paragraph (a), will aid Fort Carson and the DoD in meeting the requirements. All Contractor plans, policies, and procedures shall incorporate ISO 14000 Environmental Management.

C.1.6.2.2. Environmental Protection Program (EPP)

The Contractor shall establish and implement an EPP for the execution of this contract. All environmental protection matters shall be coordinated with the COR and the Fort Carson Environmental Coordinator. The EPP shall comply with AR 40-5 Preventive Medicine, AR 200-1 Environmental Protection and Enhancement, AR 200-2 Environmental Effects of Army Actions, AR 200-3 Natural Resources: Land, Forest and Wildlife Management, AR 200-4 Cultural Resources Management, AR 200-5 Pest Management, and with all Federal, State, and local laws, regulations, and Fort Carson plans regarding pollution control, clean air, clean water, toxic substances control, resource conservation and recovery, and other environmental matters. The Contractor shall prepare a comprehensive EPP to the KO and COR 15 days prior to contract start and 10 days prior to change describing policies and procedures IAW C.8, report C.1.6.2.2.

The Contractor shall coordinate with the DPW-ED Cultural Resource Manager to assure compliance when necessary. These areas are the three incinerator buildings 3850, 3851, and 3852 on the main cantonment, any work in the Turkey Creek Ranch area, to include road and landscape maintenance.

C.1.6.2.3. Citations

The Contractor shall resolve all citations and violation notices issued by regulatory authorities arising from its operations or the facilities for which it is responsible under this contract, unless it is established that the Government or its personnel or any previous Contractor or its personnel are at fault in the underlying violations. The Contractor shall notify the COR immediately of any requirement by environmental regulatory officials to cease any performance under this contract and within one working day of any other notice of citation or violation. The Contractor shall not agree to resolution of any environmental violation cited by an agency other than Fort Carson without prior written approval of the KO.

C.1.6.2.4. Fines and Penalties

The Contractor shall be solely responsible for any penalties levied for noncompliance in performance of the contract. The Contractor shall also be responsible for Subcontractor compliance with the environmental requirements of this contract and any fines or penalties levied as a result of the Subcontractor's work or actions in performance of the contract. Such fines and penalties shall not be reimbursable costs under this contract.

C.1.6.2.4.1. Payment

The Government will pay fines or penalty charges associated with Government violations. For violations involving both Contractor and Government fault, if the parties cannot agree on respective responsibility, the KO shall issue a final decision on the matter, which shall be subject to appeal under the Disputes Clause 52.233-1 of this contract.

C.1.6.2.4.2. Contract Action for Negligence

Environmental fines or penalty charges, assessed to the Contractor, associated with citations issued by Federal, State, or local officials for faulty or negligent operation or maintenance practices by the Contractor does not limit the Government's right to take other appropriate contract action.

C.1.6.2.5. Information Requests from Environmental Regulatory Agencies

The COR is the Government's point of contact for environmental regulatory agencies. The Contractor shall submit information requested by environmental regulatory agencies to the KO and COR NLT one day after receipt of request with requested information NLT five days after request IAW C.8, report C.1.6.2.5.

C.1.6.2.6. NetZero Initiatives

The Contractor will comply and enforce Fort Carson energy and water policies and procedures established through annual energy and water policy letters. These policies are based on and supported by Army Regulation 420-1 for Army Facilities Management, the Energy Independence and Security Act of 2007 and the Energy Policy Act of 2005. All facilities operated by the contractor will have a facility manager assigned to ensure best management energy and water practices are implemented. Each facility manager must take the DPW Facility Management class within three months of assignment.

C.1.6.2.6.1 Refuse

The Contractor shall provide dumpsters required for construction-type debris generated by repairs and miscellaneous services performed under this contract. The Contractor may place trash (as defined in FC Regulation 200-1), excluding recyclable material, in the dumpsters placed around facilities and Post. However, emphasis shall be placed on recycling all material to reduce amount of material going to landfills.

C.1.6.2.6.2 Recycle

The Contractor shall recycle all materials generated during contract performance to include office paper, cardboard, plastics, and metals supporting established NetZero initiatives. Any and all activities will be conducted with NetZero initiatives for Waste, Water, and Energy incorporated into operations.

The Government will provide recycle containers for inside of facilities. The Contractor is required to deliver inside containers to building 155, Fort Carson Recycle Center, to empty. The Government will provide recycle dumpsters for outside the facility when material generation exceeds three cubic yards per week. The DPW Recycle Program Manager can assist in developing a robust recycle program.

C.1.6.2.6.3 Energy

The Contractor shall minimize energy waste by turning off computers at night and computer monitors when not in use; turn off interior and exterior lights when not needed; keep bay doors and exterior doors closed when heating or cooling systems are on; and eliminate the use of personal refrigerators and space heaters.

C.1.6.3. Safety

C.1.6.3.1. General

It is Army policy that Government safety and health responsibilities in Contractor facilities and Contractor operations on Government property are limited to helping ensure the safety of Government-owned equipment, protecting the production base, protecting Government property and on-site Government personnel from accidental losses, and protecting the public. The Contractor is responsible for the safety and health of its employees, including, but not limited to, Subcontractors, and protection of other personnel, including the public, at Contractor facilities and worksites. The Contractor is directly responsible to Federal or State OSHA for the safety and health of its employees. The Army does not exercise direct control over the safety and health of Contractors, except in those instances when Contractor activities create the potential for accidental injury or property damage to the Government or public. Conversely, it is in the interest of the Army that its Contractors have strong safety and health programs since this will help ensure the quality of work performed and protection of Army personnel and property. The specific areas of safety concern pointed out in the following paragraphs are not intended to be all-encompassing. The Contractor is responsible for formulating and executing a comprehensive safety program for its activities under this contract.

C.1.6.3.2. Safety Program Management

The Contractor shall comply with and be responsible for initiating and maintaining a safety and health program that complies with 29 Code of Federal Regulations (CFR) 1910 Occupational safety and Health Standards, 29 CFR 1926 Safety and Health Regulations of Construction, Engineer Manual (EM)-385-1-1 Safety, and US Army and Fort Carson safety and health requirements.

C.1.6.3.3. Safety Plan

The Contractor shall develop and submit a comprehensive Safety Plan to the KO and COR 15 days prior to contract start and 10 days prior to change IAW C.8, report C.1.6.3.3. In cases where standards or regulatory guidance conflict; the stricter requirement shall apply.

C.1.6.3.4. Procedures for OSHA Inspections of Contractor Operations and Facilities

The Contractor shall respond to any citations issued by Federal or State OSHA officials for violations. The Contractor shall provide to the KO full information regarding citations issued to the Contractor or its Sub-contractors involving GFE, facilities, or other property. Unannounced inspections by Federal or State OSHA officials can be anticipated during the performance of this contract. The Contractor shall be responsible for resolving issues related to citations or requests for delays, variations, tolerances, or exemptions of applicable OSHA standards. The Contractor shall provide verbal notification to the COR within one hour of OSHA visit following with a written report to the KO and COR the day following the visit by IAW C.8, report 1.6.3.4.

C.1.6.3.5. Escort

Federal and State OSHA officials may be accompanied on their inspections and investigations by representatives from the Fort Carson Safety Office and the COR (as appropriate) and, where requisite security clearances are verified, may be accompanied by representatives of the Contractor and its employees.

The Contractor shall provide escorts in hazardous areas where the Contractor has prime work responsibility (e.g., vaults and equipment rooms) for Government and other Contractor personnel who are performing work in these areas.

C.1.6.3.6. Contractor Identified Hazardous Condition

If, at any time during performance of this contract, hazardous conditions not previously identified are discovered by the Contractor or made known to it by any Federal or State agency responsible for enforcing OSHA, or if any alternate safety measures are determined to be inadequate by any Federal or State agency, the Contractor shall report such conditions or determinations within 72 hours of identification IAW C.8, report C.1.6.3.6.

C.1.6.3.7. Fire Safety

The Government will provide fire prevention and protection services. However, the Contractor shall safeguard and maintain all Government property and provide for the safety of Government and Contractor personnel within the areas and facilities assigned to Contractor operations. The Contractor shall comply with NFPA Codes.

C.1.6.3.7.1. Fire Prevention and Protection Plan

The Contractor shall establish and maintain a comprehensive Fire Prevention and Protection Plan IAW AR 420-1. This plan shall complement the Fort Carson Fire Prevention and Protection Program. The Contractor shall implement the plan in each Government-furnished, Contractor-operated facility. Submit to plan to the KO and COR 15 days prior to Contract start and 10 days prior to changes IAW C.8, report C.1.6.3.7.1.

C.1.6.3.8. Reporting Requirements

C.1.6.3.8.1. Accident or Incident Reporting

The Contractor shall comply with AR 385-10, OSHA, and other regulatory agency requirements for record keeping and reporting of accidents or incidents in the course of Contractor performance, which result in death, trauma, occupational disease, property damage, or environmental damage. The Contractor shall provide a verbal report to the COR within one hour of occurrence and an AGAR report within three days IAW C.8, report C.1.6.3.8.1.

C.1.6.3.8.2. Third Party Accident Claims

If any claims are made by a third party against the Contractor as a result of an accident or incident which occurs in connection with Contractor performance, the Contractor shall submit a full written report to the KO and COR within 24 hours of receiving claim IAW C.8, report C.1.6.3.8.2.

C.1.6.4. Hazard Communication Program

The Contractor shall develop and implement a Hazard Communication Program IAW 29 CFR 1910.1200 Toxic and Hazardous Substance – Hazardous Communication 1926.59, FC 200-1 Health and Environmental Controls – Hazardous Communication, and FC Regulation 385-10 The Army Safety Program.

C.1.6.4.1. Hazardous Material Inventory/POL Inventory

The Contractor's inventory of hazardous materials stored on Fort Carson as required in FC Regulation 200-1 and shall be maintained at levels not to exceed 30 day usage requirements. Exceptions to the 30 day usage requirement will be identified in the Environmental Operating Permit and approved by the KO. The Contractor shall prepare a monthly report of its inventory of hazardous materials issued and provide to the KO and COR by the 5th business day of each month IAW C.8, report C.1.6.4.1.

C.1.6.4.2. Hazardous Material Spill Plan

The Contractor shall develop a detailed spill plan with site map for each contractor assigned facility that stores hazardous materials IAW FC Regulation 200-1 and the Fort Carson Spill Prevention Control Countermeasures Plan. The plan shall be submitted to the KO and COR 15 days prior to Contract performance start and 10 days prior to changes IAW C.8, report C.1.6.4.2.

C.1.6.4.3. Energy Conservation Program

The Contractor shall comply with AR 420-1, Army Facilities Management, and FC Regulation 200-1 Environmental Protection and Enhancement. The Contractor shall instruct its personnel in energy conservation practices, and require them to operate utilities in a manner that precludes wasteful use of energy. The Contractor shall only purchase Energy Star rated appliances in performance of this contract. Supporting Fort Carson's NetZero energy initiatives and goals is required during the performance of this contract. Refer to C.1.6.2.6.

C.1.7. SECURITY

The Contractor shall implement all plans required in this section on contract performance start date. This section identifies security of Information Technology, personnel, property and security clearances for Contractor efforts during the performance of this contract.

C.1.7.1. Information Systems Security (ISS) Compliance

The Contractor shall comply with all DoD, DA, FORSCOM and local Installation Automation Manager (IAM) instructions and regulations. The Contractor shall ensure that there is full compliance with Deputy Secretary of Defense Memorandum, "Disposition of Unclassified DoD Computer Hard Drives," dated June 4, 2001. Compliance with AR 25-2 Information Assurance must be met within 15 days prior to contract start. An Accreditation Document and Security Plan, as identified in AR 25-2 will be submitted to Network Enterprise Center (NEC) for approval. Network connectivity will not be granted until this task has been completed. For Government equipment on the Fort Carson network backbone, the National/DoD/Army Security Requirements under AR 380-5, Department of the Army Information Security Program; AR 25-1, Army Information Management; and AR 380-53, Information Systems Security Monitoring, are applicable.

C.1.7.2. Position Sensitivity

DoD 5200.2-R DoD Personnel Security Program, requires DoD military and civilian personnel, as well as DoD consultant and Contractor personnel, who perform work on Fort Carson systems and networks are to be assigned to positions which are designated at one of three sensitivity levels (e.g., Automated Data Processing (ADP) ADP-1, ADP-11, or ADP-111). These designations equate to Critical Sensitive, Non-critical Sensitive, and Non-Sensitive. The Contractor shall ensure that individuals complete the appropriate forms for needed background investigations. The required investigation shall be completed prior to assigning individuals to positions requiring sensitive duties. The Government retains the right to require removal of Contractor personnel from performance of duties under the contract, regardless of prior clearance or adjudication status, whose actions, while assigned to this contract, clearly conflict with the interests of the Government. The reason for removal will be fully documented in writing by the KO.

C.1.7.3. In and Out-processing Contractor Personnel

C.1.7.3.1. Obtaining Contractor Personnel Security Clearances

Security clearances are not required for Contractor employees during performance of work identified in this PWS. Access to areas that would require a security clearance must be scheduled in advance with the customer for proper clearing of the room and for escorts as appropriate.

C.1.7.4. Physical Security Plan (PSP)

The Contractor shall develop a PSP addressing physical security aspects to protect Government-furnished facilities (GFF) occupied by the Contractor and Contractor property. The PSP shall describe how the Contractor shall prevent unauthorized access, vandalism, pilferage, larceny, sabotage, arson, and active shooter situations. The PSP shall also provide for internal safeguards for the security of all GFF. The following regulations shall be followed when developing the PSP: AR 190-13 The Army Physical Security Program; and AR 190-51 Security of Unclassified Army Property (Sensitive and Non-Sensitive). The Contractor shall ensure that the PSP includes any special security requirements. The Contractor shall make the PSP available to the KO and COR at least 15 days prior to contract start date. The Contractor shall update the plan as changes occur.

C.1.7.4.1 The PSP shall include instructions for Contractor staff on addressing an Active Shooter situation. Annual training for employees shall be required and tracked by the Contractor. The Contractor may use DPW SOP #64 Active Shooter Emergency Action Plan as a guide in addressing this requirement. This SOP will be provided if requested.

C.1.7.5. Reserved

C.1.7.6. Security of Unattended Property and Records

The Contractor shall ensure all facilities under the direct control of the Contractor, including lockable containers located within joint use areas are properly locked and secured during non-working hours and when not occupied by the Contractor. The Contractor shall ensure that all records and information of personnel and personnel actions are removed from desktops and general access areas when not in use, and are properly secured at the close of each working day.

C.1.7.7. Key Control

The Contractor shall develop a Key Control Plan (KCP) and make it available to the KO and COR at least 15 days prior to contract start date. The Contractor shall update the plan as changes occur. The KCP shall include the establishment and implementation of a key control system that prevents the loss, duplication, or use of keys by unauthorized persons. The KCP shall address the key control requirements of AR 190-51.

C.1.7.8. Lost Keys

The Contractor shall notify the COR of any occurrence of lost keys to assigned facilities within 30 minutes of discovery of loss. The Contractor shall maintain records of all details of lost key occurrences. The records shall contain the key number, location(s) accessed by the key, date the key was discovered missing, and the name of the person signed for the key. Re-coring buildings due to lost keys will be a reimbursable cost to the Government.

C.1.7.9. Unauthorized Users

Use of Government keys issued to the Contractor by any person other than authorized Contractor employees is prohibited. The Contractor shall not permit entrance to locked areas by any person

other than Contractor personnel engaged in performance of work in those areas without written authorization by the KO. The KO or designated representative(s) shall have access to any Government-owned property under control of the Contractor.

C.1.7.10. Restricted and Prohibited Areas

The Contractor shall comply with Government security access requirements to restricted areas including, but not limited to, arms rooms, ammunition supply point (ASP), Sensitive Communication Information Facilities (SCIF), ranges, and training areas IAW AR 190-11 Physical Security of Arms, Ammunition, and Explosives, AR 190-13 The Army Physical Security Program, and AR 190-16 Physical Security.

C.1.7.11. Vehicle Parking

Parking for Contractor personnel is available adjacent to most buildings on a first-come, first-served, space available basis. Parking is subject to restrictions imposed by building occupants in buildings with joint Contractor and Government occupancy. The Government reserves the right to make change in parking arrangements to accommodate new or changing requirements. Contractor employees shall park POVs in designated parking areas and adhere to the installation laws and regulations.

C.1.7.12. Weapon and Firearm Control

Contractor personnel shall follow FC Regulation 210-8, Standards of Conduct, and AR 190-11, Physical Security of Arms, Ammunition and Explosives, and shall not, while on Fort Carson, transport, possess, or use privately owned dangerous weapons including, but not limited to, all forms of firearms, missile-throwing devices, switch blade or automatic blade knives, knives with blades in excess of three inches, ammunition and pyrotechnics, fireworks, bludgeons, brass knuckles, shooting pens, and similar devices. Unauthorized dangerous weapons will be confiscated and impounded by Government law enforcement and security personnel. A state concealed weapons license is not valid on Fort Carson.

C.1.7.13. Foreign Visitors

The Contractor shall not invite or bring foreign visitors (Non-US Citizens, persons in the U.S. illegally, or non-US permanent resident aliens) onto Fort Carson without prior approval of Headquarters, Department of the Army (HQDA) and the Commanding General, 4ID and Fort Carson.

C.1.7.14. Visitor Passes

Fort Carson no longer issues visitor passes. All non-commercial, non-DoD ID card holders need to enter through Gate 1. Once authorized by the security personnel to enter, they can proceed to their destination. If they leave the post, they must follow the same procedures to re- enter Fort Carson again.

C.1.7.15. Employee Terminations

When an employee is terminated from employment on the contract for any reason, the Contractor shall have an out processing checklist that identifies the Government issued CAC card and Installation Access Badge/Pass has been turned in to the COR and property items accounted for and back in proper custody. A signature shall be required on the checklist from

the appropriate person acknowledging receipt of the items. Checklists shall be retained for the duration of the contract.

C.1.8. FINANCIAL REQUIREMENTS

C.1.8.1. Accounting and Reporting Requirements

The Contractor shall provide its voucher system to the KO and COR 30 days prior to contract start IAW C.8, report C.1.8.1.

C.1.8.2. General Fund Enterprise Business System (GFEBS)

The Contractor shall use GFEBS for some financial transactions to include the following track and monitor labor (direct and indirect), materials, equipment, and subcontracting costs used in performance of this contract. Minimum information tracked shall include but not limited to costs required by the Contractor to maintain by facility number, type of work, contract line number, time, employee, customer, General Ledger (GL) Account, shop, task, and work document number.

C.1.8.3. Vouchers

Costs shall be estimated, vouchered and given one General Ledger (GL) Account per CLIN as determined by the Government. Costs shall be differentiated by Government fiscal year. The Government accounting codes will be defined by using the Installation Status Report (ISR) as provided in C.7, TE-1.8.3A and TE-1.8.3B. The ISR is reviewed on a yearly basis and the Contractor shall comply with changes requested by the Government and update the manpower matrix accordingly. The current ISR funding breakout is identified in C.7, TE-1.8.3.

C.1.8.3.2. For cost-reimbursable efforts, the Contractor shall submit requests for reimbursement no more than twice a month, on invoices and provide sufficient backup documentation to support all such requests. A briefing on financial status shall be included in the weekly status briefing as stated in C.1.4.1.2.

C.1.8.3.3. The Contractor shall have two separate GL accounts in the Management Administration area of the contract; one for Labor costs the other for ODCs. Labor costs shall include exempt (Supervisors), Union, and hourly working in the Management Administrative area. The ODC account shall be divided into subdivisions including the following:

- Main Heading for Direct Materials subdivided the same as above, but also including Direct Travel (to be used for travel directly related to the contract but not connected to training);
- Subcontracts (subdivided the same as above, but including software support);
- Postage and freight (subdivided the same as above);
- Legal Fees, Insurance Expense (separate divisions for auto and liability);
- Office Supplies (subdivisions for Office Equipment, Office Supplies Consumable, Computer Supplies, Computer Equipment), Tools and Equipment (subheading Equipment Rental); and
- Main Heading for ODCs with subdivisions including the following: Training (subdivided for costs incurred for travel and school), Drug Testing, Employment Advertising, Medical Screening, Safety, Telephone/Pagers, Criminal Background Investigation; and a Miscellaneous Account.

The Contractor shall include in their backup of the main area name (Direct Cost and Indirect Cost), the subdivision name, Voucher Number, Vendor Name, PO Number, Short Description of the Transaction, and the amount.

C.1.8.4. Reports

C.1.8.4.1. The Contractor may, if warranted by changed requirements or funding restraints, request funding adjustment to CLINS based on the agreed to Estimated Contract Values (ECVs), for the upcoming option period. Any such requests must be submitted to the KO and COR NLT three months prior to end of current option period IAW C.8, report 1.8.4.1. An example of the report is in C.7, TE 1.8.4.1. Such requests are subject to negotiation and in the absence of agreement, will be subject to final decision by the KO and appeal under the Disputes Clause of this contract. No requested change will be effective until contained in a modification to the contract issued by the KO.

C.1.8.4.2. The Contractor shall submit a financial obligations report to the KO and COR twice per month October – June, weekly July – September IAW C.8, report C.1.8.4.2 from Contractor’s non-GFEBS financial system. An example of the report is in C.7, TE 1.8.4.2.

C.1.8.4.3. The Contractor shall track Subcontract procurements and supply a report to the KO and COR NLT the 5th business day monthly IAW C.8, report C.1.8.4.3. The Contractor shall receive KO approval for any subcontract as stated in C.5.1.4.6.1. An example of the report is in C.7, TE 1.8.4.3.

C.1.8.4.4. The Contractor shall provide to the Government a quarterly report listing the Contractor’s approved purchase arrangements. This report shall be submitted to the KO and COR NLT the 5th business day following the end of the quarter IAW C.8, report C.1.8.4.4. The Contractor shall also include a copy of the purchase arrangements in the purchasing file for each purchase order.

C.1.8.4.5. The Contractor shall submit a “Burn Rate Report” to the KO and COR bi-weekly, October – June and weekly July – September IAW C.8, report C.1.8.4.5. An example of the report is in C.7 TE 1.8.4.5.

C.1.8.5. Limitation of Safety Supplies

C.1.8.5.1. The Government will reimburse the costs of one pair of prescription safety glasses for each Contractor employee who needs such glasses for the performance of work under this contract up to \$150.00 for standard lenses and \$240.00 for bifocal lenses. However, if an employee’s lens prescription changes during the course of the contract, the KO may approve and the Government may reimburse the cost of a new pair of safety glasses for that employee upon submission of proof (e.g., a new prescription).

C.1.8.5.2. The Government will reimburse no more than \$125.00 for safety boots/shoes on an annual basis for each Contractor employee who requires safety footwear in the performance of its contract responsibilities if the current safety shoes are no longer serviceable. If special safety footwear is needed based on the duties of an employee, the KO must provide approval for that special need based on contractor justification.

C.1.8.5.3. The Contractor shall provide all, PPE, and safety equipment for its employees required for use by State or Federal law, code, or regulation under this contract.

C.1.8.5.4. The Government will not provide items such as; Gatorade, bottled water, cold weather gloves, cold weather caps, jackets/coats, winter boots, etc. to contractor employees.

C.1.8.6. Relocation, Travel, and Per Diem Costs

Allowable relocation, travel, and per diem costs are generally covered in FAR Part 31. The Contractor shall not incur any relocation costs or costs associated with travel (travel includes overnight stays to PCMS or other TDY requirements), to include class registration fee, airfares, and per diem in connection with work performed on the contract, without prior written approval of the KO.

C.1.8.6.1. The Contractor shall submit requests for approval of these costs to the KO and COR NLT 14 business days prior to travel IAW C.8 report C.1.8.6.1. The Contractor shall plan all travel a minimum of 14 in advance in order to take advantage of discounted air, hotel, and rental car rates. Requests for travel to PCMS must be made no less than five days in advance. Emergency travel to PCMS must obtain verbal approval with written request to follow the NLT next business day. The Contractor will be authorized travel expenses consistent with the substantive provisions of the Joint Federal Travel Regulation (JFTR). The Contractor will be reimbursed at cost and shall provide receipts and other appropriate documentation required with invoice to support all travel costs incurred. The Contractor shall not be reimbursed for any indirect/overhead costs associated with travel.

C.1.8.7. Legal Fees

Before incurring legal fees that will result in a direct charge to the contract, the Contractor shall obtain approval of the KO. Any fees incurred without such prior approval shall not be reimbursable under this contract.

C.1.8.8. Contractor Manpower Reporting (CMR)

The Contractor will report ALL Contractor labor hours (including subcontractor labor hours) required for performance of services provided under this contract for the Department of the Army via a secure data collection site. The contractor is required to completely fill in all required data fields using the following web address: <http://www.ecmra.mil>, and then click on “Department of the Army CMRA” or the icon of the DoD organization that is receiving or benefitting from the contracted services.

Reporting inputs will be for the labor executed during the period of performance during each Government FY, which runs from October 1 through September 30. While inputs may be reported any time during the FY, all data shall be reported no later than 31 October of each calendar year, beginning with 2016. Contractors may direct questions to the help desk by clicking on “Send an email” which is located under the Help Resources ribbon on the right side of the login page of the applicable Service/Component’s CMR website at <http://www.ecmra.mil>. As part of its submission, the contractor will also provide the estimated total cost (if any) incurred to comply with this reporting requirement.

C.1.8.9. Financial Performance Evaluation

The Contractor shall include in their QC plan, evaluation of this section as stated in C.1.5.1.2.

C.1.9. SPECIAL FACILITIES

C.1.9.1. Hospital

Evans Army Community Hospital (EACH), Building 7500, is excluded from the coverage of this contract except for utility distribution systems as listed in Section C.5.

C.1.9.2. Family Housing

Family Housing is privately owned at Fort Carson. This includes all housing units, exterior facilities (to include play areas), utility service laterals leading from the point of connection to the facility, and facilities used directly for the owner's commercial or administrative benefit. Water quality to housing units is included in this PWS. All other work, in or for family housing, is excluded from this contract.

C.1.9.3. Multi-Purpose Range Complex (MPRC)

The real property facilities located at the MPRC in the southeast corner of Fort Carson are included in this contract to include the lagoons, CL2 station, pump house, deep well, storage tank, recirculating pumps, septic tanks, and utility distribution to the site.

**PERFORMANCE WORK STATEMENT
SECTION C.2
DEFINITIONS, ACRONYMS, AND ABBREVIATIONS**

INDEX

C. 2. DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

C.2.1. DEFINITIONS

C.2.2. Acronyms and Abbreviations

This section lists many of the terms and acronyms that may be used during the execution of this contract. The lists are provided for the convenience of the Contractor. In the event that the definitions listed in this section differ from those (either explicit or in context) in other provisions of the PWS, other clause and provisions of this contract, or any references included in this contract, the latter shall govern.

Word	Definition
Acceptable Quality Level	The maximum percent defective, maximum number of defects per hundred units, or the number of defects that can be considered satisfactory on the average. AQLs may be used by the Government (e.g., Quality Assurance Surveillance Plan) or by the Contractor (e.g., Quality Control Plan) as one means of measuring performance, much like other statistical decision-making tools. An AQL does not imply that the Contractor may knowingly perform in an unsatisfactory way, but does imply the Government recognizes unsatisfactory performance sometimes happens unintentionally.
Alarm Condition	A condition where an initiating device activates and puts the fire alarm control panel in alarm, sets off the building horn strobes, activates the fires suppression system, and transmits a signal to the fire department.
Account Process Code (APC)	A designated 4-character alpha-numeric code that uniquely identifies each cost account, which must be used by DPW in accounting for expenditures under this contract.
Advisory Documents	Those directives, which the Contractor may use for information and guidance but not binding for compliance.

Word	Definition
Appurtenances (Transportation Systems)	Pavement and bridge appurtenances consist of such items as ditches, culverts, and storm sewers; traffic signs, signals, markings; right of way or snow fencing; unstabilized shoulders; curbs; guard rails; cattle guards; tank crossings; and supporting embankments. Appurtenances for airfields include overrun areas, aircraft arresting gear, and tie downs in addition to the items listed above. Railroad appurtenances include roadbeds, road crossings, tracks, ballast, culverts, other drainage structures, signs, switches, lamps, safety devices, track scales and all other features and items necessary to meet operational and safety requirements. Railroad communications systems and rolling stock are excluded.
As-built Drawings	As-built drawings are the final version of construction drawings that reflect the actual construction of the system or facility.
As-is	The existing condition of a real property facility, M&S equipment or shop tools and equipment at the time of inspection inventory by the Contractor and the Government.
Bare Pavement (snow removal)	A pavement surface that bears no snow cover higher than ¼ inch above the pavement surface at any point on the pavement.
Best Management Practices (BMP)	Methods, measures, or practices to prevent or reduce the contributions of pollutants to U.S. waters. Best management practices may be imposed in addition to, or in the absence of, effluent limitations, standards, or prohibitions.
Business Hours	Contractor hours of operation during work days to perform PWS requirement. Historically these have been 0700-1530.
Bypass (waste water treatment)	The intentional diversion of waste streams from any portion of a treatment facility.
Breakdown	The stoppage or collapse of equipment of a facility, or a component thereof, that requires corrective action to restore it to an operating condition
Cannibalization	The authorized removal, under specific conditions, of serviceable and unserviceable parts, components and assemblies from material authorized for disposal. Cannibalization supplements supply operations by providing assets not immediately available through the Army supply system.
Cantonment Area	Developed portion (city like area) of Fort Carson. Area of Installation where administrative buildings, barracks and motor parks are located.
Catch Basins	A chamber or well used with storm or combined sewers as a means of removing grit which might otherwise enter and be deposited in sewers.
Central Vehicle Wash Facilities	More commonly referred to as “wash racks,” a structure designed to clean large numbers of tactical or transportation motor pool vehicles. Also refers to a large facility designated for the washing of military vehicles.

Word	Definition
Clean Fill	Fill dirt resulting from excavation work and Vector truck operations from areas containing clean ground water and soil typically associated with potable water line breaks and repairs; and potholing work.
Compliance/Compliant	A state where conditions and limits are within acceptable parameters set forth.
Connected Facilities	Those facilities connected to centralized systems including central heating, water distribution, sewage collection, etc.
Contract Discrepancy Report (CDR)	A report used to document unsatisfactory Contractor performance. The CDR requires the Contractor to explain, in writing, why performance is unsatisfactory; how performance shall be returned to satisfactory levels; and how recurrence of the problem shall be prevented in the future.
Contract Start/Contract Start Date	The first date of the base year or base period of the contract, as identified in Section B.
Contracting Officer (KO)	An individual appointed, IAW procedures prescribed by the Federal Acquisition Regulation, with the authority to enter into and administer contracts and/or terminate contracts and make determinations and findings with respect thereto.
Contractor	The Contractor, its subsidiaries and affiliates, joint ventures involving the Contractor, or any entity with which the Contractor may have merged or any individual or entity that assisted or advised the Contractor in the preparation of proposal under this solicitation. The term Contractor as used herein refers to both the prime Contractor and any Subcontractors. Contractor's Subcontractors shall comply with the provisions of the contract.
Critical Equipment and Facilities	Items of equipment or facilities that must operate continuously or throughout the respective season in order to support critical missions, the failure of which may affect the health and welfare of personnel or damage Government equipment or properties. Emergency or urgent service calls are often required to restore the critical equipment to optimum operating condition and provide the output required. Examples are: Computer Facilities, Fire Prevention and Protection Facilities, Hospitals, Electrical Plants or Systems, Water Systems, Wastewater Plants or Systems, Heating Plants or Systems and Dining Facilities.
Days	Calendar days unless otherwise specified.
Delineators	Light-reflecting devices mounted on the roadway, in series, to indicate roadway alignment. Light-reflecting units should be capable of clearly reflecting light under normal atmospheric conditions from a distance of 1,000 feet when illuminated by the upper beam of standard automobile lights. Also includes reflective glass beads that are placed on freshly painted street lines, and are able to reflect a vehicle's light within a short distance.

Word	Definition
Deluge (Fire Suppression System)	A system employing open sprinklers attached to a piping system and connected to a water supply through a valve that is opened by the operation of a detection system installed in the same areas as the sprinklers. When this valve opens, water flows into the piping system and discharges from all sprinklers attached thereto.
Direct Labor Hours	Unit of work that represents the productive effort of one person in one hour
Discharge (waste water treatment plant)	Any commercial or industrial operation, which results in the disposal of solid or liquid waste material into a sewerage system, operated by a municipality which discharges into the public waters of the state.
Downrange	Areas of Fort Carson not in the cantonment area
Dry Fire Systems	A system employing automatic sprinklers attached to a piping system containing air or nitrogen under pressure, the release of which (as from the opening of a sprinkler) allows the water pressure to open a valve. The water then flows into the piping system and out the opened sprinklers.
Duty Day	Monday through Friday, excluding Federal holidays
Effective Braking Traction	The coefficient of friction between vehicle tires and pavement surface as determined by any approved friction-measuring device.
Effluent	Wastewater (treated or untreated) that flows out of a treatment plant, sewer, or industrial outfall. Generally refers to wastes discharged into surface waters.
Emergency Asbestos Removal	An unplanned asbestos removal operation necessitated by an unexpected event that results in a public health or safety hazard, requires the need to protect equipment and vital utilities, or minimizes property damage. The encounter of unidentified asbestos containing material may precipitate an emergency asbestos removal to avoid imposing an unreasonable burden or delay on a repair, project or military-sensitive function.
End of Snow Storm	Snow storm shall be determined to be ended when the weather recording station at Butts Army Airfield has reported less than 1/10 inch (in) additional snowfall per hour for a period of two consecutive hours and blowing and drifting has stopped.
Established (Grounds)	The length of time necessary for turf or new plantings to become established varies according to species, method of planting, and site conditions. In most cases a reasonable establishment period for turf is three months or until it has been mowed three times. Establishment includes watering, weeding, and cultivating, fertilizing, control of diseases and insect pests, protection from damage by small animals, and pruning.
Expendable	Government property that is consumed in use or loses its identity in use and is dropped from stock record accounts when issued or used.
External Hot Inspection (Boilers)	Boiler is online and hot. Inspector will check for safety overrides and proper readings

Word	Definition
Fair Wear and Tear	The loss or impairment of the appearance, effectiveness, worth or utility of an item that has occurred solely because of the normal and customary use of the item for its intended purpose.
Fire (Doors)	Doors that consist of a fire door assembly, including frame and hardware, and are capable of providing a specified degree of fire protection when closed. The Fire Closing Device (e.g., Fusible Link) is usually an automatic closing device.
Fiscal Year (FY)	The Government FY begins every October 1 and ends the following September 30. The FY is designated by the calendar year in which it ends. For example, FY 2003, abbreviated FY03, begins October 1, 2002 and ends September 30, 2003.
Grounds, Improved	Acreage on which intensive grounds maintenance activities must be planned and performed annually as fixed requirements. Activities include mowing, irrigation, fertilization, cultivation, aeration, seeding, sodding, spraying, pruning, trimming, weed, dust and erosion control; drainage, planting for landscape effect, wind and sound abatement, and other intensive practices.
Grounds, Parade	Manicured areas used for ceremonies. Activities include mowing, trimming, policing, edging, irrigating, thatching, aerating, over seeding, and fertilizing. Shrub beds are maintained and pruned as needed.
Grounds, Semi-improved	Areas on which periodic recurring grounds maintenance is performed but to a lesser degree than on improved grounds. Practices normally include such cyclic variables as soil sterilization, weed and brush control, drainage maintenance, mowing for fire protection and major land repair/restoration as may result from mission activities.
Grounds, Unimproved	Acreage (including water areas, areas under buildings and surfaced areas) not classified as improved or semi-improved. Practices and intervals of attention are generally unpredictable such as might evolve from flood, fire, insects, or disease epidemics.
Hazardous Material	Any material defined as such by the Environmental Protection Agency (EPA).
Hazardous Waste	Any hazardous material no longer suited for its original intended use
High Pressure Steam (Boiler)	High pressure steam systems are those operating at pressures greater than 15 psi.
High Voltage	Electrical systems, both interior and exterior to facilities, which are greater than 600 volts.
Historic Buildings, Structures, And Landscapes	A category of Historic Property or Resource defined by the NHPA.
Hydronic	Steam and water piping systems that are part of a heating or cooling system.

Word	Definition
Installation Status Report (ISR)	A report designed to give the installation, Major Army Commands (MACOM), and Headquarters Department of the Army (HQDA) commanders and senior leaders a macro-level overview of the facility status and environmental status of Army installations.
Installed Equipment	Items of equipment or furnishings including materials for installing, which are required to make a facility usable and are affixed as a permanent part of structure; these items include, but are not limited to, plumbing fixtures and equipment, fixed fire protection systems, elevators and escalators, overhead-crane runways, lavatory counters, cabinets, fans, air conditioners, furnaces, and similar fixed equipment. Machine tools, production and research equipment, and their foundations are excluded.
Low Temperature And Pressure (Boiler)	Low temperature and pressure hot water boilers are those operating at a maximum pressure of 30 psi and a maximum temperature of 180 degrees F.
Low Voltage	Electrical systems that are 600 volts or less.
Low-Pressure Steam (Boiler)	Low-pressure steam boilers are those operating at pressures not exceeding 15 psi for steam.
Maintenance	Work required to preserve and maintain a facility, system or component in such condition that it may be effectively used for its designated functional purpose.
Mobilization	The process of assembling and placing in a state of readiness for war the manpower and material resources of a nation.
Non-Duty Days	Any Sat, Sun or Federal holiday
Non-Duty Hours	The hours after business hours during a normal duty day, and the entire 24 hour period of each non-duty day.
Normal Duty Days	Monday through Friday except for Federal holidays.
Normal Duty Hours	Established business hours, except plant operations whose normal duty hours are 24 hours a day 365 or 366 days per year.
Pest Management	The prevention and control of disease vectors and pests that adversely affect the DoD mission or military operations; the health and wellbeing of people; or structure, material, or property.
Pesticide	Any substance or mixture of substances, including biological control agents, that may prevent, destroy, repel, or mitigate pests and are specifically labeled for use by the EPA. Also, any substance or mixture of substances used as plant regulator, defoliant, desiccant, disinfectant, or biocide. Note: the AFPMB does not review or approve disinfectants or biocides per the DoD instruction.
Plant (utility)	A facility whose use is to generate or provide a utility for wide distribution to many facilities.

Word	Definition
Plant Equipment	Government-furnished property of a capital nature consisting of equipment, furniture, vehicles, machine tools, test equipment and accessory or auxiliary items, excluding special tooling and special test equipment, used or capable of use for any administrative or general plant purposes
Plant Operation (utility)	Operating utility systems and equipment to provide a continuous permit-compliant utility service. Plant operators or attendants monitor and maintain plant operations through maintenance and repair to prevent equipment and system failure. Include operational tours, etc.
Preaction (Fire Suppression System)	A sprinkler system employing automatic sprinklers attached to a piping system containing air that might or might not be under pressure, with a supplemental detection system installed in the same areas as the sprinklers. After the valve has been actuated, it opens, and water flows into the sprinkler system to be discharged from any sprinklers that are open.
Property Administrator	An individual designated in writing by the Contracting Officer to perform evaluations concerning GFP assigned to a Contractor. Acts as a technical advisor to the Contractor in these areas
Quality Assurance (QA)	Actions taken by the Government to inspect or check goods and services to determine that they meet or do not meet requirements of the contract.
Quality Assurance Evaluator (QAE)	That person appointed by the KO for Contractor surveillance
Quality Assurance Surveillance Plan (QASP)	An organized written document used by Government for quality assurance surveillance. The document contains sampling/evaluation guides and checklists.
Quality Control Program	Contractor's system to control the equipment, systems, or services so that they meet the requirements of the contract.
Quarter (FY)	Government quarters of a fiscal year run as follows: 1 Oct -31 December, 1 January – 31 March, 1 April – 30 June, and 1 July – 30 September.
Real Property	Lands and improvements to land, buildings, and facilities, including improvements and additions, and utilities systems. It includes equipment affixed and built into the facility as an integral part of the facility (such as heating systems), but not moveable equipment (such as plant equipment).
Repair Work	Work to restore a real property facility, system, or component to such a condition that may effectively be used for its designated functional purpose.
Respond (Response)	Contractor workforce is at work site with adequate personnel, tools, parts, and equipment, prepared to begin work within established time frames.

Word	Definition
Rework	Rework is a repair or maintenance effort which is required to be done based on rejection by Government Quality Assurance (QA) personnel or Contractor Quality Control. Rework is performed to resolve the identified deficiencies discovered during QA or QC inspections or validated customer notification.
Sanding (roads)	The application of sand to obtain effective braking traction.
Shall	The word “Shall” is used in connection with the Contractor and specifies that the provisions are mandatory.
Snow Berm	A deposit of snow greater than 4-inch in height.
Snow Pickup	The act of removing snow or snow berms from a roadway or parking area and depositing removed snow at a designated snow dump.
Snow Removal	The comprehensive act of removing snow from a road, parking area, sidewalk, or runway by plowing, grading, sweeping, blowing and/or picking up.
Snow Storm	Precipitation of snow at rates exceeding 1/10-in per hour, blowing and drifting snow, or a combination thereof.
Surfaced Areas	Covers all graded, paved, or stabilized (other than grass) areas used for vehicular, aircraft, track vehicle, or pedestrian traffic such as roads, streets, service drives, walks, parking areas, open storage areas, and airfield paved areas, including base and subbase courses.
Tenant/Other Organizations	Those activities or elements housed on Fort Carson not assigned to the 4 th Infantry Division, Division West, or Fort Carson in a direct or general support role (e.g. DRMO, CID, USAA, DECA, Schools, EACH, Banks, Credit Union, Colorado National Guard, Army and Navy Reserve Centers).
Wet Fire Systems	A sprinkler system employing automatic sprinklers attached to a piping system containing water or glycol and connected to a water supply so that water discharges immediately from sprinklers opened by heat from a fire.
Will	Means mandatory
Work Request	DA Form 4283 Facilities Engineering Work Request is used by Public Works customers to request work, and record work requirements.
Working Day	Monday through Friday, except Federal holidays.
Year	Fiscal Year (starting October 1) unless otherwise specified

C.2.2. Acronyms and Abbreviations

For the purpose of this contract, the acronyms in AR 310-50, authorized abbreviations and brevity codes and those listed below shall apply:

Abbreviation	
AAA	Army Audit Agency
AAFES	Army, Air Force Exchange Service
AAR	After Action Review

Abbreviation	
ACOR	Alternate Contracting Officer's Representative
AHA	Ammunition Holding Area
ANSI	American National Standards Institute
APC	Account Process Code
AQL	Acceptable Quality Level
AR	Army Regulation
ASHRAE	American Society of Heating, Refrigerating and Air Conditioning Engineers
ASP	Ammunition Supply Point
ASME	American Society of Mechanical Engineers
AST	Aboveground Storage Tank
ASTM	American Society for Testing and Material
BAAF	Butts Army Airfield
BDE	Brigade
BMP	Best Management Practices
BN	Battalion
BPA	Blanket Purchase Agreement
BTU	British Thermal Unit
CDOT	Colorado Department of Transportation
CDPHE	Colorado Department of Public Health and Environment
CDR	Contract Discrepancy Report
CFE	Contractor-Furnished Equipment
CFP	Contractor Furnished Property
CFR	Code of Federal Regulations
CLIN	Contract Line Item Number
COB	Close of Business
CRD	Camp Red Devil
CVWF	Central Vehicle Wash Facility
CW	Chill Water
DA	Department of the Army
DBA	Davis-Bacon Act
DENTAC	Dental Activities Command
DFARS	Defense Federal Acquisition Regulation System
DFAS	Defense Finance Accounting Service
DoD	Department of Defense
DODI	Department of Defense Instruction
LRC	Logistic Readiness Center
DOT	Department of Transportation
DPTMS	Directorate of Plans, Training Mobilization and Safety
DPW	Directorate of Public Works
DRMO	Defense Reutilization and Marketing Office
EA	Environmental Assessment
EACH	Evans Army Community Hospital

Abbreviation	
EMCS	Energy Monitoring Control System
EMS	Environmental Management System
EO	Executive Order
EOC	Emergency Operations Center
EPA	Environmental Protection Agency
EPO	Environmental Protection Officer
FAR	Federal Acquisition Regulation
FC	Fort Carson
FC Pam	Fort Carson Pamphlet
FC Regulation	Fort Carson Regulation
FOD	Field Officer of the Day
FY	Fiscal Year
G2	Installation Security Division
G&A	General and Administrative
GFE	Government-Furnished Equipment
GFF	Government-Furnished Facilities
GFP	Government-Furnished Property
GPS	Geopositioning System
GSA	General Services Administration
HQ	Headquarters
HTHW	High Temperature Hot Water
HVAC	Heating, Ventilation and Air Conditioning
IAW	In accordance with
IDG	Installation Design Guide
IH	Industrial Hygiene
IPBO	Installation Property Book Officer
ISR	Installation Status Report
ISO	International Organization of Standards
IWTP	Industrial Wastewater Treatment Plant
J	Cost Account Code for Utilities
K	Cost Account Code for Repair
L	Cost Account Code for New Work
LPS	Lightning Protection System
M	Cost Account Code for Support Costs
MEDDAC	Medical Department Activities
MICC	Mission and Installation Contracting Command
M&S	Maintenance and Service
NACE	National Association of Corrosion Engineers
NCOIC	Noncommissioned Officer-In-Charge
NEC	National Electric Code / Network Enterprise Command
NESC	National Electrical Safety Code
NFPA	National Fire Protection Association
NLT	No later than

Abbreviation	
NPDES	National Pollutant Discharge Elimination System
NSN	National Stock Number
NTE	Not to Exceed
O&M	Operation and Maintenance
OSHA	Occupational Safety and Health Administration
P2	Pollution Prevention
PCMS	Pinion Canyon Maneuver Site
POC	Point of Contact
POL	Petroleum, Oil and Lubricants
PPE	Personal Protective Equipment
PWS	Performance Work Statement
PX	Post Exchange (under AAFES)
RF	Radio Frequency
RFP	Request For Proposal
RP	Real Property
RSU	Remote Switch Unit
S&A	Supply & Administration
SCA	Service Contract Act
SCADA	Supervisory Control and Data Acquisition
SCIF	Sensitive Communication Information Facilities
SFG	Special Forces Group
SOP	Standard Operating Procedure
SOW	Statement of Work
STP	Sewage Treatment Plant
TB MED	Technical Bulletin-Medical
TCR	Turkey Creek Ranch
TDA	Table of Distribution and Allowances
TM	Technical Manual
TUAV	Tactical Unmanned Aerial Vehicle
UPS	Uninterruptible Power Supply
USACE	United States Army Corps of Engineers
UST	Underground Storage Tank
WRC	Wilderness Road Complex

**PERFORMANCE WORK STATEMENT
SECTION C.3
GOVERNMENT-FURNISHED PROPERTY AND SERVICES**

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C.3.1. GOVERNMENT-FURNISHED PROPERTY (GFP) AND SERVICES

C.3.1.1. Property Provided By the Government

The Government will provide the facilities, utilities, equipment, parts, supplies, and materials described herein as GFP for use solely to perform the requirements of this contract. GFP consists of Government supplies and materials, Government-Furnished Equipment (GFE) and Government-Furnished Facilities (GFF) placed in the Contractor's custody. The GFP provided shall not be construed as being sufficient to meet all of the requirements of this contract. The Contractor shall not remove GFP from Fort Carson or other supported areas without prior written approval of the Property Administrator (PA). All property furnished under this contract shall remain the property of the Government.

C.3.1.2 Reduction in GFP

The Contractor shall be expected to limit the amount of GFP to the minimum amount necessary to support accomplishment of required tasks under this contract when replacing unserviceable tools and equipment. For example, replacement for lost, destroyed, or worn-out GFP will not be sought when current inventory of same GFP is sufficient. Also, the Contractor shall examine whether rental rather than purchase of replacement GFP is more economical and make appropriate recommendation to the Government.

C.3.1.3. Transfer of Accountability

The Contractor shall become accountable for GFP when it is issued to and signed for by the Contractor from the Logistics Readiness Center (LRC) at the Consolidated Installation Property Book Officer (CIPBO). The Contractor shall use Property Book Unit Supply Enhanced (PBUSE) for property accountability as read only. The Contractor cannot make changes to items in PBUSE. This PBUSE system is the Army Property System of Record (APSR).

C.3.1.4. Property Control Plan

The Contractor shall develop and implement a plan describing how Government-Furnished Property (GFP) will be managed throughout the duration of this contract to the KO and COR NLT 10 days after contract start IAW C.8, report C.3.1.4. This plan shall contain at a minimum the following items:

- a. Acquisition
- b. Receipt (includes Identification)
- c. Records
- d. Physical Inventory
- e. Subcontractor Control
- f. Reports
- g. Relief of Stewardship Responsibility/Liability
- h. Utilization (includes Consumption, Movement, and Storage)
- i. Maintenance

j. Property Closeout (includes Disposal)

The plan shall also describe methods of communication between the Contractor and the Property Administrator (PA) and address preventive measures and corrective action related to instances of fraud, waste, and abuse of GFP in the Contractor's custody; and structured to assure procedures relating to GFP are independent from other parts of the Contractor's written directives. The Contractor shall coordinate with the Installation Property Book Office (IPBO) and accurately utilize PBUSE for inventory purposes. The Contractor shall make records pertaining to GFP available to the PA for review at any time during the period of this contract.

C.3.1.5. Government Property Report

When additional or replacement property is purchased by the contractor, with appropriate authorization, the Contractor shall provide a GFP Master Report (MR) identifying Contractor Acquired Property (CAP). The type of Government property shall be detailed by Government Property Classification (i.e. Equipment, Material, Special Test Equipment and Special Tooling). An example of the report is in C.7 TE C.3.1.5. Submit the report to the KO, PA, and COR quarterly IAW C.8, report C.3.1.5. Once the CAP is received and added to the appropriate property book account, the CAP will then be identified as GFP.

C.3.2. Initial Inventory Procedures

The Contractor shall attend Phase-In GFP transfer and inventory meetings with the Government. Meetings will be scheduled by the Government prior to contract start.

C.3.2.1. Contractor Phase-In

At least 10 days prior to end of phase-in, the contractor shall conduct a 100% joint inventory of all GFP and GFE in accordance with (IAW) AR 710-2 *Supply Policy Below the National Level*, AR 735-5 *Property Accountability Policies*, and Department of the Army Pamphlet (DA Pam) 710-2-1 *Using Unit Supply System (Manual Procedures)*. This inventory shall include, but not be limited to, facilities to include keys; property on hand received from the LRC IPBO; and supplies and stockage. This inventory will identify items, when appropriate, by National Stock Number (NSN), Part Number, or Federal Supply Class, nomenclature, present storage location, unit of issue, unit cost, quantity on hand, and authorized allowances. The operational or conditional status of all GFF and GFE shall be determined. The LRC, IPBO and the Contractor shall certify the joint inventory as accurate. The Contractor shall keep the inventory listing current.

C.3.2.2. Transfer Accountability

The Contractor shall sign all required official Government property records. The Contractor shall continuously update property custody records to maintain accountability requirements of AR 735-5.

C.3.2.3. Records Maintenance

The Contractor shall establish and maintain records of GFP in use by the Contractor. The records shall be maintained IAW the functional guidance for the automated system in use IAW FAR Clause 52.245-1. The records system will be reviewed by the PA and shall become the Contractor's official GFP control system, which shall be in effect thereafter until expiration or termination of the contract.

C.3.2.4. Security

The Contractor shall be responsible for the physical security of GFP and installed equipment and shall secure all GFP when not occupied by Contractor personnel.

C.3.2.5. Change of Status

When GFP is no longer serviceable for intended use, or has reached the end of its economic life, the Contractor shall first coordinate through the KO then prepare required transfer documentation and submit to the PA for approval and disposition directions.

C.3.2.5.1. When new equipment is acquired either by transfer or purchase, the Contractor shall provide the KO, the property administrator, and IPBO with the changes of GFP status information.

C.3.3. Government Furnished Property

The Government will provide or make available to the Contractor, property as specified in section C.7, TEs 3.3-A and 3.3-B, for use on work in this contract. The Contractor shall not use this equipment for any other work.

C.3.3.1. Records

The Contractor shall maintain a record of all Government Furnished Property (GFP), in the format of C.7 TE's. Appropriate TE's shall be updated as new property is received or turned in to DLA Disposition Services. Provide updated information, to the Government as stated in C.3.1.5 for CAP or to correct deficiencies discovered on any TE electronically to the COR for TE corrections.

C.3.3.2. Operation and Use

The Contractor shall be responsible for the proper operation, and use, of GFP identified in TE-3.3-A., and TE-3.3-B. The Contractor shall provide trained and licensed personnel to operate GFE as required by Federal, State, local laws and regulations. It is expected that operator level maintenance is performed as required by the operations manual for the specific equipment. All maintenance and repairs to the equipment in the TE's will be accomplished by Fort Carson Logistic Readiness Center (LRC) contract.

C.3.3.2.1 The contractor shall input and maintain operator qualifications, dispatch equipment (Other than GSA) and close out dispatches through the GCSArmy system. The contractor is responsible for ensuring all operators are qualified and proficient in the safe operation of the equipment they are responsible for. The contractors training/ certifications of operators' qualifications must be documented separately by the contractor and provided to the government upon request.

C.3.3.3. GSA Vehicle Fleet

The Government will provide GSA vehicles for the contract staff based on the number of employees on staff needing vehicles. The current GSA fleet inventory is identified in C.7, TE-3.3.3. and will be adjusted based on number of vehicles agreed that are needed.

C.3.4. Government Furnished Property-Facilities

C.3.4.1. Furnished Facilities

The Government will provide or make available to the Contractor Government Property-Facilities as specified in section C.7, TE-3.4. The Contractor shall not relocate activities or operational units within assigned facilities unless approved by the KO. The Contractor shall manage GFP and real property IAW AR 405-70 Utilization of Real Property, and AR 420-1 Army Facilities Management.

C.3.4.2. Maintenance and Minor Repairs

The Contractor shall maintain and perform minor repairs of assigned facilities the same as any other real property facility or system.

C.3.4.3. Alterations or Improvements

Any alterations or improvements requested to assigned facilities by the Contractor shall be requested in writing to the KO. If the KO approves the Contractor's justification for the request, the COR will submit a DA 4283 through DPW for approval and accomplishment once funded.

C.3.4.4. Change in Facilities

The Government reserves the right to reallocate and relocate assigned facilities during the term of the contract.

C.3.5. Utilities

The Government will furnish utilities as currently installed in assigned facilities. Not all facilities receive the same utility services. The Contractor shall not change or modify any utility system or component without prior KO review and written approval unless accomplished as a repair and maintenance task in C.5 of this contract. The Contractor shall not connect any Contractor-furnished equipment (CFE) or system without prior KO review and written approval.

C.3.6. Calibration Standards

The Contractor shall provide a point of contact for each major functional area to coordinate calibration services; transport Contractor-operated equipment to and from calibration site; and ensure all tests, measurement, and diagnostic equipment in the Contractor's possession is labeled to indicate condition and latest calibration date.

C.3.7. Government Furnished Material

C.3.7.1. Inclusions

Material includes, but is not limited to, supplies, parts, subassemblies, raw materials, and other components and end items used to accomplish work or services described in this contract, this paragraph, and associated subparagraphs, unless otherwise directed by the KO.

C.3.7.2. Fuel

Unless otherwise specified within this contract, the Government will furnish fuels such as heating oil and natural gas for permanently installed equipment including, but not limited to, boilers, furnaces, infrared heaters, radiant gas heaters, and hot water heaters. The Contractor shall be responsible for the distribution of Government-furnished operating fuels for emergency generators, fire pumps, as well as GFP or CFE used in performance of services specified in this contract. Follow ordering procedures in C.5.17.1.2 for ordering fuel for AST's located at 8200, 1860 and PCMS.

C.3.8. Government Furnished Manuals, and Forms

C.3.8.1. Technical Reference Library

Regulations and publications are identified in Section C.6 of this contract. Publications, Regulations, and Codes available on the World Wide Web as identified in Section C-6 will not be maintained in the technical library. The Contractor shall maintain an account through the Publications Distribution Office and shall request supplements, updates, and other publication requirements in direct support of this contract. Requests for official publications shall be prepared IAW US Army Publishing Agency (APA) web page instructions (www.usapa.army.mil). When requesting publications requiring classified access, the Contractor shall submit the request to the COR. Maps of utility locations will be transferred at initial contract Phase-In or may be viewed in the technical library.

C.3.8.2. Operating and Maintenance Manuals

The Contractor shall maintain the inventory of O&M manuals received from other construction contracts. These O&M manuals will assist the contractor in providing preventive maintenance and repairs to facility systems identified as real property. They will also provide information for matching facility components when they fail.

The Contractor shall maintain the existing library inventory and add new facility O&M's as construction is completed and documents are received from the USACE or DPW Engineering for the life of the facility. As systems fail and are replaced, the library shall be updated by the contractor with the new information so it is accurate.

C.3.8.3. Forms

The contractor shall obtain necessary Government forms from the following websites: <http://www.carson.army.mil/dhr/DHR/ASD/Installation.html> or <http://www.apd.army.mil/>. The Contractor may develop contract specific forms as necessary following applicable ISO standards. At the end of this contract, the Contractor shall turn over to the Government or the successor contractor all electronic forms developed.

C.3.9. Government Furnished Services

All services described below as being provided by the Government shall be used solely to perform the requirements of this contract.

C.3.9.1. Emergency Services

The Government will furnish emergency medical and rescue services to prevent loss of life, limb, or undue suffering. Evans Army Community Hospital, a medical treatment facility at Fort Carson, will be the first point of emergency medical care. Transfer to facilities other than Government medical treatment facilities will occur as soon as possible and as determined by attending medical authorities. Medical billings will be sent directly to the employee or the Contractor's medical insurance company as appropriate. The Government will perform fire inspections and firefighting services. The Government will provide the services of the Fort Carson Military Police for law enforcement on Fort Carson.

3.9.2. Telephone Service.

3.9.2.1. The Government will not reimburse the cost of unofficial telephone service such as telephone service not incidental to performance of the contract.

3.9.2.2. The Government will provide required land line telephone services, instruments, and communication lines, for official use only for local and long distance commercial calls.

3.9.2.3. The Government will install, maintain and remove, as necessary, Government-furnished telephones.

3.9.2.4. The Contractor shall comply with AR 25-1, Army Knowledge Management and Information Technology, chapter 6, and DA Pam 25-1-1 Installation Information Services and any subsequent regulations pertaining to Information Management.

3.9.2.5. The Contractor shall obtain prior Government review and written approval before connecting or disconnecting any Contractor-furnished equipment to Government-furnished communications systems, lines, or equipment.

C.3.9.3. Bills for Unofficial Phone Calls

The Contractor shall understand the use of the telephone system is consent to monitoring for misuse. Bills for unofficial phone calls will be forwarded to the Contractor by the KO for payment to the Defense Finance and Accounting Service (DFAS). Charges will include Government processing and handling fees.

3.9.4. Local Area Network (LAN)

The Government will provide limited access to the existing Fort Carson LAN to include E-Mail capability. The Contractor shall not use the LAN for purposes other than for work required under this contract. The Contractor shall obtain prior DPW and Network Enterprise Center (NEC) review and written approval before purchasing any computer equipment, and connecting any hardware and software to the Government-furnished LAN.

3.9.5. Computer and Computer Systems

The Government will provide all computers and computer systems and computer support to systems connected to the DoD computer network supported by the Network Enterprise Center (NEC). Contractor shall have a CAC for access the DoD computer network as described in C.1.4.11.2. The Government will not supply computers or network connectivity for contractor specific (Corporate) software or access requirements.

C.3.9.6. Custodial Services

The Government provides limited scheduled custodial services for the Contractor's operational areas. The custodial services currently being provided by the Government are identified in the table below. The custodial contract does not enter offices, remove office or break room trash, or clean break room equipment and surfaces above the floor. The custodial contract does not provide paper products or soap.

Facility	Frequency	Effort
214	Twice weekly	Clean restrooms
1225	Twice weekly	Clean restrooms
1225	Twice monthly	Clean common Areas.
8010	Twice weekly	Clean restrooms
8010	Twice monthly	Clean common Areas.
8110	Twice weekly	Clean restrooms
8110	Twice monthly	Clean common Areas.

The provided service can change based on funding limitations.

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The Contractor will be responsible for cleaning areas not performed by the custodial contractor and all paper products and soaps for restrooms/break rooms, and cleaning supplies for their cleaning efforts.

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SECTION C.4
CONTRACTOR FURNISHED PROPERTY AND SERVICES**

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C.4.1. GENERAL

The Contractor shall furnish all materials, equipment, and services necessary to perform this contract that are not specifically identified as Government-furnished in Section C.3 of this contract. This shall include, but not be limited to; supplies, equipment, tools, vehicles, identification badges, and data processing equipment.

C.4.1.2. Removal of Contractor Property

The Contractor shall remove all Contractor-owned supplies, equipment, tools, vehicles, and other items from Fort Carson within five working days after expiration or termination of this contract. The Government will not be responsible for any Contractor owned property left after contract expiration or termination. If the Contractor does not remove said property from Fort Carson within the stated time, Government will consider it as abandoned and dispose of it accordingly.

The Contractor shall furnish at contract start all tools, equipment, motor vehicles, Automated Data Processing Equipment (ADPE), and administrative equipment, not furnished by the Government, but required for performance of work required under this contract. All property purchased by the Contractor for which the purchase cost is reimbursed by the Government shall be considered Government-property unless otherwise designated by the KO.

C.4.1.3. Marking

Contractor-owned equipment and vehicles including, but not limited to, Basic Construction Equipment and Material Handling Equipment shall be clearly and permanently marked with the Contractor's name, telephone number, and the vehicle identification number on both sides. Government-owned equipment and vehicles shall be clearly and permanently marked with a DPW number and Government licenses plate as provided by the COR. The Contractor shall maintain DPW emblem stickers on all Government-owned or leased vehicles authorized by DPW. The Contractor shall mark Government-owned or leased vehicles that are Contractor-operated IAW AR 58-1 Management, Acquisition, and Use of Motor Vehicles.

C.4.1.4. Maintenance and Repair

The Contractor shall maintain all Contractor-Furnished Equipment (CFE) in a safe and serviceable condition suitable for its intended use. The Contractor shall repair inoperable and unserviceable Contractor-furnished equipment or items. The Contractor shall remove equipment from Fort Carson that is not repaired within 30 days. The Contractor may request an extension when the delay is due to long lead-time for a replacement part that the Contractor has on order.

C.4.1.5. Fuel

The Government will not provide fuel used for transport of Contractor employees between their domiciles and their work, or any other personal use, and the cost of such fuel shall not be reimbursable under this contract. Any Government issued or funded fuel is solely for use in performance of tasks outlined in the contract terms, the PWS and associated technical exhibits.

C.4.2. SUPPLIES AND MATERIALS

At the contract start date, the Government will turn over to the Contractor all existing stocks of supplies and materials. The Contractor shall furnish all additional supplies and materials necessary to meet the requirements of the contract. Supplies and materials provided by the Contractor shall be of equal or better quality than those being replaced by the maintenance, repair, or servicing action. All supplies and material shall meet applicable codes or manufacturers' specifications.

C.4.2.1. Stock Levels

The Contractor shall maintain a sufficient quantity of on-hand materials and supplies to perform all work required under this contract. Any failure on the part of the Contractor to provide sufficient quantities and quality supplies and materials shall not be cause for reduction in any service or performance.

C.4.2.2. Quality of Materials

Contractor-furnished, Contractor-replaced, or Contractor-acquired materials, supplies, and replacement parts shall meet or exceed specifications or standards listed in this contract and shall be of equivalent design and equal to or better quality than the original item in terms of durability, efficiency, and maintainability as the item being replaced.

C.4.3. SERVICES

C.4.3.1. Communication Services

The Contractor shall obtain those communication services required to perform work specified in this contract that are not Government-furnished. Contractor communication services shall be subject to standard monitoring requirements of the Government communication networks.

C.4.3.2. Cellular Telephone Service

The Contractor shall provide cellular telephone service with local and long distance capability, call forwarding and voice mail and KO determined Government personnel for official contract use only. Contract personnel shall be able to be contacted by authorized Government personnel through the cellular telephone system. Intrinsically safe radios/telephones may be required in areas where explosive hazards are present. Communication equipment operations shall be IAW AR 25-1.

C.4.4. Cleanliness

The Contractor shall maintain GFF, equipment, vehicles, supplies, and material in a safe, organized, and clean condition including facilities, yards, compounds, parking areas, and ancillary work areas. The Contractor shall maintain all outdoor areas assigned to the Contractor

for its use (including 50 feet around any Contractor-assigned facility) free of trash, debris, undesirable vegetation, and leaves. The Contractor shall also be responsible for the removal of snow and ice from walkways of Contractor-assigned facilities.

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- C.5.5. TRANSPORTATION SYSTEMS MAINTENANCE AND REPAIR
- C.5.5.1. Schedule of Maintenance and Progress Report
- C.5.5.2. Roads, Parking Lots, Running Trails, and Motor Pools
- C.5.5.2.1. Paved Areas
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- C.5.5.4. Snow Removal and Ice Control
- C.5.5.5. Utility Infrastructure Repair Support
- C.5.6. EQUIPMENT FUELING REQUIREMENTS
- C.5.7. ELECTRICAL SYSTEMS MAINTENANCE AND REPAIR
- C.5.7.1. Primary Electrical Distribution
- C.5.7.2. Traffic Control Systems
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- C.5.7.4. Facility Electrical Systems
- C.5.7.5. Emergency Generator Systems
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- C.5.7.7. Uninterruptible Power Supply (UPS) Systems
- C.5.7.8. Fire Alarm Systems
- C.5.7.9. Lightning Protection System (LPS) and Static Ground Points Resistance Testing
- C.5.7.10. Inspect and Maintain Facility Mass Notification System
- C.5.7.11. Inspect, Test and Repair Solar PV Systems
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- C.5.7.13. Electric Vehicle Charge Equipment
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- C.5.8.1. Heating Systems
- C.5.8.1.1. Central Heating Plants and Distribution Systems, HTHW Generators and Boilers (Individual and System)
- C.5.8.1.2. Boiler Fuel
- C.5.8.1.3. Boiler Opacity

- C.5.8.1.4. Corrosion Control
- C.5.8.1.5. Boiler Inspection
- C.5.8.1.6. Operator Certification
- C.5.8.1.7. Furnaces and Gas/Oil Fired Unit Heaters
- C.5.8.2. Refrigeration, Ventilation, and Air Conditioning
 - C.5.8.2.1. Air Conditioners
 - C.5.8.2.2. Chillers
 - C.5.8.2.3. Central Cooling Plant and Distribution System
- C.5.8.3. Exhaust Fans and Radon System Fans
- C.5.8.4. Compressors
- C.5.8.5. Air Dryers
- C.5.8.6. Heating, Ventilation and Air Conditioning Systems (HVAC)
- C.5.8.7. Evaporative Coolers
- C.5.8.8. Seasonal Heat Turn On and Turn Off
- C.5.8.9. Specialized Equipment Maintenance and Repair
 - C.5.8.9.1. Food Service Equipment
 - C.5.8.9.2. Inspect, Service and Clean Kitchen and Grill Hoods
 - C.5.8.9.3. Special Services for Building 8000
 - C.5.8.9.4. Maintenance of Above Ground and Underground Storage Tanks (AST's and UST's)
 - C.5.8.9.5. Gate Arm Operation
 - C.5.8.9.6. Inspect and Service Mechanical Rooms
 - C.5.8.9.7. Other Mechanical Services
- C.5.8.10. Energy Management Control Systems (EMCS)
- C.5.8.11. Fire Training Tower
- C.5.8.12. Vertical Transportation Systems (VTS)
- C.5.9. PLUMBING SYSTEMS
 - C.5.9.1. Potable Water and Waste Vent and Drains
 - C.5.9.2. Hot Tubs, Swimming Pool Circulation, Heating and Filtration System
 - C.5.9.3. Water Free Urinals
 - C.5.9.4. Water Features and Fountains
 - C.5.9.5. Fire Suppression Systems
 - C.5.9.5.1. Dry Fire Suppression System
 - C.5.9.5.2. Wet Fire Suppression System
 - C.5.9.5.3. Clean Agent Fire Suppression System
 - C.5.9.6. Fire Pumps
 - C.5.9.7. Winterize/Dewinterize Facilities
 - C.5.9.8. Propane Tanks
 - C.5.9.9. Wash Racks
- C.5.10. NATURAL GAS DISTRIBUTION AND REGULATION SYSTEM
- C.5.11. ENVIRONMENTAL SERVICES
 - C.5.11.1. National Environmental Policy Act (NEPA) Program Support
 - C.5.11.2. Hazardous Waste and Spill Response
 - C.5.11.3. Environmental Protection Officer (EPO)
 - C.5.11.4. Asbestos and Lead Based Paint

- C.5.12. INDUSTRIAL WASTEWATER TREATMENT PLANT (IWTP) AND
CENTRAL VEHICLE WASH FACILITY (CVWF)
 - C.5.12.1. Industrial Wastewater Treatment Plant (IWTP)
 - C.5.12.2. Central Vehicle Wash Facility
- C.5.13. SANITARY WASTEWATER TREATMENT AND COLLECTION SYSTEM
 - C.5.13.2. Sewage Collection
 - C.5.13.3. Reclaimed water
 - C.5.13.4. Grease Traps/Interceptors
 - C.5.13.5. Sewage Lagoons
 - C.5.13.6. Sewage Drying Beds
 - C.5.13.7. Oil/Water Separators
- C.5.14. STORM SEWER COLLECTION SYSTEM
- C.5.15. POTABLE WATER SYSTEMS AND DISTRIBUTION
 - C.5.15.1. Water Distribution and Quality
 - C.5.15.2. Fire Hydrants
 - C.5.15.3. Backflows
- C.5.16. OPERATION OF THE COLLECTION POINT and CLEAN FILL SITE
- C.5.17. WAREHOUSING AND SUPPLY
- C.5.18. CATHODIC PROTECTION
- C.5.19. FACILITY MAINTENANCE AND REPAIR
 - C.5.19.1. Facility Maintenance
 - C.5.19.2. Facility Repairs
 - C.5.19.3. General Requirements for Painting
 - C.5.19.4. General Requirements for Welding and Metal Work
 - C.5.19.5. Locksmith Services
 - C.5.19.6. Signs

C.5.1. GENERAL PERFORMANCE WORK STATEMENT (PWS) REQUIREMENTS

Section C.5 describes the tasks, and standards associated with work to be performed by the Contractor. The items in this Section C.5.1 present background information and requirements that apply to the execution of these tasks.

The Contractor shall do the following:

C.5.1.1. Critical Systems

Keep the following critical systems operational 24 hours a day, seven days a week, as detailed in subsequent sections of C.5. Critical systems are:

- Airfield lighting;
- Air conditioning in medical facilities, NOC's, SCIF's and computer rooms;
- Boiler plants and heating;
- Cold Storage Facility;
- Communication Nodes (CN);
- Electrical systems to mission essential vulnerable areas;
- Emergency lighting;
- Energy Management Control System (EMCS);
- Fire detection and suppression;
- Refrigeration units and cold storage;
- Repair work order reception;
- Road and rail network;
- Standby generators;
- Security;
- Supervisory Control and Data Acquisition System (SCADA);
- Traffic signals;
- Utility infrastructure and distribution systems; and
- Wastewater treatment plants and associated collection.

C.5.1.2. Codes and Specifications for Work

Work performed under Section C.5 shall be IAW current versions of all laws, regulations, directives, and codes applicable to such work or followed by knowledgeable and experienced practitioners in the various trades and crafts involved, such as the OSHA, NEC, National Fire Protection Association (NFPA), Unified Facility Criteria (UFC) including 3-600-2, International Building Code, International Mechanical Code, International Plumbing Code, American Society of Mechanical Engineers (ASME), Manual on Uniform Traffic Devices, National Electrical Manufacturers Association (NEMA), American Welding Society (AWS), Americans with Disabilities Act (ADA), Fort Carson Installation Design Standards, Operation and Maintenance (O&M) manuals, manufacturer's specifications, and all references cited in Sections C.5 and C.6. The COR shall coordinate with the DPW-ED Cultural Resource Manager to assure compliance

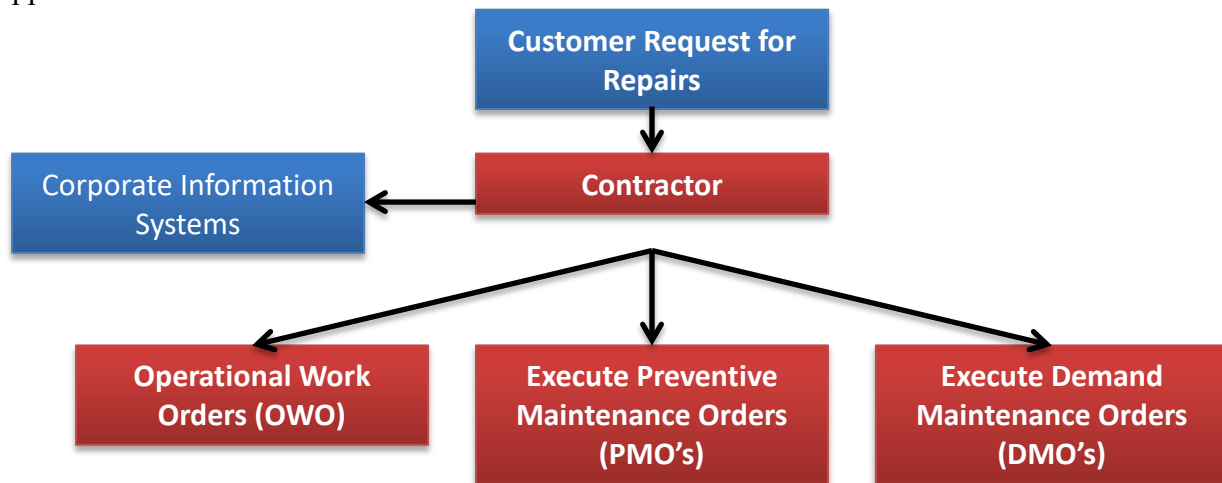
when necessary. Where codes and specifications are specifically referenced in Section C.5 paragraphs, they appear in descending order of precedence. The order of precedence for codes and specifications are: Federal law, State law, AR, NFPA, uniform codes and specifications published by national organizations such as the ASME, and manufacturers' operation and maintenance manuals.

C.5.1.3. Government Use of Facilities

Be aware that Government use of facilities is anticipated while the work under this contract is being performed. Plan and accomplish the work so that there will be a minimum of interference and inconvenience to organizations and activities in the vicinity and to other personnel working in affected facilities. Coordinate with facility occupant prior to blocking building exits or driveways.

C.5.1.4. Concept of Operations

Be aware that the following chart depicts how work will be received from the customer and the relationship that will exist between the DPW and the Contractor to accomplish work. The Government reserves the right to have work performed by other Contractors or Government agencies. The work performed may require coordination with the DPW for reviews and approvals of work as described in each Section C.5 task.



C.5.1.4.1. The DPW is responsible for planning and programming the overall DPW mission including Government and fiscal approvals, and the performance of DPW services not covered by this PWS. See AR 420-1 Army Facilities Management for a list of DPW plans and programs. This contract does not include Military Construction, Army, or Family Housing plans and programs or substantial construction projects that are not an integral part of the operations and maintenance functions required by this contract.

C.5.1.4.2. The DPW is responsible for creating the Annual Work Plan, which is the approved work of the upcoming fiscal year IAW AR 420-1. The Contractor shall provide input to the COR for the Annual Work Plan based on day-to-day operations and maintenance in

execution of the PWS. The Contractor will execute approved and funded tasks described in this PWS.

C.5.1.4.3. Crosswalk of Repair Order Terminology

Common Terminology		Contract Terminology	
Service Order	SO	Demand Maintenance Order	DMO
Preventive Maintenance	PM	Plant Maintenance Order	PMO
Work Order	WO	Project Work order	PWO
Standing Operating Order	SOO	Operational Work Order	OWO

C.5.1.4.4. The Contractor shall receive requests for any aspect of real property maintenance and repair, information, services, technical assistance, and environmental service issues. Requests shall be received electronically, verbally, by telephone, ARMA submittals or in person. This includes requests that will be routed to Colorado Springs Utilities for repair responsibility.

The customer requesting these services may be anyone on Fort Carson to include Contractor employees, DPW, the COR or the KO. When customer requests are received by the Contractor that are not covered by the scope of this PWS, the Contractor shall refer those requests to the KO for resolution.

C.5.1.4.5. Execute recurring services including utility plant operation; road, railroad, and grounds maintenance; and Preventive Maintenance Orders as required.

C.5.1.4.5.1 All work received from customers or generated as a result of contractor findings shall be entered by the contractor, in the ARMA system as the first point of entry. The contractor shall enter, triage and push finalized requests into GFEBS notifications within 2 hours of receipt.

C.5.1.4.5.2 Customer submitted ARMA requests: Customers shall be afforded 48 hrs to respond to questions, clarifications or corrections to facilitate triage, prior to requests being cancelled. Requests missing adequate information to create a complete DMO shall be cancelled after 48 hrs so as to prevent system generated GFEBS notifications. Cancellations shall contain a specific reason in the resolution portion of the notes.

C.5.1.4.5.3 ARMA Cancellations: The Contractor shall cancel any requests submitted that involve subjects not covered under this contract. The contractor shall provide guidance in ARMA, back to the customer as to cancellation reason and POC info for the appropriate area.

C.5.1.4.5.4 No DMO work shall be executed without an ARMA entry and subsequent GFEBS notification.

C.5.1.4.5.4 All requests for Service involving Electrical Distribution or Natural Gas Distribution shall be opened as a DMO initially. Efforts that have been determined to be an IGSA responsibility shall be referred to Colorado Springs Utilities (CSU) and the DMO closed unless existing stock materials are transferred to CSU. Once those materials have been transferred and annotated within the DMO, then the DMO may be finalized.

C.5.1.4.6. Limitations to Repair Authority

The table below describes the repair authority for the effort in this PWS. If a repair is estimated to exceed \$100K the KO may determine that the effort will not be completed in this contract.

C.5.1.4.6.1. At the direction of the KO, there may be requirements for single scope; Command Emphasis; Life, Health, Safety (LHS); or code compliance efforts requiring the Contractor to perform. The Contractor can anticipate 150 of these Miscellaneous Requirements each year.

Description	Authorized repair value	Requires KO approval
Repair order for real property facility or system	< 32 hours	≥32 Hours
Subcontracted repairs		≥\$25K
Miscellaneous Requirements		All

C.5.1.4.6.2. When repairs are necessary to non-real property items, funding under ISR 400 must be used. Reference TE-1.8.3A. The items included in this category are government installed modular trailers, Latrine/Shower trailers, computer rack cooling, sheds assigned to the contractor and other items. Reference section C.7 for TE-5.1.4.6.2. for a list of known non-real property requiring repair and maintenance. For items not identified, the contractor shall proceed with repairs at the direction of the KO.

C.5.1.4.6.2 IGSA: Maintenance and Repair responsibility for Electrical Distribution and Natural Gas distribution will be de-scoped from this contract, except for Government directed efforts, to Colorado Springs Utilities effective 30 December 2022. While the intent is that CSU will begin this effort in its entirety on that date, the effort will remain on this contract for repair response only and only at the direction of the COR. All preventative maintenance efforts in these areas is suspended. TE-C.5.1.4.6.2 and TE-C.5.1.4.6.2A provides the current IGSA responsibilities.

- No regular and re-occurring maintenance efforts for these two areas shall occur on this contract. All Preventative maintenance efforts for these two areas shall be suspended effective 30 December 2022.
- Efforts in these two areas remain unchanged for Pinon Canyon and Turkey Creek Recreation area requirements within this PWS.

C.5.1.4.7. Repair versus Construction

This PWS is for maintenance and repair of real property to include operations of plants. This may include repair by replacement of existing facility systems or components subject to approval thresholds as outlined above. It does not include construction effort that would be identified as capital improvements such as remodeling of any office space or building.

C.5.1.4.7.1. Individual DMO's which require less than 32 work-hours will be considered to be maintenance subject to the Service Contract Labor Standards. No service order shall exceed 32 work-hours without following criteria described in C.5.1.4.8. below.

C.5.1.4.8. Wage Rate Requirement (Construction) Statute

C.5.1.4.8.1. Individual DMO's that will exceed the 32 work-hour limit and are subject to the Service Contract Labor Standard, will be subject to Contracting Officer approval prior to being authorized to continue. If previously classified to be subject to the Service Contract Labor Standards, a cost estimate will be developed to substitute the appropriate Construction Wage Rate Requirements General Decision labor categories and rates for each construction trade working on that repair. An example of this submittal is in TE 5.1.4.8

For priority 2 and 3 DMO's, the Contractor shall notify the Contracting Officer immediately upon identification that the work required to complete the DMO will exceed 32 work-hours and meet the criteria for CWR work and cease all work on the DMO until the above process is completed, and the Contracting Officer provides written approval to proceed. Submit detailed cost estimate identifying required labor, material, and equipment to perform the repair if repair is expected to exceed 32 hours IAW C.8, Report C.5.1.4.8.1.

For priority 1 DMO's, the Contractor shall continue working until the emergency is resolved or the utility service is restored. If additional work is required, the process described above must be followed. If a priority 1 DMO exceeds 32 hours and meets the CWR criteria above, the Contractor shall provide documentation NLT 1200 the next business day IAW C.8, report C.5.1.4.8.1A explaining the situation resulting in the emergency repairs exceeding 32 hours.

C.5.1.4.8.2. The Contractor shall submit a weekly report by 0900 Monday, or the next business day if Monday is a holiday, to the KO and COR IAW C.8, report C.5.1.4.8. Identifying status of KO approved repairs meeting the construction statute. The following shall be included:

- DMO number
- DMO description
- Estimated cost of material and labor, by trade
- Date of KO approval
- Estimated labor hours by trade and performance period
- Estimated completion date

C.5.1.4.8.3. In the event the Contractor elects to obtain services from a subcontractor to perform repairs under the requirements of the PWS, performance and payment bonds for the contract shall be secured for firm fixed-price subcontracts.

C.5.1.5. Condition Assessment

Continually assess the condition of facilities, systems, and equipment and identify needed repairs or replacement to the COR. Identify systems or specific equipment, e.g., boilers, air handlers, water heaters in need of replacement and submit information to the KO for development of separate project to program their replacement. This assessment shall consider age, to date replacement cost, and reliability of equipment.

C.5.1.6. Management Information System

The Contractor shall do the following:

C.5.1.6.1. Plan, program, coordinate, estimate, respond to Government schedule, and evaluate resources for work requests for performance by the Contractor.

C.5.1.6.2. The Government provided Management Information System is General Fund Enterprise Business System (GFEBS). It is the requirement of the Contractor to be proficient in the use of GFEBS.

C.5.1.6.3. Prepare, input, and maintain Contractor- generated data in GFEBS. Contractor data includes operational data to provide complete and accurate outputs/reports required herein. Work management shall include maintaining a suspense system with appropriate registers and files, and routing and distributing reports.

C.5.1.6.4. All facility data prior to this contract is located in a Government owned MAXIMO database located in the Contractor's facility. The Contractor may access this data for historical data purposes.

C.5.1.6.5. Input accurate data into GFEBS when received, 24hours a day, seven days a week.

C.5.1.7. Response and Completion of Demand Maintenance Orders (DMO's) and Plant Maintenance Orders (PMO's)

DMO and PMO work is authorized and accomplished using appropriate orders in the Government authorized Management Information System (MIS). Response to DMO's shall be completed within the timeframes specified below in Sections C.5.1.7.1.1 thru .6. Accomplish PMO work on authorized Maintenance Service Orders IAW manufacturer's O&M requirements, the Contractor's maintenance plans, and existing schedules. For accounting purposes, the cost of DMO's and PMO's shall be captured to the correct and specific WBS code. All completed DMO's shall be signed as accepted by either the customer point of contact or other responsible party. Maintain the signature of the customer on file until contract completion. The COR may request copies of the signature files at any time during the contract or require the Contractor turn over the files at Phase-out.

DMO's assigned to the respective real property facility or system shall be used to accomplish all work. Work performance shall not be accomplished using an overhead account. A DMO shall not be closed and a new DMO created due to lack of response.

C.5.1.7.1.1. Assignment of Priorities, Required Completion Times, and Response Time: The Contractor shall assign appropriate priorities to DMO's based on the situation reported and responses to questions asked. Consider Life, Health, Safety, Security, and Property issues when prioritizing the reported failure. Further guidance for priorities is located in AR 420-1, Section 3-56, Priority system for service order maintenance. Technical Exhibit 5.1.7.1.1 lists facilities identified as High Priority. Repair on these facilities will be priority one or two only, based on the failure. The Government reserves the right to change assigned priority. Complete work when the Government provides a required completion date.

C.5.1.7.1.2. Priority 1, Emergency: Respond within one hour of customer notification and has priority over other work. Work shall be continuous until such time as the emergency is arrested, must be completed within 24 hours. A separate DMO shall be opened, if necessary, to make permanent repairs as a result of the emergency.

C.5.1.7.1.3. Examples of emergency repair work include, but are not limited to the following:

- Any dining facility problem that inhibits operation (e.g., inoperative dishwasher, failed refrigerators);
- Broken water line (to include interior plumbing water lines);
- Downed electrical wire;
- Exposed or shorted electrical wiring;
- Fire or security alarm malfunction;
- Lockout to living areas;
- Unsecured buildings, arms rooms, vaults, living quarters;
- No heat in any structure containing water or steam, exterior temps below 40 degrees F;
- Obstruction on a road that blocks a lane of traffic;
- Rail road Repairs during movement;
- Plugged sewer line which results in sewage backing up;
- Plugged toilets (when only one is available);
- Any utility system leak or unplanned outage;
- Any Gas odor complaint, follow procedures in FC Reg. 420-24, Fort Carson Emergency Gas Plan; and
- Security fence damage.
- Water remediation efforts to prevent further damage and facilitate repairs.
- Remediate and properly dispose of mold contaminates of no more than 100 SF as necessary to facilitate repair efforts of real property

C.5.1.7.1.4. Priority 2, Urgent; Respond within 24 hours, complete within seven calendar day. Respond and begin repair within 24 hours of customer notification. Work shall be continuous until such time as the urgent condition is terminated, must be complete within seven calendar

days. A separate DMO shall be opened, if necessary, to make permanent repairs if urgent condition cannot be completed within seven calendar days.

C.5.1.7.1.5. Examples of urgent repair work include, but not limited to the following:

- No electric power in an office or living quarters;
- No heat to a room, parts of, or section of, a building, structure or facility when pipes are not in danger of freezing;
- All repair work in buildings identified as High Priority, unless issue qualifies as a Priority 1;
- No hot water in a barracks or dining facility (other than kitchen operation);
- Water leaks not controlled by user-accessible shut off valves; and/or
- Whole building air conditioning failures.

C.5.1.7.1.6. Priority 3, Routine Repair Work; typically is an inconvenience or an unsightly condition. Respond to and complete routine repair work within 30 calendar days after receipt of the DMO. The Contractor shall provide justification with notification to the COR whenever circumstances arise that will prevent the Contractor from completing Priority 3 service orders within the 30 day time period (i.e. awaiting materials).

C.5.1.7.2. Refer to flow chart for Responding to Water Contamination Threat if a call is received identifying a possible water contamination issue. Reference section C.7 for TE-5.1.7.2.

C.5.1.7.3. One PMO shall be opened for the building where facility maintenance is being performed as required in C.5.19.1. If a repair is identified during the maintenance cycle, a DMO shall be opened in ARMA, for the repair.

C.5.1.7.4. If material is ordered against a DMO, the priority of the material order shall not be less than the DMO priority.

C.5.1.7.5. Repairs to barracks shall be completed ahead of other competing DMO's. The average completion time for barracks DMO's shall not exceed 15 calendar days. The average will be determined by adding days open from received to completion and dividing by total number completed for any time period.

C.5.1.8. Repair and Replacement Material Specifications

All repair and replacement materials shall meet or exceed Original Equipment Manufacturers (OEM) specifications and shall be of equivalent design and equal to or better quality than the original item in terms of durability, efficiency, and maintainability. The repaired or replaced material shall perform and look as intended when new as far as design, material, strength, appearance, and texture. Repairs shall be consistent with surrounding facility or grounds appearance. The Contractor shall receive KO approval prior to installation of parts or materials deviating from OEM specifications.

C.5.1.9. Temporary Repairs to Critical Systems

When availability of repair parts or materials prohibits the repair to a critical system, the Contractor shall make temporary repairs to make the critical system operational until such time as correct repair parts or materials are available. Repairs shall be completed and the critical system restored to normal operating condition within one day of receipt of the correct repair parts or materials unless an extension is approved in writing by the KO.

C.5.1.10. Moving Equipment to Accomplish Work

Items not fastened to facilities including, but not limited to, furniture, partitions, and portable equipment (except electronic equipment) shall be moved by the Contractor, protected from damage, and replaced to original position upon completion of work. If work required will not allow furniture and portable office equipment to be replaced to its original position, new locations will be identified by the COR, in coordination with the customer.

C.5.1.11. Utility Outage Notification

Utility outages may occur on electrical distribution, sewage and storm water collection, steam and high temperature hot water heating distribution, domestic hot water, water, gas distribution and facility heating systems. Outages are classified as either emergency or planned.

Unscheduled outages are system failures that pose a threat to life, safety, and health. Unscheduled outages require a one hour response and shall be worked on until corrected. Unscheduled outages shall be reported to the COR and other affected agencies such as Fire and Emergency Services and Military Police (MP) within one hour of occurrence.

Planned outages may be required to perform maintenance and repair work, and shall be scheduled and organized such that the number and duration are minimized. Planned Outages shall be in accordance with FC Regulation 420-19, Fort Carson Scheduled Utility Outage Requests (SUOR). The Contractor shall properly prepare the outage request form in FC Regulation 420-19 SUOR.

The Contractor shall do the following:

C.5.1.11.1. Provide outage request applications, FC Form 420-E, to government construction representatives, military units, or commercial Contractors in possession of a DPW approved DA Form 4283 or active USACE project number unless otherwise stated in FC Regulation 420-19.

C.5.1.11.2. Process applications and send out the completed outage form to the requester, required customers, and required government staff via outlook meeting maker within twenty-four (24) hours after form completion.

C.5.1.11.3. Provide notification at least five work days in advance to affected occupants and senior management in the affected building(s). If a scheduled utility service outage is expected

to last more than four hours, provide applicable information to the COR at least seven days in advance for email notification and coordinating for newspaper notification in *The Mountaineer* as applicable.

C.5.1.11.4. Provide a situational report (sit rep) to the COR and others determined by the COR, for all unscheduled utility outage within 60 minutes of discovery. The report shall provide outage details known at that time with estimated completion to restore. An updated sit rep shall be sent out every 60 minutes until the utility service is restored detailing actions and efforts to restore the utility.

C.5.1.11.5. Develop a tracking tool to track all dig permits from request to expiration, by contractor. All data shall be kept for contract duration and available to the Government when requested and at the end of the contract.

C.5.1.12. Repairs to Buried Utilities

Repairs to underground utilities including, but not limited to, gas, sewer, water, steam, high temperature hot water, electrical and cathodic protection, shall be inspected and documented with photos and a detailed narrative describing repairs and conditions by the Contractor QC prior to backfilling. Also, coordinate with DPW GIS office for GPS coordinate locations prior to burying utility. This includes work performed after normal duty hours. The Contractor shall maintain the inspection information and provide files to the Government when requested and at the end of the contract.

C.5.1.13. Traffic Control for Planned Maintenance and Repair

Traffic control may be required in order to perform maintenance and repair work, and shall be scheduled and organized such that the number and duration of planned traffic control outages are minimized. The Contractor shall follow requirements of the Manual of Uniform Traffic Control Devices (MUTCD) part six for all traffic deviations or closures. The Traffic Control Plan shall be prepared and submitted to the KO, COR and others provided by the COR 14 days prior to event IAW C.8, report C.5.1.13.

C.5.1.14. Utility system Mapping and GIS Information

Existing utility mapping is available for use on this contract. The Contractor shall have access to a DPW share drive where the most current maps are available. The maps will be provided in Bentley MicroStation .dgn electronic Computer Aided Design (CAD), Portable Document File (PDF) or Shape File (SHP) file format. This contract shall utilize this utility system mapping during the daily performance of work specified in the contract. The accuracy of this mapping may vary and may not always show existing or correct conditions.

The Contractor shall provide updated mapping information when new utility infrastructure is installed or activated; when utility or other map discrepancies or inaccurate information is

identified; when utility lines are extended, disconnected or abandoned; or lines are removed. These updates shall include a redline drawing to show the changes needed to the existing utility mapping along with GIS Spatial Data Standards for Facilities, Infrastructure and Environmental (SDSFIE) attribute information that identifies system equipment information. There are up to 30 GIS SDSFIE attribute information points that will typically be required. Surface feature locations and GIS attributes shall be provided in a Trimble electronic GPS format post process corrected within three meter accuracy, to the DPW GIS section within seven calendar days of any change. An example of an as built drawing is provided in C.7, TE-5.1.14.

C.5.1.14.1. The Government will provide Bentley View or Bentley Microstation software for use to view and print utility mapping.

C.5.1.15. New As-Built Drawings and Technical Manuals

As-built drawings, design documents, and technical manuals for most facilities are available for review at the GIS Office or Technical Library. The Contractor is advised that existing record drawings and maps of facilities vary in accuracy in showing as-built or actual status. Since some record drawings may not be completely accurate, the Contractor shall perform field verification of drawing accuracy prior to performance of work then provide any identified updates to the COR and others provided by the COR NLT five days after identification IAW C.8, report C.5.1.15.

C.5.1.16. Asbestos and Lead Containing Material

Asbestos and lead containing material may be found in many of the facilities involved in this contract. Asbestos containing materials may be found behind walls, on piping and ducts, in attic and crawl spaces, on siding, window putty, roofing, and in tile and sheet flooring. Some suspect buildings have been surveyed, and the results are available for review in the DPW Environmental Division. Work under this contract may be required in these areas.

Any such work that requires disturbing asbestos containing materials shall be accomplished IAW 29 Code of Federal Regulations (CFR) 1926.1101; 40 CFR 61 subpart M Section 61.140 through 61.157; and 40 CFR 763, and Colorado Regulation # 8 and Colorado Regulation # 19. However, any repair or maintenance task involving removal of asbestos in excess of 150 square feet or 250 linear feet is not within the scope of this contract. The Contractor shall notify the COR of such task that may exceed these limits so that the work may be completed under a separate contract. All asbestos work shall be coordinated with the DPW Environmental Division for inspection and information tracking of removed materials. Materials for disposal shall be coordinated through DPW Environmental Division. Provide copies of all material and clearance sample results to the COR and DPW Environmental Division NLT 72 hours after receiving results IAW C.8, report C.5.1.16.

C.5.1.17. Procurement of Energy Efficient Products

All products purchased under the contract shall be IAW Section 507 of Executive Order (EO) 12902 *Energy Efficiency and Water Conservation at Federal Facilities*. The Contractor must maintain supporting documentation of meeting the EO in the purchasing file and in the technical library O&M record.

C.5.1.18. Heating Equipment Fuel Changeover

The primary fuel for heating equipment on Fort Carson is natural gas. The gas distribution source is Colorado Springs Utilities (CSU). During periods of extreme cold weather or increased fuel demands, heating equipment fuel changeover may be required from the utility company. Notification will be made by phone call to the Contractor's customer service section, 526-5345. Change over heating equipment in the central heating plants 1860 and 9609 from natural gas to fuel oil. Notify the COR within one hour of a heating equipment fuel changeover requirement.

C.5.1.19. General Standards and Tasks

Comply with all requirements in this PWS, whether specifically stated as such or reasonably implied from other language, and its proposal, if incorporated as part of this contract and in doing so, meet the following standards for all tasks under this PWS:

- Complete work IAW all applicable specifications and standards;
- Complete work within specified time criteria, with minimum disruption to the mission performance of Fort Carson and its servicing agencies and other Contractors;
- Complete work, without requiring rework;
- Complete work in a cost-effective manner, including recommending improved alternatives for completion of task objectives and suggesting deletion of unnecessary requirements;
- Complete work using common business practices of scheduling daily employee work and making appointments with customers for repair and preventive maintenance activities;
- Utilize effective business strategies and practices to keep employees engaged in repair activity throughout the work day;
- Utilize authorized Management Information System to the fullest extent possible to include but not limited to: Technical Exhibit data, accurate report development, technological efficiency gains for employee production, material procurement and warehouse inventory; and
- Must provide customer notification of completed DMO's.

C.5.1.20. Inadvertent Discovery of Archaeological, Cultural, or Paleontological Materials during Excavation

If, while performing any type of ground disturbance and/or excavation as part of accomplishing a task under this contract, the Contractor encounters any indication of historical or archeological items, it shall cease work and notify the COR, mark the area to avoid inadvertent disturbance, and comply with the requirements in FC Regulation 200-1, Chapter 8. In emergency operations, the Contractor shall comply with this requirement to the extent possible without jeopardizing such operations.

C.5.1.21. Storm Response

After storms, the Contractor shall be required to identify downed tree branches, downed trees, standing water in storm swales, clean-up debris from facility damage, etc. within 24 hours of storm end or next duty day. The Contractor shall open DMO's for all identified damage for necessary repairs. This response shall be part of the Contractor's contingency plan.

C.5.1.22. Warranty Records

The Contractor shall track and manage all warranties of Government-owned property, vehicles, and equipment, which is under warranty and used, managed, or supported under this contract IAW FAR and DFAR requirements and FC Reg. 420-22. The Contractor shall maintain a record copy of all warranties and make them available to the Government upon request.

The Contractor shall do the following:

C.5.1.22.1. Maintain a spreadsheet tracking warranty records to enable execution of its enforcement. This responsibility shall extend to GFE, real property facility systems, warranties arising from construction projects under contracts completed by USACE Contractors and other DPW Contractors, including any extended warranties, such as roof warranties. Receive notification of project completion for warranty tracking by warranty letter and other means.

C.5.1.22.2. Maintain the spreadsheet continuously that tracks all repair calls made as warranty until completed. The warranty call information shall be kept throughout the warranty period to help determine systemic problems. This data will be kept for six months after the warranty expires.

C.5.1.22.3. Allow Government access at any time to the spreadsheet. Rights to both the data and the systems shall be available for turnover to the Government or a successor Contractor upon expiration of this contract.

C.5.1.22.4. Warranty Enforcement

C.5.1.22.4.1. Ensure warranties of material, equipment, and construction are utilized and enforced. Determine whether there are any applicable warranties that might apply to the situation before making repairs on any facility or equipment.

C.5.1.22.4.2. Ensure there is proof of warranty coverage for all purchases and add that proof to the records maintained under Section C.1.7.1.

C.5.1.22.4.3. Report to the COR difficulties encountered in the enforcement of warranties and instances in which the costs of enforcement would exceed the benefits of warranty enforcement.

C.5.1.22.5. Exceptions

C.5.1.22.5.1. Repair warranty items without recourse to the warranty if the failure or defect results from human abuse, or from improper or inadequate maintenance, as determined by the Government, or if the Government directs such repair. Existence of a warranty does not alleviate the Contractor of its responsibility to perform work needed to prevent potential damage to personnel and property or to prevent unnecessary shutdown of facilities or functions.

C.5.1.23. Preventive Maintenance Orders (PMOs)

Perform preventive maintenance on installed real property equipment. This includes, but is not limited to: HVAC coils, HVAC air handlers, furnaces, pumps, fans, transformers, sub stations, overhead doors, various utility lines and components and equipment.

The Contractor shall do the following:

C.5.1.23.1. Ensure the authorized MIS is used for all PMO activity. A list of current PM tasks is identified in C.7, TE-5.1.23.1.

C.5.1.23.2. Perform equipment level Preventive Maintenance (PM) to minimize future repair requirements and keep each real property system or component operating efficiently and properly.

C.5.1.23.3. Develop a cycle schedule by PM task to perform the effort according to manufactures recommendation for requirement and frequency. The schedule shall clearly differentiate between each functional area and shall reference the specific equipment item being maintained by equipment identification number or building number, equipment name and equipment location.

C.5.1.23.4. Ensure the correct frequency/frequencies are established and PM is performed based on the equipment, manufactures recommendation, applicable code or other established criteria. In the event frequency is not known or established, provide a recommendation to the KO for determination.

C.5.1.23.5. Open a separate DMO if a repair requirement is identified during the PM activity.

C.5.1.23.6. Maintain records of completed PMs for the duration of the contract. Include date of PM, effort during PM, hours expended and PMO number and technicians name. Make the records available to the COR upon request and provide to the Government upon contract expiration/termination.

C.5.1.23.7. Attach a sticker, where applicable, to the equipment item (or in the immediate vicinity thereof) indicating the equipment number/item, the date of the maintenance, and the initials and identification number of the Contractor employee who performed the work.

C.5.2. CUSTOMER SERVICES

Customer Services include, but are not limited to, operating a work reception and customer service counter, processing DMOs , PMOs, excavation support, sending updates to the DPW Technical Library and GIS office, and military unit support maintaining their real property.

C.5.2.1. Work Reception and Customer Service

The work reception and customer service counter is the customer focal point for DPW. This section shall receive and process repair requests and provide status updates to customers. As an extension of DPW, the assigned personnel shall also provide DPW information packets, professionally respond to customer questions, and properly route requests for services. It is expected these personnel understand the PWS requirements to prevent misinformation to customers and delays in repair response. Requests for customer service shall be received via telephone, electronically, in writing, or in person.

The Contractor shall do the following:

C.5.2.1.1. Staff the customer service counter with personnel familiar with the functions provided under this contract and other DPW and Fort Carson services. Respond to requests for information on the functions of DPW, O&M Contractor responsibilities, locations of various DPW offices, and criteria for repair priorities. Provide points of contact for other services provided by DPW.

C.5.2.1.2. Ensure that customer service personnel are respectful, courteous, calm, responsive, and provide correct information to customers. Requests from customers shall be responded to first come first served. There shall be no preference to customers on phone or in person.

C.5.2.1.3. Within six months into the base contract period, establish a means by which both Contractor employees and customers may check electronically on the status of Repair Service Orders.

C.5.2.1.4. Monitor utility SCADA systems and Energy Management Control Systems (EMCS). The service call center may be utilized as an alternate DPW Emergency Operation Center in the event of an exercise or real world disaster, such as a natural disaster.

C.5.2.1.5. Ensure that after normal business hours customer service accepts and inputs all repair requests into the MIS database, dispatch appropriate personnel for emergency calls, monitor the status of personnel called out, and update DMO status for work responded to by technicians.

C.5.2.1.6. Ensure the utility SCADA and EMCS systems are monitored regularly for potential facility problems. Ensure necessary corrections/repairs are made when a facility alarms within EMCS. .

C.5.2.1.7. Maintain a log describing calls received and action taken by the operator during non-duty hours and submit the daily log to the COR NLT 0730 Monday through Friday IAW C.8, report C.5.2.1.7.

C.5.2.2. Demand Maintenance Orders (DMO's)

DMO's are used to approve emergency response and scheduled routine repairs as defined in DA Pam 420-6 *Directorate of Public Works Resource Management System* and DA Pam 420-11 *Project Definition and Work Classification*. DMO's are generated for repair work only. Upon receipt of a DMO, the Contractor shall determine cost of the repair and follow requirement of C.5.1.4.6 for Limitation to Repair Authority.

The Contractor shall do the following:

C.5.2.2.1. Input new DMO's into MIS database when request is received without duplicating existing effort. Prioritize DMO's per C.5.1.7.1.

C.5.2.2.2. Perform repairs on all real property facilities listed in section C.7, TE-5.2.2.2-A, except those listed as non-real property facilities in TE-5.2.2.2-B. There are three Technical Exhibits (TE) providing historical repair services data including type and quantities of typical repairs and Miscellaneous Requirements. See section C.7, TE-5.2.2.2 FY 11, TE-5.2.2.2 FY 12, TE-5.2.2.2 FY13 and TE-5.2.2.2 FY14.

C.5.2.2.3. Perform comprehensive maintenance and repair on real property facilities that have been vacated by deployed soldiers as directed by the KO. This will include, but not limited to: barracks, Supply & Administration (S&A), motor pools, and headquarters. Refer to C.5.19.1.6 for requirements in performing this effort.

C.5.2.2.4. Follow established requirements of the MIS database to closeout DMO's. Prior to closing, ensure labor hours, equipment and material costs are included on the DMO.

C.5.2.2.4.1. Identify labor (activity type) used to perform DMO's by trade used to complete the repair. Example; if a laborer and plumber were used to complete a repair, than the DMO will identify separate labor line items for the laborer and another for the plumber.

C.5.2.2.5. Obtain guidance from DPW Environmental when doing routine repairs to or within the historical properties identified in TE-5.2.2.5. If making an emergency repair, contact DPW Environmental within 24 hours after emergency repair has been accomplished.

C.5.2.3. Excavation Permits/Utilities Locates

Excavation permits (utility locates) are issued to military units for training exercises and to commercial Contractors (to include this Contractor) for their tasks that require digging.

Prior to performing any excavation on Fort Carson, military unit and commercial Contractors are required to coordinate with utility owners to locate buried utilities (steam and high temperature hot water distribution, potable and non-potable water, electrical distribution, irrigation, sewage, storm water collection, cable systems, telephone and natural gas systems).

The Contractor shall do the following:

C.5.2.3.1. Process and issue excavation permit applications (FC Form 86-E) with control number and coordination points of contact as requested by military units or commercial Contractors in possession of a DPW approved DA Form 4283 or active USACE project number IAW FC Reg. 420-20 dated 3 January 2011.

C.5.2.3.2. Identify and locate underground DPW owned utilities upon receipt of a permit application IAW FC Reg. 420-20 and American PW Association (<http://www.apwa.net/>) *Color Code and Marking Guideline* and industry standard. Ensure utility locators initiate a repair service order when damaged or non-functional tracer wires are discovered.

C.5.2.3.3. Process submitted applications and issue completed excavation permit number with effective dates. Maintain and update a digital file of issued permits. The data shall contain 12 months of information.

C.5.2.3.4. Provide copies of underground utility locates to DPW when actual locates do not match current base utility drawings. Refer to procedures outlines in Section C.5.1.14 of the PWS. Keep a data base of drawings turned into the DPW to include date, time, area affected and utility affected.

C.5.2.4. Geographic Information System (GIS) Office

The DPW GIS Office located in building 1219 is the repository for all construction as-built drawing and utility drawings. Many of the documents included in the GIS Office are “one-only” and accountability of these documents is critical to the maintenance and repair of the facilities and infrastructure of Fort Carson. The contents of the GIS Office are in either hard copy or electronic format. Updates to utility drawings, via CD, shall be provided to the O&M Contractor monthly to update their computer database.

The Contractor shall do the following:

C.5.2.4.1. Provide updates to drawings to the GIS office for updating as they are identified following guidance in C.5.1.4.6.

C.5.2.4.2. Follow GIS office procedures for requesting drawings to perform requirements of this contract.

C.5.2.5. Special Events

Special events are planned morale, recreational, training exercises or other types of events that require support from the Contractor. Support required includes, but is not limited to: electrical, fencing, trench digging, bleacher relocations, temporary barricades and water hauling or distribution.

A list of these events is in TE C.5.2.5.

The Contractor shall do the following:

C.5.2.5.1. Provide support for Special Events. Start and completion dates, along with duration of work, will vary by requirement and will typically be identified on OPORD's.

C.5.2.5.2. Capture all effort supporting the Special Events on DMO's.

C.5.3. SUPPORT SERVICES

DPW support services include but are not limited to, developing maintenance plans, updating maintenance plans as necessary, supporting data for Installation Status Report (ISR), utility data, meter maintenance, facilities assessment, and animal manure removal.

C.5.3.1. Management Information System (MIS)

The Authorized MIS program is a corporate tool used to manage infrastructure maintenance and repair services. It holds current information on equipment, facility, safety, and plan inventory; automatically schedules work; tracks repair and maintenance work orders from "cradle to grave"; tracks repair and maintenance information and labor reporting; tracks material levels and charges; and provides management reports.

The Contractor shall do the following:

C.5.3.1.1. Input, maintain and update data in the MIS. Record actual expenditures for each DMO and PMO to allow for DMO close out. The Government has the ability to run reports and queries from the MIS, however, the Contractor may be asked to provide reports for Government validation. Submit the reports to the COR within two business days of request IAW C.8 report C.5.3.1.1. The Contractor may receive up to 15 report requests monthly.

C.5.3.1.2. Keep MIS data current and accurate. DMO and PMO data shall be accurate within one working day of receipt of status change and shall accurately reflect current status.

C.5.3.2. Operation and Maintenance (O&M) Plans

Operation and maintenance plans are used as a guideline in the performance of system operation and preventive maintenance on equipment, systems, and facilities. Operation and maintenance plans include, but are not limited to: proposed operation of programs, sequencing, cycle time, estimated time standards, seasonal requirements, safety plans, hazards, and precautions.

The Contractor shall do the following:

C.5.3.2.1. Review and update existing plans to ensure provided plans are current with industry and O&M requirements and shall be submitted electronically or on CD to the KO for Government review. TE 5.3.2. identifies all the required O&M plans. Submit the plans to the KO and COR NLT 30 days after contract start IAW C.8, report C.5.3.2.1.

C.5.3.2.2. Prepare/develop additional O&M plans as determined by the KO.

C.5.3.2.2.3 Make appropriate updates to O&M plans to ensure the information for the applicable process is accurate and up to date with existing regulations, equipment, and operations.

C.5.3.2.2.4. Review plans annually and update as necessary based on that review. The Operation and Maintenance Plans shall become a part of the contract. Submit plans as required in C.5.3.2.1.

C.5.3.2.2.5. Ensure the required maintenance plans do not include Contractor legends (headers). They will not be identified as proprietary as the plans identify how the Contractor will maintain the various Government systems. Therefore all the plans are considered Government property.

C.5.3.3. Additional Support Requirements

The ISR is an assessment of the condition of Fort Carson facilities and is prepared using the ISR program. The DPW is responsible for preparing the ISR; the Contractor is only responsible for supporting that effort by providing accurate supporting data. CLS reports are quarterly requirements the Government must submit to Installation Management Command (IMCOM). Provide supporting MIS data for accurate CLS reporting.

The Contractor shall do the following:

C.5.3.3.1. Perform ISR infrastructure and facility inspections when necessary. Requirements for inspection will be provided by the COR with a five work day suspense after receipt. There are currently no more than 30 data reports needed quarterly.

C.5.3.3.2. Provide data supporting CLS reporting. There are currently no more than 30 data reports needed quarterly based on IMCOM requirements which change frequently.

C.5.3.4. Meter Management

Utility suppliers invoice Fort Carson for electric, natural gas, and water. The DPW processes these suppliers' invoices and sends the original to Defense Finance Accounting Service (DFAS) for payment and one copy to Fort Carson Contracting Office to hold for information. DPW bills

Reimbursable Customers (customers DPW is not funded for) for utilities purchased for their use as well as those utilities provided by Fort Carson (sewer, steam, High Temperature Hot Water (HTHW)). Charges are based on actual or estimated consumption. Installation of meters is also a management enhancement tool to identify energy cost savings attributed to conservation projects, energy system maintenance activities, and energy load management. Meters can also be used to identify potential problems with existing infrastructure and equipment prior to system failures.

The Contractor shall do the following:

C.5.3.4.1. Operate and maintain all Fort Carson utility meters. Specific tasks include, but are not limited to, collecting data through the Energy Management Control System (EMCS) along with repair by replacement and maintenance of electric, water, and natural gas meters. The KO may identify the installation of up to 30 utility meters annually to meet Army requirements.

C.5.3.4.2. Ensure there is connectivity for all electric, water, and natural gas meters that transmit data into the EMCS system. Coordinate with the NEC for final connection into the EMCS system. Verify that each meter is transmitting accurate data. Maintain a record of all meters installed for a period not less than five years, and available to the Government upon request. The report shall indicate the location, type, EMCS working status, and scheduled completion date for connection into the EMCS system for each meter.

C.5.3.4.3. Ensure that meters shall provide data at least daily and measure at least hourly consumption of electricity and gas. Use the following units of measurement: Electric= Power (kilowatt), average demand over 15 minute intervals. Energy (kilowatt-hours), Natural Gas= Cubic Feet per hour, Water-Gallons per day. Provide this data to the DPW so historical and real time building consumption rates and trends can be determined. Ensure EMCS meter data is relayed and connected into the Army's web-based Meter Data Management System (MDMS) and compatible with the MDMS software. Manually read meters for reimbursable utility customers if their meters are not connected or reports to EMCS according to section C.5.3.4.4.

C.5.3.4.4. Develop and maintain a master meter spread sheet showing location, type, make, model and size of each meter. Indicate the past months and current months meter reading for all reimbursable customers. (The DPW, Operations and Maintenance Division will provide a list of reimbursable facilities.) Included meter reading for the Photo Voltaic (solar) systems if readings cannot be provided from the EMCS system. Read meters monthly and provide the data to the COR NLT the 17th of each month IAW C.8, report C.5.3.4.4. Exclude Fort Carson Family Housing and MEDDAC as they report their own meter readings.

C.5.3.4.5. Perform repair and replacement of existing meters as needed. Perform initial field verification on each newly installed electric and rotary or turbine gas meter to verify meter is working correctly. Additional field verification testing shall be required when an anomaly is identified. Perform meter calibration when required. Keep maintenance and calibration reports on file for each meter and make them available to the DPW when requested. Ensure that each

meter has a unique asset identification tag to identify the meter type, location, and other pertinent information.

C.5.3.4.6. Set up and program electric meters to read on the faceplate at a one to one ratio. If this is not possible due to meter capability, identify on the meter the CT (current transformer) and PT (potential transformer) ratio and multiplier. Affix a permanent label or tag to the meter can or enclosure in a location that is clearly visible when reading the meter and listed in the meter data base.

C.5.3.4.7. Test electric meters onsite initially to verify actual load being measured equals the meter readings. Ensure all meters have an irreversible type pad lock seal with an individual seal number to identify tampering, unauthorized access or demand values reset.

C.5.3.5. Facility Assessment

Facility assessments are made by Fort Carson to determine the condition of facilities after natural or man-made disasters occur including, but not limited to, earthquakes, tornado's, aircraft crashes, acts of terrorism, flooding, or strong winds that could affect activities or operations of Fort Carson. They are also accomplished in preparation for mobilization and as part of reoccurring safety or maintenance inspections. As required to support Fort Carson, the KO may require the Contractor to perform an assessment after an event to identify damage or requirement for engineer evaluation.

The Contractor shall do the following:

C.5.3.5.1.

Perform all infrastructure and facility assessments as requested by the KO. Ensure all damage assessments are accurately identified and completed by the date required. Provide assessment results to the COR within one day of completing the assessment.

C.5.3.6. Animal Manure Removal

Animal manure includes bird, raccoon, mice, or other animals. The effort may be required in crawl spaces, above ceilings or in wall cavities on all Fort Carson real property.

The Contractor shall do the following:

C.5.3.6.1. Remove and properly dispose of animal manure (feces and droppings) that are a nuisance or pose health or safety issues.

C.5.3.6.2. Disinfect and deodorize affected and surrounding areas to control odors.

C.5.3.6.3. Refer to C.7, TE-5.3.6 for US Army Public Health Command (USAPHC) Technical Information on clean-up of rodent contamination

C.5.4. GROUNDS MAINTENANCE

Grounds maintenance includes grounds, related features and appurtenances on Fort Carson. Work shall include, but is not limited to: maintenance of turf, trees, shrubs, playgrounds, outdoor recreational areas, removing trash from receptacles in pavilions and parks, training ranges and running trails. The Contractor shall provide work to maintain grounds and related features and appurtenances to standards specified herein. Care shall be taken to protect existing trees, shrubs, and ornamental plants from harm during grounds maintenance activities. Trees, shrubs, and ornamental plants damaged or destroyed shall be repaired or replaced with the same plant of equal or better size and condition within five days. Weed control shall not harm non-target plants, water, fish, or animals. Additional grounds maintenance for special events and other requirements shall be as determined by the KO. All grounds on Fort Carson cantonment, Turkey Creek Ranch, and Down Range are considered to be Improved Grounds if they are mowed. The mowed areas will be distinguished as Irrigated or Non-Irrigated Improved Grounds.

The Contractor shall do the following:

C.5.4.1. Irrigated Improved Grounds

Irrigated improved grounds include approximately 330 acres located within the cantonment area of Fort Carson and PCMS. These areas include but are not limited to, parade fields, parks, and athletic fields and established irrigated turf around buildings. The irrigation system utilized on Ft Carson is Hydro Point data system “Weather Trak”. It is a smart irrigation system that is real time and climate controlled.

C.5.4.1.1. Mowing and Trimming

The Contractor shall do the following:

C.5.4.1.1.1. Maintain mowing and trimming equipment to assure clean and even cutting, prevent rutting of ground area, avoid scalping of vegetation, and avoid damaging trees and shrubs. Areas for performance of mowing and trimming can be found in section C.7, TE-5.4.1.1.1. Provide to the Government a seasonal mowing schedule to the COR NLT 1 May annually and three days prior to change IAW C.8 report C.5.4.1.1.1.

C.5.4.1.1.2. Mow and trim all irrigated turf to a uniform height of no less than 3 inches and no taller than 4 inches. Direction may be provided for the mowing of athletic fields to a height of (2-1/2) inches if an official tournament is to take place.

C.5.4.1.1.3. Perform out of cycle mowing to areas in support of Command events. Anticipate 400 acres over eight events during the mowing season.

C.5.4.1.1.4. Expect additional mowing outside established frequencies based on Command Emphasis requirements provided by the Government. Anticipate an additional 200 acres of irrigated turf mowing for this requirement.

C.5.4.2. Edging and Sweeping

The Contractor shall do the following:

C.5.4.2.1. Edge approximately 25,450 linear feet of turf along sidewalks, driveways, and curbs in the cantonment area (administrative and other high visibility areas) an additional 250 feet at PCMS. Properly edged turf shall not extend over curbs, sidewalks and driveways. Irrigated areas shall be edged where turf and hard surface meet and will present a clean squared appearance with no grass growth onto the hard surface

C.5.4.2.2. Edge non-irrigated areas to ensure no weed or grass grow over street curbs. Grass and weeds shall be trimmed to the backside of curbs.

C.5.4.2.3. Sweep approximately 150 acres of irrigated turf after each mowing period to remove grass clippings and debris. This area encompasses parade fields, Athletic Fields, Division HQ, Garrison HQ, and Heritage Row. Clippings shall only be swept away if cut clippings are visible after mowing.

C.5.4.2.4. Grass clippings shall be disposed of as compost. A roll off container will be provided by the Post Refuse and Recycle contract. Contact the COR of the Post Refuse and Recycle contract at 526-1695 for container replacement and haul off.

C.5.4.3. Non-Irrigated Improved Grounds

Non-irrigated improved grounds include, but are not limited to: roadsides in the cantonment area, ammunition storage areas, semi-improved athletic fields, playgrounds, camp grounds, Turkey Creek Ranch, and Downrange training ranges on Fort Carson and PCMS Including uninhabited ranch houses at PCMS.

The Contractor shall do the following:

C.5.4.3.1. Schedule mowing and trimming, maintain mowing and trimming equipment to assure clean and even cutting, prevent rutting of ground area, avoid scalping of vegetation and avoid damaging trees and shrubs.

C.5.4.3.2. Mow grass/vegetation to a uniform height no less than five inches or more than six inches unless specifically stated elsewhere in the PWS. Do not allow non-irrigated areas identified on the TE to exceed 10 inches in height, roadside mowing is not included in this specification and is addressed elsewhere in the PWS.

C.5.4.3.3. Mow and trim approximately 200 acres of roadside to the same height requirements stated in C.5.4.3.2. Roadside mowing shall extend one mower width (approx. 10 feet) on either side of the road or dirt shoulder. Approximately 1,100 acres of ammunition storage areas, semi-improved athletic fields, playgrounds, and camp grounds shall be mowed and trimmed monthly.

C.5.4.3.4. Furnish labor and supervision necessary for the performance of all downrange mowing operations incidental to type I, type II, and type III mowing requirements to keep the ranges, training facilities, and downrange roadsides of Fort Carson and PCMS maintained. The locations, frequency and performance requirements of the work will be in accordance with detailed specifications as identified in C.7, TE-5.4.3.4 and TE-5.4.3.4A. Coordination for this effort will be made with DPTMS, Range Control to ensure range/training area clearance is approved.

C.5.4.3.5. Maintain the Butts Army Airfield (BAAF) non-irrigated grass as depicted in C.7, TE-5.4.1.1.1. to a uniform height not less than six inches and not to exceed 12 inches in compliance with AR 95-2, Airspace, Airfields / Heliports, Flight Activities, Air Traffic Control, and Navigational Aids.

C.5.4.3.6. Mow grounds areas around selected facility areas in October to reduce/prevent tumble weed hazards. Adjust mower deck height so grass blade length is not less than six inches or more than 10 inches after cut. The areas impacted by this requirement are west of Butts Army Air Field, west of the firing Ranges along Butts road, up to 200 feet around the ASP and AHA. The mowing of these areas will not exceed 500 acres for a one time effort each year.

C.5.4.3.7. Expect additional mowing outside established frequencies based on Command Emphasis requirements provided by the Government. Anticipate an additional 500 acres non-irrigated mowing for this requirement.

C.5.4.4. Fertilization

The Contractor shall do the following:

C.5.4.4.1. Develop a plan and properly fertilize all irrigated areas of improved grounds to maintain healthy lawn and soil, to promote growth and reduce weeds. The soil type and appropriate testing of the irrigated areas shall determine the frequency and type of fertilizer. The application of soil amendments or conditioners may be used to aid in maintaining healthy grass. Colorado State University (CSU) Extension, <http://www.ext.colostate.edu/index.html>, shall be used as a primary source for information to aid in properly maintaining irrigated grass on Fort Carson.

C.5.4.5. Aeration/Dethatching and Sweeping

The Contractor shall do the following:

C.5.4.5.1. Aerate, dethatch, and sweep approximately 315 acres of improved grounds. This is a separate task from sweeping previously described in Improved Grounds. Perform these tasks annually after the soil thaws and turf has begun growth, usually in April, and annually after grass becomes dormant, usually in September. Areas for performing aeration/dethatching and sweeping are all irrigated areas.

C.5.4.5.2. Thoroughly remove all turf residue. See C.5.4.2.2.4 for disposal.

C.5.4.6. Irrigation

The Contractor shall do the following:

C.5.4.6.1. Operate and maintain all potable and non-potable irrigation systems on Fort Carson (TE-5.4.6.1A) and PCMS (TE-5.4.6.1B), with the exception of the golf course and family housing. Maintain the non-potable pumping system at the Sewage Treatment Plant (STP) and all non-potable systems and distribution. Areas for irrigation are at section C.7, TE-5.4.6.1A.

C.5.4.6.2. Irrigate approximately 315 acres of improved grounds. The irrigation cycle shall provide that vegetation is irrigated at a rate not to exceed ½ inch of water in a 24-hour period, not more than four cycles per week, and not more than two inches of water in a one week period. These rates can vary year to year due to water restrictions set forth by CSU. Ensure vegetation is not stressed to advanced stages (wilting, brown, curled leaves) due to lack of watering or over watering. Imposed water restrictions during identified drought conditions may cause damage to vegetation and will not be the fault of the Contractor.

C.5.4.6.3. Provide maximum water conservation and comply with any Fort Carson water use restrictions. During periods of restricted water use, irrigation system use will be minimized or turned off as determined by the Government.

C.5.4.6.4. Manage the irrigation system to avoid interference with military ceremonies or other events, as required.

C.5.4.6.5. Activate the irrigation system in the spring, checking for leaks and malfunctions, and making necessary repairs; deactivate the system in the fall; and protect the system from freezing. Purge sprinkler head systems and store portable systems during winter months.

C.5.4.6.6. Hand water vegetation not covered by an irrigation system. Hand water on improved grounds at same intensity as specified in for irrigation cycle above. Hand watering of trees and plants shall be based on COR direction with contractor and DPW Forrester input.

C.5.4.6.7. Perform winter watering when air temperatures are above freezing. Apply water during the day to allow it to soak in before freezing. Perform winter watering during long periods with little or no snowfall. Winter watering of turf and identified plants shall occur as needed to maintain turf and plant health. Disassemble portable irrigation systems after each day they are used during winter months.

C.5.4.6.8. Maintain all components of the irrigation systems, including, but not limited to: Weather Trak irrigation control system, approximately 62,000 heads, 154 installed systems and 1544 zones on approximately 315 acres, seven other installed systems on approximately 20 acres, and automatic trickle irrigation systems totaling approximately 30,000 linear feet in length. Components used for repair/replacement shall be of equal design and capacity to the original. The irrigation systems shall not be modified without prior written approval by the KO. Section C.7, TE-5.4.6.8, identifies sprinkler component types.

C.5.4.6.9. Systems activation shall be once annually, systems deactivation shall be once annually. Military and special events shall be responded to as required.

C.5.4.6.10. Perform watering of trees and shrubs as needed to support healthy growth from new planting until establishment. Plant establishment may take one to three years based on the plant type, species and drought conditions. Ensure new plantings (trees, shrubs, etc.) are watered during the non-growing season to prevent plant failure, as temperatures allow. Follow guidance from local nursery's or CSU Extension office.

C.5.4.7. Top Dressing and Over Seeding

The Contractor shall do the following:

C.5.4.7.1. Top dress and over seed improved athletic fields and other areas as needed to keep the turf established. The Contractor will be required to coordinate with DFMWR prior to closing off any athletic field. Determine when to apply and which areas require top dressing and over seeding so irrigated areas are maintained in a healthy plush condition. Ensure top dressing soil mix meet specification in section C.7, TE-5.4.7.1. The Contractor may be required to use sod in lieu of over seeding as identified by the COR.

C.5.4.7.2. Apply top dressing uniformly at the recommended rate for the specific turf being covered. Use State certified seed of the latest season's crop, delivered in original sealed packages bearing the producer's guaranteed analysis for percentages of mixture, purity, germination, weed-seed content, and inert material. Maintain copies of State-certification for not less than 12 months after seed purchase and make available to the Government upon request. The date of testing shall be within one year of the time of planting. Ensure seed is labeled in accordance with U.S. Department of Agriculture rules and regulations under the Federal Seed Act and applicable state seed laws. Seed that has become wet, moldy or otherwise damaged shall not be acceptable. Seed shall be adapted to southeast Colorado. The pure live seed mixture to be used shall be as indicated in section C.7, TE-5.4.7.2.

C.5.4.8. Irrigation Ditches

Irrigation ditches are indicated in section C.7, TE-5.4.8-A.

The Contractor shall do the following:

C.5.4.8.1. Maintain approximately 43,000 linear feet of irrigation ditches/ pipelines. Maintenance includes, but is not limited to: ditches, culverts, pipelines, head gates, and other control components. Proper maintenance shall result in irrigation ditches with an unimpeded flow of water, free of silt and other obstructions, and repaired of leaks and washouts.

C.5.4.8.2. Properly dispose of removed materials IAW permits and regulations.

C.5.4.8.3. Inspect, and operate the gate valves at Teller Dam and North Side Dam to allow free movement of gate quarterly and lubricate as necessary Assuring all gates, valves and other controls operate smoothly.

C.5.4.8.4. Ensure earthen dams are properly maintained. This includes, but is not limited to: vegetation removal from dam face, toe and groins; ensuring spillways are clear of debris and properly flowing; outflow structures are lubricated and properly working.

C.5.4.8.5. Make repairs to dam and dam structures as identified from inspections by state or USACE dam inspectors

C.5.4.8.6. Maintain a stock pile of emergency repair materials on site of various sized aggregates to prevent dam breach from excessive rains.

C.5.4.8.7. Reserved.

C.5.4.8.8. Maintain the following real property assets that lie outside the boundary of Fort Carson, west of Highway 115. These assets include but may not be limited to head-gates, valves, risers, culverts, diverter boxes, flumes, measuring devices, pipelines, surface ditches, and any other associated conveyances and water works are operated and maintained through legal easements across private property. All operations and maintenance activity to be conducted off the Fort Carson Installation shall commence only pursuant to specific approval by the KO. The assets are part of the installations water right locations at: Gale Ditch, which extends approximately one mile west of Highway 115 along Rock Creek; Ripley ditch, which extends approximately one quarter mile west of Highway 115 along Little Fountain Creek; and Womack Ditch, which extends approximately one half mile west of Highway 115 along Little Fountain Creek.

C.5.4.9. Running Trails

The running trails consist of approximately 64,000 linear feet of unpaved trails and 24,000 linear feet of paved trails. Running trails are identified in TE 5.4.1.1.1

The Contractor shall do the following:

C.5.4.9.1. Mow all trailside vegetation one mower deck width on either side of the trails and exercise points including each piece, leg and sign to a height not less than five inches. Trim

neatly and consistently all vegetation at exercise points and timber edging at same time as mowing.

C.5.4.9.2. Maintain the unpaved trail surfaces to eliminate any eroded surfaces and possible tripping hazards.

C.5.4.9.3. Grade trailside ditches to maintain drainage flow from trail surfaces allowing positive flow to main drainage ditches and eliminate safety hazards.

C.5.4.9.4. Maintain trail exercise points, benches, water fountains and all other trail appurtenances free of defects.

C.5.4.10. Pruning and Removal of Trees, Shrubs and Stumps

Tree and shrub pruning and removal shall be limited to; areas around buildings, recreation parks, roadsides where there is an obstruction issue, security issues, and tree growth affecting overhead power lines as identified in C.5.7.1.12.5. Develop and manage a pruning plan IAW ANSI A300 Pruning Standards As part of the C.5.4. O&M plan. The plan shall identify a pruning cycle that encompasses the contract term.

The Contractor shall do the following:

C.5.4.10.1. Prune select trees, sprouts and shrubs located on Fort Carson in the cantonment area, Turkey Creek Ranch and BAAF (housing is excluded) to remove dead, dying, diseased, and other trees. To ensure Migratory Bird Treaty Act compliance, utilize the NEPA review process prior to tree/shrub removal. The trees can vary in size; most will be less than 18 inch diameter but in some cases may exceed 18 inch diameter.

C.5.4.10.2. Ensure all logs greater than eight inch and less than 24 inch diameter are cut to 16 to 20 inch lengths and delivered to an on Post location provided by the COR.

C.5.4.10.3. Remove stumps of all removed trees. Use alternate methods if stump grinding or digging cannot remove them. Match surrounding areas after the stump is removed.

C.5.4.10.4. Remove stumps to a depth of six inches below ground level. Remove root flares and suckers within two feet of the stump to a depth of four inches below ground level.

C.5.4.10.5. Prune trees and shrubs to establish symmetrical growth. Brace and cable trees and shrubs as required. Work areas shall be cleared daily.

C.5.4.10.6. Ensure utilities are unobstructed and real property facilities are clear of branches.

C.5.4.10.7. Hazardous conditions caused by trees and shrubs shall be mitigated.

C.5.4.10.8. Repair damage to trees and shrubs caused by storm events to prevent further damage or disease or personnel safety.

C.5.4.10.9. Support Fort Carson's Tree City USA recertification requirements by providing information as identified below:

Number of trees planted;

Number of trees pruned;

Number of trees removed;

Cost associated with these activities;

Tree planting and initial care, such as drip irrigation or hand watering;

Tree Maintenance, irrigation repair, fertilizing, winter watering;

Management activities, NEPA reviews, meeting, site visits, etc;

Utility line Clearance or bucket work;

Other activities related to the Urban Forest

Provide information to the COR NLT 1 December annually IAW C.8, report C.5.4.10.9.

C.5.4.11. Weed Control and Ornamental Pest Control

The Contractor shall apply pesticides on Fort Carson and the PCMS for the purpose of controlling weeds in areas of turf, hardstands, utility pits, firing ranges and decorative rock for improved and maintained areas. In addition, the Contractor may treat insects and other plant pathogens to ensure a healthy urban forest. Pest and pathogens include but are not limited to Tussocks Moths, Ips beetles, Saw flies, and fungal or bacterial infections. Provide a safe, effective, and environmentally sound Integrated Pest Management (IPM) program. The IPM program shall control identified pests and weeds that may adversely affect Fort Carson's mission or military operations the health and well-being of its people, structures, materiel, property, and grounds. Manage and issue approved pesticide products for issue to soldiers and civilians for self-help use IAW PWS Paragraph C.5.17.2. Follow all required OSHA, EPA, State and local regulations and those identified in C.6 applicable to Pest Management.

The Contractor shall do the following:

C.5.4.11.1. Integrated pest management philosophy

Employ preventive measures to limit conditions conducive to pests. Monitor pest populations to determine if and when corrective measures are needed and practical. Use pesticides when no other feasible and effective physical, mechanical, cultural, biological, or educational control tactics exist. When using pesticides utilize the most effective product approved with the least hazard to humans and the environment.

C.5.4.11.2. Pest Management Plan

Comply with applicable provisions of the Fort Carson Integrated Pest Management Plan (IPMP) unless otherwise approved, in writing, by the IPM Coordinator. Develop a Pest Management Plan IAW DODI 4150.7 and the IPMP. This plan shall be separate from the Operations plan identified C.5.3.2. Submit the plan to the KO and COR 10 days prior to contract start IAW C.8, report C.5.4.11.2.

C.5.4.11.3. Pesticide Reduction and Minimization

Make all reasonable efforts to reduce the use of pesticides to minimize the impact to the human and natural environments to the greatest extent possible and practical. Fort Carson is committed to reducing pesticide usage and whenever and wherever possible and practical using the least environmental toxic pesticides to control the identified target pest(s).

C.5.4.11.4. Pesticide Application Certification/Licensure

Ensure at least one Qualified Supervisor who currently possesses a valid certificate/license in the following Colorado Pest Control Categories: 106 (Forest Pest Control), 109 (Industrial and Right-of-Way Weed Control), 206 (Right-of-Way Weed Control), and 207 (Turf and Ornamental Pest Control) to meet State requirements. Ensure all contractor applicators have current valid Certified Operator certificates/licenses in all of the same pest control categories as the Qualified Supervisor. Ensure applicators only apply pesticides in categories in which they are certified or licensed. Submit to the COR current copies of employees' certifications/licenses NLT 1 February annually and 10 business days of new hires IAW C.8, report C.5.4.11.4.

C.5.4.11.5. Implementation

Implement a program that follows the plan described in section C.5.4.14.2. Begin survey and control efforts by 1 March of each year.

C.5.4.11.6. Prevention and Control

The Contractor shall do the following:

C.5.4.11.6.1. Survey areas and determine if preventive or control measures are needed. Implement the correct measure as required to reduce pests to the level specified in the IPMP.

C.5.4.11.6.2. Broad-Leaved Weeds in Turf: Regularly survey the turf for weeds or conditions conducive to weeds. Turf shall be maintained as grassy lawn free of weeds.

C.5.4.11.6.3. All other Weeds

Survey the hardstands, utility pits, natural/unimproved areas, decorative rock areas and adjacent cracks and crevices for weeds and maintain these areas weed free.

C.5.4.11.6.4. Ranges

Perform services on small arms and target ranges which may include bare ground or broadleaf weed control.

C.5.4.11.6.5. Tree spraying

Perform services for the control of destructive species of insects and pathogens such as, but not limited to Tussocks Moths, saw flies, Ips beetles fungal or bacterial infections located within the cantonment of Fort Carson or PCMS.

Estimate of Treatment Areas

	Minimum	Maximum
Improved Areas Cantonment	500 acres	2000 acres
Ranges (locations to be determined)	100 acres	300 acres
Tree Spraying (Cantonment area: Coordinate with DPW Forestry)	1400 trees	3800 trees
PCMS Improved Areas Cantonment	40 acres	150 acres
PCMS Ranges	5 acres	25 acres

C.5.4.11.7. Response to Requests

Respond to weed control service orders within five work days, and shall determine if preventive or control measures are needed and implement the appropriate measure(s).

C.5.4.11.8. Records and Operating Reports

Prepare and submit to the COR monthly NLT 5th business day, all required operating records, reports, and pesticide usage and surveillance data using the electronic format provided by the IPM coordinator.. IAW C.8, report 5.4.11.8. See C.7, TE-5.4.11.8 for report example.

Provide a monthly report of pesticide product stored in Building 3708 to the COR NLT 5th business day IAW C.8, report C.5.4.11.8-A.

Provide daily report of pesticide product utilized the previous day to the COR NLT 0800 Monday – Friday IAW C.8, report C.5.4.11.8-B. See TE-5.4.11.8-B for report example.

C.5.4.11.9. Equipment

Ensure all vehicles and equipment used in these operations are identified with appropriate labels and signage preventing contamination to others.

C.5.4.11.10. Pesticides

Procure, handle, store, and apply pesticides in strict accordance with the EPA registered pesticide label and the Fort Carson IPMP. Only certified applicators shall operate pesticide application equipment. All pesticides shall be used in accordance with Federal, Colorado, local laws, and installation regulations.

Use the approved pesticides listed in section C.7, TE-5.4.11.10. The Contractor shall not deviate from the approved list of pesticides without prior approval from the Integrated Pest Management Coordinator (IPM Coordinator) and Army Environmental Command Pest Management Consultant (AEC PMC) through the COR. Use the Out of Cycle Pesticide Use Request form (TE-5.4.11.10-A) for pesticide approval submitted to the COR annually NLT 1 Oct IAW C.8, report C.5.4.11.10.

C.5.4.11.11. Pesticide Storage, Mixing, and Disposal

Store and mix pesticides only at Government designated locations following guidance to prevent contamination of water supply with pesticides. All pesticides, rinse water, and containers shall be disposed of in accordance with label directions and Fort Carson hazardous waste regulations.

C.5.4.12. Fencing

The Contractor shall do the following:

C.5.4.12.1. Repair, install, and remove protective physical barriers, security type chain link fencing, barbed wire fencing, razor ribbon, concertina wire, steel rod bar fencing, cable wire and all types of gates. Utilize materials which match existing material as applicable.

C.5.4.12.2. Repair all security fencing IAW Priority one work. Enclose restricted areas, interior security buildings, and enclosures, motor pools, water catchments, structures, ammunition dumps, and other limited and excluded access areas, with security chain link fence. Maintain drainage along and through fences.

C.5.4.12.3. Repair livestock-proof fencing where livestock are getting through the fence at PCMS and the south boundary of Fort Carson. Fence repairs bordering livestock areas will have priority.

C.5.5. TRANSPORTATION SYSTEMS MAINTENANCE AND REPAIR

Transportation systems on Fort Carson and PCMS includes paved surfaces (bituminous and concrete), to include paved running trails; unpaved shoulders; culverts; paint striped/lettered areas other unpaved surfaces, such as, but not limited to: gravel, earth, soil aggregate, stabilized surfaces (roads, tank trails, fire breaks, open storage areas, parking lots); traffic signs; and general purpose rail system.

C.5.5.1. Schedule of Maintenance and Progress Report

The Contractor shall do the following:

C.5.5.1.1. Determine needs for preventive maintenance and develop a weekly schedule of all preventive maintenance to be performed. The schedule shall include by day and activity what task is to be completed and the location of the task. The weekly schedule shall be maintained for a period no less than one year and made available to the Government upon request.

C.5.5.1.2. Comply with all environmental requirements for performing the above work. Appropriate NEPA review will be completed on all soil disturbance work.

C.5.5.1.3. Colorado's nuisance emission limitation guideline applies to paved and unpaved roadways. The no off-property transport emission limitation guideline shall apply to on-site haul roads and the nuisance guideline shall apply to off-site haul roads. All available practical methods that are technologically feasible and economically reasonable to minimize fugitive dust emissions must be employed when using haul trucks. The no off-property transport emission limitation guideline applies to haul trucks when operating on Fort Carson. When operating off Fort Carson, the applicable guideline is no off-vehicle transport of visible emissions.

C.5.5.1.4. Use low volatile organic compound paint, which is also lead-free, free of ozone depleting chemicals, and low in hazardous air pollutants. Obtain DPW Environmental approval for paint prior to use. Track paint usage by brand, type, color and gallons used and submit report monthly NLT 10th business day to the COR for the DPW Environmental Air Program for the previous month IAW C.8, report C.5.5.1.5.

C.5.5.2. Roads, Parking Lots, Running Trails, and Motor Pools

Roads, parking lots, running trails, and motor pools may be paved or unpaved. Components include, but are not limited to, roadways, driveways, tank trails, sidewalks, shoulders, ditch lines, drainage culverts, contraction and expansion joints, bridges, guard rails, appurtenances, curbs and gutters, parking bumpers, jersey barriers, traffic signs and bollards.

C.5.5.2.1. Paved Areas

The Contractor shall do the following:

C.5.5.2.1.1. Inspect surfaces for defects and permanently repair asphalt roadway and potholes with a hot mix asphalt repair. Temporary patches may be placed if weather does not permit a permanent patch. The permanent patch must be placed as soon as the weather allows.

C.5.5.2.1.2. Not exceed 150 square yards for any asphalt patch. Go to section C.7 to reference TE-5.5.2.1.2 for a map of current paved roads and parking lots.

C.5.5.2.1.3. Ensure potholes are patched in compliance with Section 403, Hot Bituminous Pavement, of the Colorado State Department Highways Standard Specifications for Road and

Bridge Construction and shall be equal in quality of workmanship and material to that of the City of Colorado Springs and El Paso county asphalt maintenance repair.

C.5.5.2.1.4. Repair of potholes more than four inches in depth shall occur within two hours of being identified and potholes less than four inches deep shall be repaired within seven working days of detection. A pothole is defined as a hole in the asphalt surface equal to or less than 10 sq. ft. and equal to or greater than 1-1/2 inch in depth.

C.5.5.2.1.5. Cut square/rectangle all potholes in non-failed asphalt before patching. Government approval of unique paving materials is not required, if in compliance with Colorado Department of Transportation specifications.

C.5.5.2.1.6. Grade and compact road shoulders so that shoulders are level with, or no more than ½ inch below the surfaced road and smooth after compaction. Grade and compact unpaved roads the full width to provide smooth surfaces that are free of ridges, ruts, holes and properly shed water. Vegetation shall be left undisturbed to prevent erosion. Grade shoulders during May through September. Follow Section 3 of the Fort Carson Fugitive Dust Management Plan during grading operations.

C.5.5.2.1.7. Pick up sweep all asphalt surfaced roadways, airfield runways and aprons, parking lots, loading dock areas and concrete surfaced tank trails. Sweeping shall remove all debris and roadway dirt from the road center traffic path and all gutters. A schedule shall be developed and maintained by the Contractor to sweep all paved primary streets at least once every month, Airfields weekly, secondary streets once every two months, and all parking lots two times a year.

C.5.5.2.1.8. Apply water during street sweeping activity to minimize fugitive dust emissions.

C.5.5.2.1.9. Sweep primary roads starting within 48 hours after last snowfall or when the roads are clear of snow and ice and complete with in three working days during winter months, when a granular product is used for a de-icing,.

C.5.5.2.1.10. Perform street sweeping of roads and parking lots during low traffic volume times, typically early morning, early evening or weekends to not restrict traffic flow.

C.5.5.2.1.11. Inspect and paint all existing roadway asphalt and concrete striping and marking. Perform airfield stripping and marking annually.

C.5.5.2.1.12. Be responsible for scheduling and painting parking lots. All paved parking lots shall be painted once during the contract term, unless directed by the Government to perform sooner. Coordinate paint scheduling with the facility occupants and closing parking lots or perform during non-business hours.

C.5.5.2.1.13. Schedule and paint all school crosswalks each year before the start of the fall school session. Work must be not begin earlier than 30 days before the first day of school and must be complete by the day before the first day of school.

C.5.5.2.1.14. Perform paint striping and marking IAW the Manual of Uniform Traffic Control devices for Streets and Highways. Ensure paint conforms to AASHTO M248 (1991I), ready mixed traffic paints, Fed Spec TT-P-1952D. Ensure reflective beads conform to AASHTO M247 and Fed Spec TT-B-1325C. Obtain approval of paint with DPW Environmental prior to use. Apply glass beads uniformly to wet paint at a rate of six pounds of glass/reflective media per gallon of paint. Paint and painting quality shall be equal to that used on the roadways of the City of Colorado Springs, El Paso County and the State of Colorado.

C.5.5.2.1.15. Perform asphalt striping only between 1 April and 30 September, unless a variance is approved by the KO. Re-stripe all roadway longitudinal markings by 30 June. Complete crosswalks, legends and stop bars by 30 September. Schedule around peak usage times to accomplish the task. Paint BAAF runway, taxiway and apron striping annually.

C.5.5.2.1.16. Use Preformed Thermoplastic tape stop bars and pedestrian crossings on asphalt surfaces that have not failed.

C.5.5.2.1.17. Install new crosswalk locations as identified by the Government for the safety of soldiers, civilians and family members

C.5.5.2.1.18. Keep a record documenting the dates that a parking lot, road, intersection, or crosswalk is painted or had Preformed Thermoplastic installed. Records of thermoplastic installation shall be kept for a period of five years and be available to the Government when requested.

C.5.5.2.1.19. Repair or replace any damaged guardrails within seven days of identification. Meet the Colorado Department of Transportation (CDOT) standards for guard rail repair.

C.5.5.2.1.20. Repair/replace damaged concrete sidewalk, hardstand, curb and gutter within 30 days of identification, to match existing in size, style, and finish.

C.5.5.2.1.21. Make repairs to bridges based on inspection from DPW Engineering or USACE. Make repairs to bridges that include bolt tightening, handrail/guardrail repairs, repairs to holes in bridge deck and minor structural repairs. Map and description of bridges located on Fort Carson is located in C.7, TE C.5.5.2.1.21.

C.5.5.2.1.22. Perform proper maintenance to installed porous pavement parking lots.to ensure proper design operation. There are three of these parking lots on Fort Carson. One behind building 1225, one at 2039 and the other at 1843. Maintenance requirements for this system is identified at C.7, TE C.5.5.2.1.22.

C.5.5.2.2. Unpaved Surfaces

The Contractor shall do the following:

C.5.5.2.2.1. Perform maintenance on the approximately 10,674,000 square yards of unpaved gravel, earth, soil aggregate and stabilized surfaces, to include roads, tank trails, road shoulders, runways, firebreaks, open storage areas and parking lots on Fort Carson and PCMS.

C.5.5.2.2.2. Grade the full width and length of downrange roads and tank trails to include roadsides and shoulders. Perform grading on unpaved roads and tank trails to repair surface defects and irregularities (holes, ruts, corrugations, and washouts). Utilize soils from parallel ditches to fill holes and restore uniformity of grade and crown. Graded roads and tank trails shall have slopes perpendicular to the centerline of about ½ inch per foot (four percent), while forming a crown at the centerline. Existing super elevated curves shall be maintained to a slope of two to four percent. Graded shoulders shall have slopes no more than one inch per foot (eight percent) away from the crown of the road.

C.5.5.2.2.3 Obtain clearance and Range Pass from Range Control in bldg. 9550 before entering downrange for repairing of tank trail, runway or road.

C.5.5.2.2.3. Grade approximately 350,000 square yards of firebreaks to keep vegetation down preventing fire from crossing Fort Carson property. Accomplish this at least twice annually between April and September. Firebreak surfaces shall be graded or dragged, ruts and holes filled and leveled to permit passage of firefighting equipment, and to prevent erosion. Width of firebreaks shall be no less than 12 feet and no more than 24 feet.

C.5.5.2.2.4. Grade, water, and compact the full surface area of approximately 5,000,000 square yards of unpaved roads, parking lots, and open storage areas in the Cantonment area annually. Reference section C.7, TE-5.5.2.1.2. Add base course material as needed to repair gravel parking lots and open storage areas. Finished surfaces shall match existing grade and drainage patterns.

C.5.5.2.2.5. Perform maintenance and repairs to known unpaved runways down range on Fort Carson (3) and PCMS (1). Obtain schedule of runway usage at least monthly from Range Control. Perform scheduled routine maintenance and repair to runway as necessary for safe operation of all aircraft.

C.5.5.2.2.5.1. Ensure Aircraft Runway surface is uniformly crowned for water shed at 2.0% (+/- .5%) from centerline, no ruts deeper than 1/2" inch and with a compacted surface.

C.5.5.2.2.5.2. Ensure the TUAV aircraft runway surface is uniformly flat with minimal crowning from the center line. There shall be not ruts deeper than ¼ inch and the surface is to be compacted. The material on the surface is to be of fine aggregate no larger than ¼ inch.

C.5.5.3. Railroad System

The Fort Carson railroad system has 13 miles of rail, and PCMS railroad system has 3 miles. Components include, but are not limited to, tie plates, bolts, washers, nuts, ties, ballast, signage, insulated joint material, rail anchors, guard rails, crossing arms, barricades, electric and other signals and gage rods. Coordinate with the Installation Transportation Officer from the LRC so maintenance will not interfere with train movements. The Fort Carson rail system extends to Kelker Junction and shall include all bridges and roadway crossings.

The Contractor shall do the following:

C.5.5.3.1. Ensure inspections of Fort Carson and PCMS track are performed by a U.S. Army Certified Inspector IAW AR 420-1, American Railway Engineering and Maintenance of Way Association (AREMA) and UFC 4-860-03. Maintain a copy of all track inspections for a period no less than three years and make available to the Government upon request.

C.5.5.3.2. Maintain all Fort Carson track to category-A status. Maintain PCMS track to category B status. Cannibalization of track shall not be allowed. Coordinate with the Installation Transportation Officer from the LRC so maintenance will not interfere with train movements. Notify both the train operating personnel and the COR immediately upon the discovery of any condition that would warrant any restriction below normal operating standards. This will be performed on Fort Carson and PCMS track and include the DPW facility manager at PCMS on issues concerning PCMS track.

C.5.5.3.3. Not delay rail movements due to action or inaction by the Contractor.

C.5.5.3.4 Have rail inspector in the Fort Carson rail yard during all rail movements. In the event of a derail immediately notify the Government, validate position of switches after the derail occurred, take photos of the incident and write a statement describing the incident for maintenance records and accident investigation. Ensure cars impacted by the derail do not move until official investigation of derail is complete. Forward the statement and supporting photos to the KO within 24 hours of the rail incident.

C.5.5.3.5 Perform repairs to track caused by derail. Repair track damage during a deployment with the highest priority preventing rail movement delays.

C.5.5.4. Snow Removal and Ice Control

Snow removal and ice control services are the plowing and removal of snow, and ice control of streets, bridges, sidewalks, rail crossings, flange ways and switches; parking lots and Airfields. A map showing snow removal, ice control, and emergency snow route areas is in section C.7, TE-5.5.4, TE 5.5.4A (PCMS). Some aspects of this requirement will be at Building 20000, 2525 Aviation Way, in Colorado Springs and at the ADAAG facilities (7330, 7314, and 7324) located on the south end of the Colorado Springs Airport. Historical snowfall data for Fort Carson is in section C.7, TE-5.5.4B. Sidewalks and parking lots are subject to the same snow depth guidelines as street snow removal.

The Contractor shall do the following:

C.5.5.4.1. Be aware of weather trends and factors that create the potential for hazardous snow or ice accumulations and shall be prepared to respond to such conditions. Utilize the weather forecaster at Butts Army Airfield (BAAF) during normal duty hours, or Fort Carson Operation Center (FCOC) after normal duty hours and/or other commercial sources for weather forecast. Weather alerts and storm warnings are disseminated telephonically by the FOD or FCOC. The Government will assure that the Contractor is listed as a recipient of such alerts and warnings.

C.5.5.4.2. Develop and submit a Snow Removal and Ice Control Plan separate from the required O&M plan, for the execution of Contractor responsibilities IAW AR420-1, 7-15 and this PWS. Include in the plan: location of snow control center, contact numbers, activation and operational procedures, identified snow equipment, QC participation, de-icing operations, etc. Submit the plan NLT 1 September annually to the COR with a briefing to follow by 30 September for the upcoming winter IAW C.8, report C.5.5.4.2.

C.5.5.4.3. Provide an AAR of snow events after each event where plowing occurred or as requested by the Government and at end of season. Include at a minimum all applicable information regarding activation, response; equipment status, material used, lessons learned, recommended changes and positive outcomes.

C.5.5.4.4. Restore unimpeded traffic movement on Fort Carson streets, sidewalks, and unobstructed routine passage of rolling stock on Fort Carson railways

C.5.5.4.5. Ensure the Snow Control Center shall be activated and radio communications with crews shall be established within 30 minutes after implementation during normal duty hours or one hour after implementation outside of normal duty hours, and shall continue operations until requirements have been met. Notify the Government Technical Monitor within one hour of snow control activation. Notify the FCOC when deicing and snow removal operations are started, and shall provide a SITREP every one hour until event has ended.

C.5.5.4.6. Initiate snow plowing when snow accumulation on traffic surfaces is expected to reach and surpass three inches. Utilize the standard measuring points of the helipad southwest of Building 1435, building 8200, Gate 20, Gate 5, and Building 9601. If any one of the points reaches the trigger level, snow removal teams shall be deployed for snow removal throughout Fort Carson. However, if significant drifts or accumulations are encountered on Fort Carson, plowing and snow removal is appropriate in that general area.

C.5.5.4.7. Perform work in priority sequence identified by the snow response plan. Accomplish snow removal of sidewalks concurrently with roads as warranted.

C.5.5.4.8. Change priorities identified in C.5.5.5 with KO approval. Respond to changing conditions, emergencies (e.g., traffic flow disruption or accident) and Government initiated redirection of priorities to protect life and property and maintain traffic flow.

C.5.5.4.9. Ensure Wounded Warriors barracks sidewalks and parking lots at Building 7492, 7493, are maintained throughout the entire winter storm event. Ensure snow removal and deicing at building 7494 and 7495 occurs continuously during business hours only. On Sundays, chapel (1500, 1350, 1850 and 2359) parking areas and sidewalks will have snow removal/ice control completed regardless of snow depth before staff or parishioners arrive for services.

C.5.5.4.10. Ensure priority of work on railroads is dependent on anticipated track usage. Remove snow and ice from railway switches, frogs, guardrails and flange ways at road crossings as a priority one task if rail movement is scheduled.

C.5.5.4.11. Plow the full width and length of all surfaces requiring snow removal during and after snowstorms until all requirements have been met. After end of snowstorm, plow to bare pavement when temperatures are trending upwards toward thawing conditions.

C.5.5.4.12. Remove snow to within 12 inches of permanent obstructions (e.g., poles) located within areas requiring snow removal

C.5.5.4.13. Traffic control signs and devices rendered inoperative during snow and ice removal shall be placed back in service within one hour after damage during normal duty hours and two hours after normal duty hours.

C.5.5.4.14. Report to the Provost Marshall any parked or abandoned vehicles obstructing plowing or sanding operations within 15 minutes after observation.

C.5.5.4.15. Do not use large equipment to plow within 50 feet of the installed final denial barriers to prevent damage to the barriers. Hand shoveling may be required to maintain proper operation of the system.

C.5.5.4.16. Apply deicing materials to remove and prevent hazardous accumulations of ice and restore effective braking traction to the following areas: intersections, dangerous curves, dangerous hills, Gates 1, 2, 3, 4, 5, 6, 19, 20, bridges and any other areas specified by COR.

C.5.5.4.17. Apply liquid de-icing material pre-storm and as required during and post storm in any location required for de-icing. Remove snow as necessary in these locations prior to applying deicing material to increase its effectiveness. Determine the necessity for applying deicing material on streets and parking areas in order to maintain effective braking traction on traffic surfaces. Notify the Government in accordance with the notification procedures for plowing operations described in Section C.5.5.4.6 when de-icing material is being applied. Not use deicing material that may hinder final denial barrier operation or corrode components of the system within 50 feet of the barriers.

C.5.5.4.18. At end of snowstorm and after traffic flow has been restored, push back snow berms along roadsides so that melting snow will drain into drainage systems or adjacent grounds rather than onto traffic surfaces. Remove snow or snow berms that hinder vehicular or

pedestrian access to buildings, fire hydrants or dumpsters restrict vehicular and pedestrian traffic; obstruct vision of traffic or hinder proper operation of traffic control devices; or hinder deliveries to Dining Facilities within eight hours.

C.5.5.4.19. Perform snow removal and ice control on secondary roads and parking lots during normal work hours unless storm is classified as a major event requiring other response.

C.5.5.4.20. Designate snow removal equipment at Phase-In and include this equipment by vehicle number in the required snow removal and ice control plan, reference C.5.5.4.2.

C.5.5.4.21. Clean and service equipment used for snow removal and ice control operations within 24 hours after completion of post-storm cleanup.

C.5.5.5. Equipment Operations Support

C.5.5.5.1. Provide support to other shops, IAW this PWS, during repairs of existing utility infrastructure failures and other required effort. This includes, but is not limited to, excavation and traffic support for repairs of utility lines, equipment transportation and material hauling.

C.5.5.5.1.1. Ensure all backfilling that is performed under a paved surface is accomplished with compaction testing. Compact soil full depth in lifts no greater than four inches. Perform compaction testing of soil full depth not to exceed 24 inch intervals. Ensure soil compaction is no less than 95% using Proctor or modified Proctor methods unless other testing method is warranted.

C.5.5.5.2. Provide Force Protection requirements for real world or exercise requirements including, but not limited to: relocating or installing traffic barricades, concrete barriers, and providing other support.

C.5.5.5.3. Haul potable water to support training requirements at camp Red Devil (CRD). Potable water hauling shall be IAW DPW SOP #60 *Fort Carson Potable Water Hauling*.

C.5.5.5.4. Ensure all equipment that may be adversely affected by weather is protected. This includes, but is not limited to: water tankers, sewer trucks, and sweepers with water pumps and hoses (both vacuum and towed).

C.5.6. EQUIPMENT FUELING REQUIREMENTS

Provide field refueling of installed and mobile emergency generators and contractor operated equipment. Diesel fuel shall be Grade Ultra-Low Sulfur No 2-D, DL2, Clear with a maximum sulfur content of 0.0015wt% diesel fuel.

The Contractor shall do the following:

C.5.6.1. Provide in field refueling of firefighting equipment as top priority when needed during firefighting operations as directed by the KO. All equipment shall be fueled to capacity and all fuel shall be captured via DMO.

C.5.6.2. Refuel all standby generators identified in section C.7, TE-5.6.2. All fixed standby generators shall be fueled to facilitate 12 – 16 hours of continuous operation and maintained per section C.5.7.5.2.

C.5.6.3. Have the capability to supply fuel as needed to the generator(s) before it runs out of fuel in the event the generator is required for an extended period.

C.5.7. ELECTRICAL SYSTEMS MAINTENANCE AND REPAIR

Electrical systems include, but are not limited to, emergency generators, emergency lighting inverters, fire alarms, facility electrical, traffic control, intrusion detection, airfield lighting, resistance testing, Mass Notification System, all poles and lights (including solar generated lights), secondary electrical distribution, solar Photo Voltaic (PV) arrays, transformers, substations and the main distribution system electric SCADA system. The Contractor shall be capable of performing work on electrical systems with the system energized.

Electrical power [34.5 kilovolts (KV)] is supplied to Fort Carson by Colorado Springs Utilities into three main substations and one 12,470 (KV) tap line off highway 115 to the 9300 area. O'Connell Substation 1 is located at the intersection of O'Connell Boulevard and Chiles Avenue; Titus Substation 4 is located on the south side of Titus Boulevard and west of Butts Road. Minick Substation 3 is along Minick east of Prussmen Boulevard. The two each power transformers in Substation 1, 2 & 4 and associated switchgear on the 15kv side to the point of common coupling (load side of CSU utility meter station) are owned and maintained by Fort Carson. The primary meter station to the 9300 area is on the load side of the fused cutouts and is also the responsibility of Fort Carson.

The demarcation point between Fort Carson and Colorado Springs Utilities is the primary meter equipment feeding the 9100 wildlife area on pole P01114. Colorado Springs Utilities will provide all equipment for meter repairs. Colorado Springs Utilities will not perform work on the load side of the fused cutout at the Fort Carson property line into 9100 or past the point of common coupling as outlined in documents on file at the Directorate of Public Works.

Electrical power is supplied to Turkey Creek Recreation area & Camp Red Devil by Black Hills Energy (Florence Station) at 13200/7620 volt. PCMS Substation is supplied by San Isabel REA 115kv line.

Electrical power is supplied to Pinion Canyon Maneuver Site (PCMS) by San Isabel Electric Association by 115kv line that feeds the PCMS power transformer and 12,470 switch gear that is disturbed throughout the containment area.

The maintenance boundary for the electrical system at Building 7500, Evans Army Community Hospital (EACH), ends at five feet outside of the facility.

The demarcation point between Fort Carson Family housing and Government maintenance is the line side of the meter or the main service panel disconnect for housing units without a meter socket.

The electrified ranges and Observation Point Short shall be the Contractor's responsibility. Targeting Systems secondary distribution electrical shall be maintained by others.

The maintenance boundaries for schools, Fort Carson Middle School (Building 6200), Abrams Elementary School (Building 600), Mountain Side Elementary School (Building 5508), Patriot Elementary (Building 6205) and Weikel Elementary school (6070) shall be at the primary connections on the schools' transformers, except emergency Priority 1 support IAW the Memorandum of Agreement with Fountain Fort Carson School District 8.

The maintenance boundary for the Post Car Wash (Building. 1070), Envision Express (Building. 735), Fuel Transfer Point BAAF (Building 9650), and Specker Ave. (Building 640) Post Self Storage (Building. 6185) shall be the load side of the cutouts or switch fuse except emergencies as directed by the KO.

The demarcation point for the Fort Carson privately owned two megawatt solar array west of the Titus #4 substation shall be the Government-owned pad mounted switches. All maintenance and repairs on the load side of the Government pad mounted switches shall be the solar array owner's responsibility. The Contractor shall maintain security and operational control of the solar array switch gear for emergency shutdown and shall insure the solar switches are kept closed to prevent loss of power. If the Contractor takes the solar array offline for more than an hour during switching the Contractor shall notify the COR. Maintain sole key control for the solar array switch gear.

The maintenance boundary for the Mates Facility (9100 area complex) is the primary side of the transformer, except for emergencies as directed by the Government.

Internal building wiring systems are 480/277 volt, 208/120 volt three phase and 120/240 volt single phase.

The Contractor shall do the following:

C.5.7.1. Primary Electrical Distribution

C.5.7.1.1. Provide operations and maintenance plan necessary for the supply of continuous power to all electrical systems and components in the substations owned by the Government. The electrical power distribution to Fort Carson facilities include, but not limited to, cantonment area, remote areas such as Tank Tables V, VII, and VIII and rifle ranges, Turkey Creek Recreation Area, Camp Red Devil, Recondo Well and wildlife area 9301. Maintenance and

repair will be in accordance with industry standards and practices to include, but not limited, to those publications identified in Section C.6.

C.5.7.1.2. Be responsible for supplying continuous power to Fort Carson and PCMS facilities. Respond and take action upon notification of an electrical emergency in the cantonment area to safeguard life and property within ½ hour during normal duty hours and within one hour during non-duty hours. In remote areas, response time shall be 1 ½ hours during normal duty hours and two hours during non-duty hours. Work until the problem is corrected.

C.5.7.1.2.1. Notify the COR of the status of the repair work, the area or building affected, and estimated down time if repairs cannot be completed at that time and ensure danger or risk is eliminated. Utilize the mobile generator when necessary to provide continuous power but shall continue repairs until completed.

C.5.7.1.2.2. Notify the COR and Technical monitor of power outage that affects an entire building or large outage areas affecting multiple buildings. Provide an outage report to the Government within 12 hours that gives a brief description of the outage, cause, and repairs made and duration of the outage. All power outages shall be reported verbally to the COR within 30 minutes and written report following within 12 hours IAW C.8, report C.5.7.1.3.2.

C.5.7.1.3. Schedule and perform outages IAW Fort Carson Regulation 420-19 and perform scheduled outages upon request of the Government.

C.5.7.1.4. Make repairs to demarcation point meter equipment on pole P01114 within two business days of notification.

C.5.7.1.5. Properly disconnect electrical from facilities identified for demolition.

C.5.7.1.6. Provide electrical support to other's with isolating an electrical line or service, opening vaults, pits or manholes or locating an electrical line,

C.5.7.1.7. High Voltage and Lock-Out/Tag-Out Program

C.5.7.1.7.1. Ensure contractor personnel working on Electrical systems below 600V or High Voltage Systems 600V are "Qualified" in accordance with requirements set forth in the National Electric Safety Code, OSHA and UFC-3-560-01, Electrical Safety O&M and National Electric Code.

C.5.7.1.7.2. Ensure all electric voltage sources de-energized for maintenance purposes have the safety switches, circuit breakers, or fuse box handles mechanically secured in the OPEN position with Lock out tag out to prevent accidental energizing while conducting maintenance IAW AR 420 1.

C.5.7.1.8. Asset Identification & Security

C.5.7.1.8.1. Install and maintain asset identification labels on all major system components not included in the CSU IGSA to include but not limited to poles, transformers, switches, vaults, manholes, capacitor banks, streetlight and walkway lighting poles, Air Handlers, Boilers.

C.5.7.1.8.2. Place new asset identification labels, update map information and update maintenance data base within seven calendar days of installation or any changes to the electrical distribution system. Provide new asset identification number for work being completed on other projects and contracts within seven calendar days of the request along with a copy to the Government.

C.5.7.1.8.3. Utilize the electric system one-line map during routine operation and maintenance and as discrepancies are found per C.5.1.15. Provide updates to the technical monitor by email.

C.5.7.1.8.4. Ensure all primary electrical equipment is secured and locked to prevent access to general public. Ensure key locks for transformers, switches, and field systems are keyed alike. Ensure substations are keyed separately and shall not match transformers, switches, and field systems. Maintain key control for substation locks and electric system equipment locks.

C.5.7.1.9. Substations

C.5.7.1.9.1. Operate, maintain, and repair electric substations. Ensure areas are secure; fences and gates are connected to ground grids or mats; high voltage warning signs are mounted on fences and gates; gates are locked except when authorized personnel are in the area; and grass, weeds and debris are removed from inside the fenced enclosure.

C.5.7.1.9.2. Maintain, test, and repair electrical substation circuit breakers, associated protective relays, meters, transfer switches, DC station power systems and battery charges and all associated substation switchgear equipment.

C.5.7.1.9.3. Visually inspect Substations 1, 2, and 4 weekly and PCMS substation monthly for any visual damage or problems. Clean dirt and grime from the substation power distribution centers at substation 1, 2, and 4 weekly and those at PCMS monthly. Ensure the connections are tight, wiring is not damaged and there is no visible damage. Record load readings, temperature, trip readings, and type of targets of all circuit breakers in substation log book.

C.5.7.1.9.4. Maintain, test and replace DC battery systems at Substations 1, 2, 4 and at PCMS IAW manufacture's recommendations to ensure mechanisms remain fully operational. Record once a month information such as voltage, specific gravity, and level of electrolyte in battery, temperature, and charger voltage, and ensure all connections and wiring are clean and free of dirt and corrosion. Records shall be maintained in the substation log book at each substation.

C.5.7.1.9.5. Utilize infrared imaging equipment during maintenance to locate and identify possible problems with substation equipment if any problems are identified during maintenance and notify the COR immediately, then take immediate corrective action.

C.5.7.1.9.6. Test electrical substations 1, 2, 4 & PCMS for proper operation one time during contract between option years two and four. Work shall include, but is not limited to, breaker and relay operation testing, insulation, grounding system testing, meter calibration, and IR inspection. The completed test result records shall remain in the Electric shop office and be turned over to successor contractor at end of contract.

C.5.7.1.9.7. Inspect, maintain and repair substation power transformers at Substations 1, 2, 4 and PCMS. Check oil levels, nitrogen pressure and nitrogen bottle level. Record tap positions and Load Tap Changer counter operations. Perform yearly Dissolved Gas Analyses (“DGA”) tests on the Substation Power transformers. Maintain records for all testing.

C.5.7.1.9.8. Operate and maintain the electric substation SCADA systems, to include the SCADA Master HMI server located in the electric shop and the SCADA remote terminal located in the 24 hours, seven days a week utility monitoring section. Maintain all SCADA equipment located in the three Fort Carson electric substations to include but not limited to RTU’s, DDIO, Positron units, modems and various other items, to ensure the system is fully functional.

C.5.7.1.10. Primary Power Distribution

C.5.7.1.10.1. Operate, test, maintain, and repair the overhead and underground electrical distribution system to include all primary electrical distribution, and secondary building services. Perform maintenance to ensure proper operation of equipment and continuity of power. Work shall include but not be limited to the inspection, maintenance, repair or replacement of overhead and underground electric system equipment.

C.5.7.1.11. Overhead Electric Distribution

C.5.7.1.11.1. Inspect and maintain Government owned poles, insulators, conductors, crossarms, lighting arrestors, fuses, capacitor banks, bolts. Inspect all poles once during the contract period. Repair or replace loose hardware to include but not limited to, braces, cutouts, grounds, guys, conduits, distribution transformers, pole top primary oil and air break sectionalizing switches, pad mounted air and oil switches from 100 to 600 amps, primary and secondary distribution conductor lines and associated components, fused cutouts, non-fused knife blade switches, voltage regulators, and reclosures as needed.

C.5.7.1.11.1.1. All connections shall be tightened. Replace damaged wood poles; straighten leaning poles, replace or install anchors and down guys to maintain structural support and to minimize stress on poles. Relocate poles as necessary for operational purposes.

C.5.7.1.11.2. Replace poles using the size and class of pole needed to support structure for overhead distribution, but not less than the size and class of the existing pole. Replace damaged or deteriorated cross arms and bent, broken or missing bolts and braces. Utilize spring-lock washer hardware on poles. Dispose of the old poles.

C.5.7.1.11.3. Remove “clearly abandoned” phone lines and fire alarm wiring from poles during routine maintenance or upon request of the government. Contact the Government for guidance regarding questions on abandoned lines or wiring.

C.5.7.1.11.4. Visually patrol all overhead primary lines at least quarterly to identify problem areas.

C.5.7.1.11.5. Trim all trees as necessary in order to maintain at least 10 feet between any portion of the tree and power lines. Maintain conductors free of debris and extraneous items (e.g., boots, kites or debris).

C.5.7.1.11.6. Follow NESC to prevent overhead lines from contacting each other.

C.5.7.1.11.7. Inspect pole top switches and equipment two times during the contract performance with 18-24 months between each inspection. Inspect for loose or damaged components, insulators, connections, operation levers and utilize infrared equipment during maintenance to locate, identify and correct possible problems.

C.5.7.1.11.8. Maintain and install raptor protection as needed on overhead power lines. The Government will provide details if raptor protection as necessary on any overhead utility pole.

C.5.7.1.12. Underground Electric Distribution

C.5.7.1.12.1. Inspect, operate, maintain, replace or repair the underground electrical distribution equipment to facilities. Equipment consists of, but is not limited to, pad-mounted switches, separable connectors, junction points, cable terminations, vaults and manholes and all related underground components.

C.5.7.1.12.2. Inspect, maintain and clean electrical vaults and manholes IAW the maintenance plan. Properly maintain records of maintenance performed, including an inspection sheet with a butterfly drawing showing entry conduits, cable location and junction points.

C.5.7.1.12.3. Inspect and perform maintenance and repair on pad-mounted switches two times during the contract performance with 18-24 months between each inspection and maintenance cycle. Keep a record of the maintenance performed on switches that includes an inspection sheet with a switch-layout drawing showing cable size to and from location, fuse size, amp readings for all lines and date of maintenance.

C.5.7.1.13. Transformers

Fort Carson has two types of Transformers; pole mounted and pad mounted types.

C.5.7.1.13.1. Inspect once during the life of the contract, maintain in accordance with industry standards, test and repair transformers as needed. Maintain a record of maintenance performed

on each transformer to include voltage and amp readings, date, time and ambient temperature. Notify the COR before replacing if the transformer is found to have a problem.

C.5.7.1.13.2. Maintain records of all transformers in and out of service. Include the Asset ID number, location, manufacturer's name, size, date installed, voltage, phase, impedance, fusing style, live or dead front, inspection dates, load information, gallons of oil and condition coordinate location. Make records available to the Government for review upon request

C.5.7.1.14. Capacitor Banks

C.5.7.1.14.1. Inspect, maintain, test and repair capacitor banks to ensure continuity of service Remove and replace a capacitor found leaking; Dispose in accordance with the Fort Carson PCB Handling and Control Procedures Plan. Reference section C.7 for TE-5.7.1.14.1.

C.5.7.1.14.2. Inspect the capacitor banks quarterly. Replace capacitor if bulging, punctured, or inoperative. Check for visual flash over damage clean and remove vegetation, weeds, spiders, rodents and debris. Verify operation of automatic capacitor banks, record operations and maintain records for all capacitor banks. Ensure control instrument light bulbs in control panels are operable.

C.5.7.1.14.3. Relocate capacitor banks as needed to maintain proper power factor of 95% to 99%.

C.5.7.1.15. Street Lights, Photovoltaic/Photocells and Poles

Streetlights, poles, school light timers and gate flashers are shown on the installation basic information maps. Maintain all street lighting and street lighting circuits at PCMS and Turkey Creek

C.5.7.1.15.1. Repair, replace, troubleshoot, and maintain poles, streetlights and school light timers IAW O&M manuals and manufacturer's specifications. Replace missing and damaged access covers on light poles. School light timers shall be set for school days, holidays, weekends and summer.

C.5.7.1.15.2. Perform lighting surveys to identify lights out and schedule repairs as needed to insure the above lighting levels are maintained per below. Ensure exterior lighting is off during daylight hours. When making repairs, all lights shall be made working in lieu of meeting percentage below.

Ammunition Holding Area (AHA)	100%
Ammunition Supply Point (ASP)	100%
Entry Control Points (ECP)	100%
Street lighting	No more than 2 in 10 continuous lights out
Walkway lighting	No more than 1 in 4 continuous lights out

Parking lot lighting	No more than 25% out in a lot, two adjacent poles shall not be out
Motor Pool (MP) lighting	No more than 25% out in a MP, two adjacent poles shall not be out

C.5.7.1.15.3. Turn off exterior lighting as an energy savings measure when directed by the KO.

C.5.7.2. Traffic Control Systems

Traffic control systems include, but are not limited to, a closed loop system of traffic signals linked to and controlled by a centralized master controller; Model 170 configured cabinets, central processing units, data communications system; load switches, traffic loops, video and node detectors, loop detectors, pre-emption circuitry, wiring and connectors, flasher controls, school crossings, signal light fixtures, signal support structures, support wires and cable, and light standards. Locations of traffic control systems are indicated on the electric base information maps.

The Contractor shall do the following:

C.5.7.2.1. Repair, replace, and maintain traffic control systems and components including, but not limited to, central processing units and cabinets; data communication components; vehicle detection components, school flashers, signal lights, signal support structures; and pedestrian crossing equipment IAW O&M manuals and manufacturer's specifications and in coordination with the DPW Traffic Engineer. Maintain a traffic control system logbook upon completion of repairs or maintenance, and store logs in each individual traffic signal control cabinet. Place defective vehicle detectors into recall mode until repairs can be made.

C.5.7.2.2. Notify the COR when Provost Marshal Office and the Military Police coordination is necessary prior to blocking part of an intersection to make repairs and again when intersection is clear.

C.5.7.2.3. Repair traffic signals totally out as priority one. Repair traffic signals that are on flash as priority two if identified during normal business hours or the next business day when identified during non-duty hours.

C.5.7.2.4. Ensure traffic control systems are fully operational. All repaired or installed items in the control cabinet shall be done to keep the cabinet compatible for Model 170 traffic controllers.

C.5.7.3. Airfield Lighting System

The airfield lighting system includes, but is not limited to, airfield beacons, apron lights, above ground threshold lights, Precision Approach Path Indicator (PAPI), Omni Directional Approach Lighting System (ODALS), runway lights, helicopter pad lights, semi flush airfield lights,

approach lights, aircraft clearance lights, tower and base operation lighting controls, various transformers, unlighted warning devices, lighted wind socks, lighting vaults, cabling, wiring, lighting constant current regulators (CCR).

The Contractor shall do the following:

C.5.7.3.1. Repair, replace and maintain airfield lighting system IAW most current Federal Aviation Administration Advisory Circular 150/5340-26 (C.7, TE-5.7.3.1), O&M manuals, and manufacturer's specifications.

C.5.7.3.2. Ensure airfield lighting system is fully operational at all times.

C.5.7.4. Facility Electrical Systems

The Contractor shall do the following:

C.5.7.4.1. Repair and replace, IAW National electric Code, O&M manuals and manufacturer's specifications, facility electrical systems and components including, but not limited to, panel boards, load centers, service entrance and distribution panels, fire exit signs, smoke detectors, breakers, wiring, raceways, grounding rods, cables, cords, photo cells, receptacles, static ground points, ground points, electrically operated cargo doors, variable frequency drives, boiler and heating control systems, lighting fixtures, switches, mechanical and programmable timers, magnetic and manual starters, contactors, rolling marquee signs and other electric signs, facility mass notification systems, electric motors, electric heaters, heat trace heat tapes and PV solar equipment.

C.5.7.4.2. Replace batteries in barracks building smoke detectors and Onity keyless locksets no less than once annually. Schedule and track battery replacement and provide information to the Government when requested.

C.5.7.4.3. Repair, test, trouble shoot and maintain specialized and proprietary electric systems and equipment. This includes, but is not limited to; advanced computer controlled lighting systems from manufactures such as Lutron Quantum, Hubbel CX and other various lighting control manufactures; UPS system from manufactures such as APC/Schneider electric, Liebert/Emerson power, Eaton PowerWare. Power control systems running; I-Fix & I-Power software; Microgrid control system running IPERC software; Relaying and control equipment from Schweitzer Engineering; Woodward Generator controls; Nova Tech RTU and DDIO; 400 Hz frequency converters by Unitron among other various systems.

C.5.7.4.4. Install additional lights, outlets and switches to support electrical requirements as approved by the KO.

C.5.7.5. Emergency Generator Systems

Emergency generator systems provide backup electrical power to support critical facilities. When an emergency generator system will be out of service due to maintenance or repairs for 12 consecutive hours or more, a temporary generator system is required. Emergency generator system components include, but are not limited to, diesel drive units, power units, automatic transfer switches, raceway systems, wiring systems, cable systems, electrical distribution panels, metering and monitoring systems, day tank monitoring systems, batteries, and electrical driven fuel transferring systems. Emergency generator information is listed in TE-5.6.3. Fueling requirements are as indicated in Section C.5.6. Natural gas and propane fueled generators are discussed in Section C.5.8.7.8. Provide a Generator hour log to the COR monthly NLT 7th business day IAW C.8, report C.5.7.5. Example of report is in C.7, TE-5.7.5.

The Contractor shall do the following:

C.5.7.5.1. Repair and maintain emergency generator systems IAW TM-5-685 & NFPA 110 *Standards on Stored Electrical Energy Emergency and Standby Power Systems*, O&M manuals, and manufacturer's specifications on all emergency generator systems. Notify the COR when an emergency generator system is unable to supply emergency power. Provide temporary emergency power in these situations based on the facility load. Check with EMCS for facility load requirements.

C.5.7.5.2. Perform method nine opacity testing during scheduled maintenance of fixed emergency generators. If the emissions from the exhaust exceed 20% opacity, immediately stop the generator and notify the COR and DPW-ED.

C.5.7.5.3. Perform operational checks after each emergency generator system repair and after natural or man-made power outages IAW O&M manuals and manufacturer's specifications. Coordinate maintenance that interrupts power with facility occupants prior to performing maintenance cycles.

C.5.7.5.4. Validate that emergency generator systems supply emergency power within the timeframes specified by the O&M manuals or manufacturer's specifications when a power outage occurs.

C.5.7.5.5. Fluid levels shall be maintained within normal operating range. Batteries shall be free of corrosion and fully charged, generator enclosures shall be kept neat and clean. Emergency generator log books shall be current and accurate, stored at the generator and protected from weather. Maintain a maintenance report record that includes generator manufacture information, generator size S/N, run hours, voltage and amperage readings date and time.

C.5.7.5.6. Perform operational checks every 10-12 hours during power outage and log in the book. Replenish fuel until power is restored. Post power outage operational checks shall be performed within one working day of restoration of power.

C.5.7.5.7. Manually test NEC communication node generators under load every two weeks for one hour after unit reaches stable temperature. Ref TM-5-692-1

C.5.7.5.8. Perform necessary maintenance and repairs to fixed emergency generators as they are classified as real property. Ensure all maintenance and repair requirements needed for the proper operation of the engine and the power generation system are performed as part of this contract. Maintenance and repair shall include fluid level check, oil changes, fan belts, air filters, etc. as required by the operations and maintenance manual. Log all maintenance activity into the generator log book.

C.5.7.6. Emergency Lighting Systems

Inventory of buildings is listed in section C.7, TE-5.2.2.2-A.

The Contractor shall do the following:

C.5.7.6.1. Replace, repair, and maintain all emergency lighting systems and components in buildings listed in section C.7, TE-5.2.2.2-A including, but not limited to, exit signs, emergency light units, inverter units, batteries, and electrical distribution panels.

C.5.7.6.2. Perform a function test on each emergency light fixture, exit sign or emergency system for two minutes once a year. Install inspection label with current date serviced and employee number, to each emergency light, exit sign or inverter after test or maintenance. A label example is located in C.7, TE-5.7.6.

C.5.7.6.3. Emergency lighting systems shall supply emergency lighting requirements IAW UFC 3-601 and NFPA 101 when a power failure or power interruption occurs. Batteries shall be fully charged, free of corrosion, and electrolyte level shall be maintained within normal operating range.

C.5.7.7. Uninterruptible Power Supply (UPS) Systems

UPS shall supply emergency power based on its size and battery capability when a power failure or power interruption occurs. UPS systems are installed in several critical building that requires uninterruptable facility power. UPS system installed supporting non-real property equipment, such as computer server rack mounted systems, are not included in this PWS unless specifically identified. UPS locations are identified on C.7, TE-5.7.7.

The Contractor shall do the following:

C.5.7.7.1. Repair, replace, test and maintain UPS systems and components including, but not limited to, control units, batteries, wiring, and associated hardware IAW manufacture recommendations NFPA 111 and O&M Manuals.

C.5.7.7.2. Log repair and maintenance actions on a UPS System Log book and store logs with each UPS.

C.5.7.7.3. Ensure batteries are free of corrosion and fully charged. Include in monthly maintenance records: recording battery terminal and pilot cell voltages; recording ambient temperatures; check for abnormal conditions and corrosion; clean cell terminals; and check for proper ventilation, voltage and amperage readings. Include in semiannual maintenance: record all cell voltages; total battery terminal voltage; record panel meter voltage and record rectifier-charger current. Perform during annual maintenance; torque cell and terminal plate nuts and bolts to manufacture recommendations.

C.5.7.8. Fire Alarm Systems

This section covers all facilities that have installed fire alarm systems. Many report to the Fire Department Control Center when alarmed. Some are local alarms that do not report to the Fire Department Control Center. These are all critical systems that must be maintained and coordinated with the Fort Carson Fire Department when issues arise. Fort Carson Fire Department Control Center or the COR will notify the Contractor of any fire alarm system problems.

The Contractor shall do the following:

C.5.7.8.1. Repair, maintain, test and replace fire alarm systems and components including, but not limited to, panels, enunciator panels, central station receivers, transmitters, transmitter interfaces, interface panels, transmitter antennae, initiating devices, signaling devices, supervisory devices, auxiliary devices, computer system, cables, wiring, and raceways.

C.5.7.8.2. Notify facility occupants, COR and Fort Carson Fire Department Control Center when a system is out of service for repair or maintenance and when a system will be out of service eight continuous hours or more.

C.5.7.8.3. Ensure fire alarm systems transmit alarms to Fire Department Control Center upon activation of an initiating device. Complete work the work day following receipt of parts if parts needed to be ordered. Ensure fire alarm systems are not left in alarm condition.

C.5.7.8.4. Systems shall be tested per guidance given in UFC-3-601-02 and NFPA requirements. Child Development Centers & Child Youth Service facilities shall be tested IAW UFC-3-601-02 requirements.

C.5.7.9. Lightning Protection System (LPS) and Static Ground Points Resistance Testing

Resistance testing is performed at the Ammunition Supply Point (ASP) Ammunition Holding Area (AHA), BAAF rail facilities, and other known facilities. A listing of all known test points and testing frequency can be found in section C.7, TE-5.7.9.

The Contractor shall do the following:

C.5.7.9.1. Perform resistance testing on LPS at the ASP and AHA IAW DA Pam 385-64 *Ammunition and Explosive Safety Standards* Table 6-1 and Appendix D. Provide resistance test results to Chief Ammunition Inspector and the COR five business days of test completion of testing.

C.5.7.9.2. Perform resistance testing on BAAF aircraft hangar buildings and parking ramps, IAW TM 1-1500-204-23-1, *Aviation Unit Maintenance (AVUM) and Aviation Intermediate Maintenance (AVIM) Manual*, POL storage, transfer locations, and electric static discharge points FM 10-67-1 *Concepts and Equipment of Petroleum Operations*. Provide resistance and static ground points test results to BAAF Base Operations Officer and COR five business days of test completion of testing.

C.5.7.9.3. Test the NEC Remote Switch Unit (RSU) IAW MIL STD 188-124B, *Grounding Bonding and Shielding*. Complete and submit resistance test prior to expiration date of previous report. Provide resistance test results to the COR annually within five business days of test completion .

C.5.7.10. Inspect and Maintain Facility Mass Notification System

The Contractor shall do the following:

C.5.7.10.1. Inspect, repair, maintain, and replace facility Mass Notification System (MNS) components. Work to include but not limited to batteries, solar panels, control boards, message boards, amplifier, and speaker drivers and other related MNS system equipment.

C.5.7.11. Inspect, Test and Repair Solar PV Systems

Maintain, test and repair solar PV systems. The solar system located west of substation 4 is not part of this PWS.

The Contractor shall do the following:

C.5.7.11.1. Inspect, test and repair PV solar system IAW manufacture's recommendations quarterly (4 x per year). Include inspection of the power inverters, solar panels mounting and wiring and overall system operation. Repair any failed components so the PV works as designed. Verify solar output is correct. Solar PV systems and locations are identified in section C.7, TE-5.7.13.1.

C.5.7.12. Microgrid Operations and Maintenance

Operate, maintain and repair the Fort Carson Smart Power Infrastructure Demonstration for Energy Reliability and Security (SPIDERS) electric microgrid. The SPIDERS microgrid is located along Specker Ave and supplies power to buildings 1435, 1445, 1550, 1528, 1526 & 2434. This microgrid utilizes one existing generator at Building 1435 and two existing generators

at Building 1550 along with supplemental power from the privately owned Fort Carson Solar 1, a two megawatt solar array.

C.5.7.12.1. Initiate operation of the microgrid from one of the (HMI) Human Machine Interfaces located in Building 221 or 1550 upon direction from the KO. Perform microgrid HMI system checks every two hours to verify proper system operation while in microgrid operation mode. Perform operational checks on the three microgrid generators every six hours during operation.

C.5.7.12.2. Perform a function test of the SPIDERS microgrid system four times a year to verify proper operation. Include in the test; generator to grid load tests for each generator and a full soft start of the microgrid. Pre-switch the electrical distribution loads from Circuit 10 to an alternate circuit prior to initiation of the microgrid soft start test to eliminate disruptions.

C.5.7.12.3. Maintain, test and repair the SPIDERS system UPS and DC batteries systems. There are five outdoor style UPS systems located at the motor operated pad mounted switches, three UPS's inside the SPIDERS breaker cabinets, and a DC battery system in the SPIDERS 15kv grid sync break. Perform preventative maintenance four times a year on each UPS.

C.5.7.13. Electric Vehicle Charge Equipment

C.5.7.13.1. Inspect, maintain and repair Electric Vehicle Charging Equipment (EVCE). Location are at buildings; 1550(1), 214(1), 300(3), 8000(1), 151(1) and at Specker St and Long St. (1).

C.5.7.13.2. Inspect the bi-directional vehicle-to-grid EVCE charge stations for proper operation annually.

C.5.7.13.3. Replace any damaged equipment and the filters.

C.5.8. MECHANICAL SYSTEMS MAINTENANCE AND REPAIR

Mechanical systems include, but are not limited to, heating, plumbing, fire suppression, and refrigeration, ventilation, and air conditioning systems. The Contractor is responsible for all maintenance and repair on mechanical systems equipment.

The Contractor shall do the following:

Follow requirements of Colorado State air quality regulations and permits numbered C-12, 291-1; C-12, 291-2; C-12, 291-3; C-11, 682-1; manufacturer's instructions, industry standards, procedures and AR-420-1.

C.5.8.1. Heating Systems

Heating systems include, but are not limited to, boilers, furnaces, infrared heaters, natural gas heaters, oil fired unit heaters, fin tube radiators, steam and water heating systems.

The Contractor shall do the following:

Be responsible for the operation and maintenance of heating systems. Some heating systems are run seasonally the COR will provide dates for seasonal turn on and off 30 days in advance. Set temperatures IAW most current Commander's Policy Letter. Provide freeze protection when a heating system is inoperable, and the outside air temperature is 32 degrees F or below. Notify the COR and provide a temporary heating source when a heating system in an occupied facility is inoperable beyond working hours. Temporary heating source shall heat the facility to the facility temperature requirement.

C.5.8.1.1. Central Heating Plants and Distribution Systems, HTHW Generators and Boilers (Individual and System) Operation

Fort Carson has three-central heating plants and associated distribution systems that are to be maintained within the scope of this contract:

Building 1860, HTHW Plant: supplies HTHW (350-375°F Max) to 135 buildings. This HTHW is used to produce heat, hot water and steam for mess halls and heat/hot water for the other buildings on the system shown on the Basic Information Maps. The fuel supply for this plant is interruptible natural gas with five day back up fuel oil supply.

Building 9609, Butts Army Airfield (BAAF) Boiler Plant: supplies steam to seven buildings for heat and hot water: 9600, 9601, 9602, 9604, 9609, 9611 and 9612. The fuel supply for this plant is firm natural gas with a 10,400 gallon back up fuel oil supply.

Building 9651, Consolidated Boiler Chiller Facility (CBCF), Located in the 1-4 brigade complex. At present the CBCF supplies Heat and Chilled water to the three new barracks (9449, 9459, 9469), new Hanger (9630), CAB HQ buildings (9622), and Control Tower (9662).

A list of equipment for these plants is in section C.7, TE-5.8.1.1.

Boilers include, but are not limited to, low pressure steam, and low temperature and pressure hot water boilers. Boiler Plants and their associated systems include, but are not limited to, High Pressure steam and High temperature, High Pressure hot water systems; and all components necessary for a functional plant and distribution system.

The required O&M plan for C.5.8 shall include boiler plant; operating schedules, detailed startup and shut down procedures, fuel transfer procedures, and Plant Maintenance for buildings 1860, 9609 and 9651 IAW C.8, C.5.3.2.1.

The Contractor shall do the following:

C.5.8.1.1.1. Operate, maintain and repair Heating Plants and distribution systems (supply and return) including: but not limited to: steam boilers and High Temperature Hot Water (HTHW) generators, pumps, motors, valves, controls and all components associated with the Central Heating Plants and distribution systems. Should a critical piece of equipment go down, the contractor shall notify the COR within two hours of equipment failure and begin repairs.

C.5.8.1.1.2. Accomplish all boiler operation, maintenance, repair, and inspection IAW applicable ASME Boiler and Pressure Vessel Code issued by the National Board of Boiler and Pressure Vessel Inspectors (NBBI); NFPA 31 *Standard for Installation of Oil Burning Equipment*; NFPA 54 *National Fuel Gas Code*; NFPA 58; NFPA 211 *Standard for Chimneys, Fireplaces, Vents, and Solid Fuel Burning Appliances*; AR 420-1 *Army Facilities Management*; National Institute for the Uniform Licensing of Power Engineers, Inc. (NIULPE) Certification; UFC 3-430-07 Operations and Maintenance: Inspections and Certification of Boilers and Unfired Pressure Vessels; ASME CSD-1-2009 Controls and Safety Devices for Automatically Fired Boilers; TM 5-650 Central Boiler Plants; National Board Inspection Code (NBIC) Parts 1,2 and 3; and manufacturer's instructions; and individual boiler work instructions located with each boiler.

C.5.8.1.1.3. Be responsible for the operation of boiler and HTHW plants 24 hours per day, seven days per week, 365 days a year, including Federal holidays.

C.5.8.1.1.4. Ensure all safety devices are properly operating at all times.

C.5.8.1.1.5. Be responsible for seasonal maintenance and repair of Central Heating Plants, plant equipment, and steam and HTHW distribution systems. Ensure that heating systems are maintained in accordance with manufacturer's recommendations.

C.5.8.1.1.6. Ensure that water, steam, or fuel does not leak, except as noted in O&M manuals and manufacturer's specifications.

C.5.8.1.1.7. Standby boilers shall be in a pressurized, hot condition. Off-line boilers shall be in a wet lay-up condition.

C.5.8.1.1.8. Maintain a Monthly Repair and Utilities Operating Log and a daily operating log. The forms shall be locally generated to fit the requirement.

C.5.8.1.1.9. Prepare and submit a boiler run time report to the COR NLT 5th business day of each month on boilers in Buildings 1860, 9609, 8300, 8000, and 9620 and 9651 IAW C.8, report C.5.8.1.1.9. Example of report can be found in C.7, TE-5.8.1.1.9.

C.5.8.1.1.10. Perform operational inspections at least once per day, excluding weekends and holidays, at Building 9609 and 9651.

C.5.8.1.1.11. Prepare and submit a boiler operating log to the COR NLT 5th business day of each month for Buildings 1860, 9651 and 9609 IAW C.8, report C.5.8.1.1.11. Example of the log can be found in C.7, TE-5.8.1.1.11

C.5.8.1.1.12. Visually check individual boiler systems for proper operation and leaks bi-weekly. A log shall be maintained in each boiler room detailing inspection along with any maintenance and repairs. Individual boilers are identified in C.7, TE-5.8.1.1.12.

C.5.8.1.1.13. Notify the Government within one hour when discovery of abnormal boiler operations occur. This may include switch from natural gas to fuel oil, taking generator off line, opacity out of spec, etc.

C.5.8.1.2. Boiler Fuel

C.5.8.1.2.1. Convert to backup fuel oil upon direction from the Government or Colorado Springs Utilities (CSU) for Building 1860. Comply with Colorado Springs Utilities (CSU) request to switch to diesel when the natural gas is being interrupted. Colorado Springs Utilities primary Fort Carson contact number to call is 719-526-5345.

C.5.8.1.2.2. Complete the conversion within two hours after notification. Notify the COR within one hour of completing conversion.

C.5.8.1.2.3. Notify the COR when converting to back up fuel, e.g., to maintain gas line pressure.

C.5.8.1.2.4. Maintain back up fuel supplies at each plant to support a five day, 24 hour operation. Place orders for fuel as identified in C.5.17. Fuel tank dipstick chart is in section C.7, TE-5.8.1.2.4.

C.5.8.1.2.5. Use only Ultra-Low Sulfur fuel (0.0015% Sulfur) and maintain copies of all fuel delivery documentation.

C.5.8.1.3. Boiler Opacity

C.5.8.1.3.1. Personnel performing opacity observations shall be certified to use EPA Reference Test Method 9: *Visual Determination of Opacity of Emissions from Stationary Sources*.

C.5.8.1.3.2. Boiler testing on fuel oil shall be IAW the Fort Carson Title V Air Operating Permit

C.5.8.1.3.3 Burner combustion exhaust emissions shall not exceed 50ppm of carbon monoxide when burning natural gas, or a No. 1 (20% density) smoke on the Ringlemann Chart.

C.5.8.1.3.4. Perform opacity observations on all oil-fired boilers IAW operating permits within 60 minutes of startups and/or daily for 24-hour continuous firing. Night restarts require opacity observations as soon as possible the following morning.

C.5.8.1.3.5. Submit to the COR an Opacity Observation Report in accordance with the Operating Permits, for all restarts, and/or 24 hour continuous runs on fuel oil. Submit report to the COR NLT 24 hours after test completion IAW C.8, report C.5.8.1.3.5.

C.5.8.1.3.6. Notify the COR and DPW-ED – Air Pollution Program Manager, within one hour of identification, when an exceedance of opacity is observed.

C.5.8.1.4. Corrosion Control

C.5.8.1.4.1. Develop and implement a corrosion control program for boiler water in Central Heating Plants and approximately 363 individual boilers listed in section C.7, TE-5.8.1.1.12.

C.5.8.1.4.2. Utilize the Fort Carson Sewage Treatment Plant lab for processing monthly water samples ensuring proper corrosion protection of boiler.

C.5.8.1.4.3. Treat boiler and HTHW systems to control corrosion and scale formation. Perform water chemistry testing once per eight hours in attended plants and once per day in unattended plants. Independent water chemistry analysis shall be delivered to the COR IAW C.8, report C.5.8.1.4.3.

C.5.8.1.4.4. Maintain water treatment standards in section C.7, TE-5.8.1.4.4 for approximately 363 individual boiler systems listed in section C.7, TE-5.8.1.1.12. These boilers shall be tested on a bi-weekly basis and recorded in the Repairing and Utility Operations Log, FC Form 4367.

C.5.8.1.4.5. Annotate the FC Form 4367 to indicate results of testing and what treatment, if any, was performed.

C.5.8.1.5. Boiler Inspections

C.5.8.1.5.1. Annually perform a summer shutdown of boiler heating equipment and HTHW equipment for maintenance and overhaul as prescribed in TM 5-650, Central Boiler Plants.

C.5.8.1.5.2. Provide support to government boiler inspector including but not limited to, cooling down the boiler or opening one up, for required certified inspection of generators at 1860, 9609, and 9651 semi-annually. When contacted by the inspector, prepare the boiler for the necessary inspection and provide any documentation requested by the inspector. Submit a copy of the report IAW C.8, report C.5.8.1.5.3.

C.5.8.1.5.3. Inspect heat plant boilers twice annually; one external hot inspection and one internal cold inspection. Unfired pressure vessels shall be internally and externally inspected once every two years. A report shall be submitted to the COR NLT five business days after

inspection is completed IAW C.8, report C.5.8.1.5.3. Inspections shall not interfere with steam or hot water production and distribution requirements

C.5.8.1.6. Operator Certification

C.5.8.1.6.1. Ensure boiler operators, and supervisors operating boilers possess and maintain National Institute for the Uniform Licensing of Power Engineers (NIULPE). Ensure supervisors maintain NIULPE Second Class Engineer certification and operators maintain NIULPE Forth Class Engineer Certification.

C.5.8.1.6.2. Ensure new operators obtain required certification NLT six months after hire date.

C.5.8.1.6.3. Ensure at least one properly certified operator is present at all times at Building 1860 central plant.

C.5.8.1.7. Furnaces and Gas/Oil Fired Unit Heaters

The Contractor shall do the following:

C.5.8.1.7.1. Repair, replace, maintain, and operate furnaces, unit heaters, infrared heaters and associated components IAW NFPA 31, NFPA 54, NFPA 58, NFPA 211, AR 420-1 *Army Facilities Management*, TM 5-642, and manufacturer's specifications.

C.5.8.1.7.2. Ensure there are no leaks to natural gas or fuel oil systems or components. Furnaces and unit heaters shall heat the space to the specific temperature requirement of the facility in which they are located in section C.7, TE-5.8.1.7.2. and TE-5.8.1.4.4.

C.5.8.1.7.3. Track by location and inspect, annually at start-up, any furnace and unit heater that is 10 years old and older for heat exchanger and malfunctioning safety controls.

C.5.8.1.7.4. Ensure unit heater burner combustion exhaust emissions do not exceed 50ppm of carbon monoxide when burning. Maintain furnace and unit heater inspections for the duration of the contract and make available to the COR upon request.

C.5.8.2. Refrigeration, Ventilation, and Air Conditioning

Refrigeration, ventilation, and air conditioning systems include, but are not limited to, air conditioners, Computer Room Air Conditioners (CRAC), chillers, cooling towers, rooftop units, package units, split systems, exhaust fans, heat pumps, temperature control systems, blending stations, air compressors, and air handlers.

Include in the C.5.8 maintenance plan per C.5.3.2.1. a refrigerant management plan to ensure compliance with 40 CFR 82 and the Fort Carson Refrigerant/ODC Compliance Plan.

The Contractor shall do the following:

C.5.8.2.1. Air Conditioners

A list of air conditioners and chillers is identified on TE-5.8.2.1.

Set air-conditioned space temperature to the specifications of AR 420-1. If the Garrison Commanders Energy Policy letter is more restrictive, follow the policy letter guidance.

The Contractor shall do the following:

C.5.8.2.1.1. Operate, replace, repair, and maintain air conditioning systems and associated components IAW NFPA 90A *Standard for the Installation of Air-Conditioning and Ventilating Systems*, NFPA 90B *Standard for the Installation of Warm Air Heating and Air-Conditioning Systems*, and AR 420-1, and manufacturer's specifications.

C.5.8.2.1.2. Perform recover, reclaim, and recycle refrigerant IAW 40 CFR 82.161

C.5.8.2.1.3. Ensure air conditioners cool the space to the temperature requirement of the facility in which they are located.

C.5.8.2.1.4. Ensure refrigerant is not purposely vented to the atmosphere during repair operations.

C.5.8.2.1.5. Change filters and clean coils annually at season start-up so systems are operating efficiently and as needed thereafter.

C.5.8.2.2. Chillers

The Contractor shall do the following:

C.5.8.2.2.1. Operate, replace, repair, and maintain chillers, associated systems, and associated IAW AR 420-1, and manufacturer's specifications.

C.5.8.2.2.2. Perform, recover, reclaim, and recycle refrigerant IAW 40 CFR 82.161.

C.5.8.2.2.3. Ensure chillers cool the water to the temperature requirement of the facility in which they are located.

C.5.8.2.2.4. Ensure refrigerant is not purposely vented to the atmosphere during repair operations.

C.5.8.2.2.5. Ensure water chemistry is IAW section C.7, TE-5.8.2.2.5.

C.5.8.2.3. Central Cooling Plant and Distribution System

The Central Cooling Plant and System includes piping, equipment, controls and components in the buildings listed above, and the distribution system (supply and return). The Fort Carson Central Cooling Plant is an electric centrifugal unit type plant utilizing two 1200-ton, one 750 ton centrifugal unit and three Class 500 cooling towers. A list of major operational equipment is found in section C.7, TE-5.8.2.3.

The Central Cooling distribution system consists of approximately 38,000 linear feet (LF) of supply line (3 to 18 inches) and 38,000 LF of return line (3 to 18 inches). The distribution system includes piping, valves and appurtenances on the system up to the first flange after the isolation valve in the building. Piping locations can be found in the Basic Information Maps.

The cooling season normally runs May through October but is adjusted based on current weather conditions.

The Contractor shall do the following:

C.5.8.2.3.1. Operate, repair and maintain the Central Cooling Plants (Building 1864 and 9651) and Chilled Water (CW) distribution system and three cooling towers (Buildings 1861, 1862 and 1863) IAW AR 420-1 and manufacturer's specifications.

C.5.8.2.3.2. Perform-annual inspection and required maintenance on all equipment at the Central Cooling Plants. Perform this requirement after season shut down so the plant is fully operational prior to the next season.

C.5.8.2.3.3. Maintain the chillers (main body and all components attached to the body) and ancillary components (equipment such as but not limited to refrigerant monitors, evacuation systems) in a manner that causes the machines to be fully functional in accordance with manufacturer's and industry standards.

C.5.8.2.3.4. Ensure the Central Cooling Plant Building 1864 is manned in conjunction with the heat plant on a 24 hour / seven day-a-week basis. Perform operational inspections at least once per day at Building 9651. Should a critical piece of equipment go down, the contractor shall notify the COR within two hours of equipment failure.

C.5.8.2.3.5. Perform hourly readings and building checks of the 1864 chiller plant. The start-up date shall be coordinated with the Government.

C.5.8.2.3.6. Maintain a Central Chiller Operations Log for building 1864 and submit log to the COR monthly NLT the 5th business day IAW C.8, report C.5.8.2.3.6. Report example is in TE C.7, TE-5.8.2.3.6.

C.5.8.2.3.7. Ensure cooling towers cool water to the temperature requirement ranges, and are cycled equally throughout the cooling season.

C.5.8.2.3.8. Ensure water chemistry is IAW TE-5.8.2.2.5.

C.5.8.2.3.9. Water level control systems shall maintain the water level required to feed the cooling tower.

C.5.8.2.3.10. Normal Operating Ranges

Meet flow requirements and cooling demand using multiple plant chillers if necessary. Operate the Central Cooling Plant within the following optimum ranges:

Chilled water supply temperature (Design 42F degrees)

Ensure supply temperature is between 40 – 44F with an optimum temperature of 42F.

C.5.8.2.3.11. Perform required seasonal maintenance and repair of the chilled water (CW) distribution systems IAW manufacturer's recommendations.

C.5.8.2.3.12. Annually inspect CW vaults and cycle distribution valves. Ensure vaults are clean, valve properly operate and no leaks in the vault.

C.5.8.2.3.13. Perform seasonal inspection on the CW distribution System, 1 March through 31 March. Maintain a file identifying inspection results and repairs. Documentation of inspections shall be maintained for the duration of the contract and made available to the COR upon request.

C.5.8.3. Exhaust Fans and Radon System Fans

These fans are designed to remove gases and unwanted odors from one space to the atmosphere. A list of exhaust fan and radon system locations and sample maintenance frequencies is in C.7, TE-5.8.3-A and TE-5.8.3-B.

The Contractor shall do the following:

C.5.8.3.1. Operate, replace, repair, and maintain exhaust fans and radon mitigation fans and associated IAW AR 420-1 and manufacturer's specifications. Exempt small restroom fans from the maintenance requirement.

C.5.8.3.2. Exhaust fans shall operate on demand, either by manual switch or automated controls, to ventilate the air in the space where they are located. Radon mitigation fans shall operate continuously.

C.5.8.3.3. Adjust flow meter as required by manufacturer's recommendation.

C.5.8.3.4. Keep radon warning sticker attached and legible to exterior piping.

C.5.8.4. Compressors

Compressors include, but are not limited to, those supplying air for control, process, and mission support air and other systems. A list of known compressors is in section C.7, TE-5.8.4.

The Contractor shall do the following:

C.5.8.4.1. Operate, maintain and repair compressors and associated IAW AR 420-1, and manufacturer's specifications.

C.5.8.4.2. Turn off compressors in the following buildings no sooner than mid-September then turned on no sooner than beginning of May: 811, 1012, 1013, and 1218.

C.5.8.4.3. Ensure air does not leak from air compressors or components. Storage tanks shall be maintained free of accumulated water. Ensure air compressors operate on demand to supply and maintain air pressure required for the application and system they serve.

C.5.8.4.4. Perform inspections and maintenance twice annually on all compressors that are on fire sprinkler systems. Notify the Fire Department prior to shutting down and again when systems are back on-line. Place a tag on each compressor indicating the date of service.

C.5.8.4.5. Perform oil changes IAW manufacturer's recommendations or annually at a minimum.

C.5.8.4.6. Perform service on hydraulic compressors twice annually IAW manufacturer's recommendations.

C.5.8.5. Air Dryers

A list of air-dryer locations and sample maintenance frequencies is located in section C.7.

The Contractor shall do the following:

C.5.8.5.1. Operate, maintain, replace, and repair compressors and associated components IAW AR 420-1, UFC 3-420-02 FA Compressed Air and manufacturer's specifications. Reference section C.7 for TE-5.8.5.1.

C.5.8.5.2. Perform maintenance on all air-dryers annually to ensure they perform continuous operations. Place a tag on dryer indicating the date of service.

C.5.8.5.3. Turn off air-dryers in the following buildings from 15 September through 1 May, then turn them on from 1 May until 15 September: 811, 812, 1012, 1013, and 1218.

C.5.8.6. Heating, Ventilation and Air Conditioning Systems (HVAC)

All facilities on Fort Carson have an HVAC system to maintain comfortable environmental conditions. These systems may be antiquated or technologically advanced. The systems can

provide space comfort in office space, warehouse space, computer rooms, bay areas, and recreational space. All systems shall properly operate as designed.

The Contractor shall do the following:

C.5.8.6.1. Operate, replace, repair, and maintain air handlers, roof top/package HVAC units, and associated components IAW NFPA 90A, NFPA 90B, AR 420-1, and manufacturer's specifications.

C.5.8.6.2. Clean or replace HVAC filters as needed to permit free flow of air. This will be accomplished annually at a minimum. Place the date on the filter indicating the date of replacement or inspection. A list of known filters is in section C.7, TE-5.8.6.2.

C.5.8.6.3. Clean HVAC coils annually to preclude breakdown and maintain system efficiency. A list of coils is in section C.7, TE-5.8.6.3.

C.5.8.6.4. Ensure HVAC units and dampers operate on demand to circulate air throughout the facility.

C.5.8.6.5. Ensure space temperature is maintained to the specifications of AR 420-1 or the implemented Fort Carson Garrison Commander's Energy Policy, if more restrictive.

C.5.8.6.6. Keep all permanent filters clean. All disposable filters shall be replaced with the proper size filter. Use flow meters to determine if airflow has been reduced to the point where the filter needs to be changed. For those systems without flow meters, change or clean the filters annually at a minimum.

C.5.8.7. Evaporative Coolers

These are installed in various locations where central cooling typically is not provided for comfort cooling. A list of known evaporative coolers is in section C.7, TE-5.8.7.

The Contractor shall do the following:

C.5.8.7.1. Operate, replace, repair, and maintain evaporative coolers and associated components IAW NFPA 90A, NFPA 90B, AR 420-1, and manufacturer's specifications. Set space temperature to the specifications of AR 420-1 or the implemented Fort Carson Garrison Commander's Energy Policy, if more restrictive.

C.5.8.7.2. Service evaporative coolers, including water treatment, at least annually to assure continuous operation.

C.5.8.7.3. Inspect cooler pads; if there is uneven wetting or open spots, assure water distribution system is flowing in evenly and the trough openings are unclogged. Correct any deficiencies.

C.5.8.8. Seasonal Heat Turn On and Turn Off

Heat turn-on historically occurs between September 15 and October 31, and heat turn-off historically occurs between May 1 and June 15. The dates will vary from year to year based on local weather conditions and will be provided to the Contractor in advance for proper scheduling of requirement.

The Contractor shall do the following:

C.5.8.8.1. Perform seasonal heat turn-on and turn-off of systems including, but not limited to: boilers, furnaces, air handlers, evaporative coolers, chillers and all associated control systems in all applicable facilities identified in TE-5.2.2.2A.

C.5.8.8.2. Turn heat on or off within the time frame and in the priority established by the Government.

C.5.8.9. Specialized Equipment Maintenance and Repair

Specialized equipment includes, but is not limited to, food service equipment, refrigerators and ice machine condensers, hoods, wash racks (not the Central Vehicle Wash Facility), POL Storage, natural gas or propane fueled emergency generators, fire training tower, and the Utility Control System.

The Contractor shall do the following:

All individuals performing refrigerant recovery operations shall be certified IAW 40 CFR 82.161.

C.5.8.9.1. Food Service Equipment

Food service equipment are non real property items used to prepare meals. This contract is responsible for providing the utility service to the equipment in DFAC's not repairing the equipment. This contract is responsible for the built-in range top burners in barracks and food service equipment in CDC's, Fire Stations and the Elkhorn Conference Center. A list of facilities with food service equipment is in section C.7, TE-5.8.9.1.

The Contractor shall do the following:

C.5.8.9.1.1. Repair and maintain barracks range tops and food service equipment in identified CDC's IAW manufacturer's specifications, NFPA 96 *Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations* and AR 420-1.

C.5.8.9.1.2. Maintain and repair all built-in, walk-in freezers and refrigerators which are classified as real property. Maintain and repair reach-in refrigerators and freezers in facilities

identified in TE-5.8.9.1. Refrigerant shall not leak and shall not be purposely vented to atmosphere during repair operations.

C.5.8.9.1.3. Maintain and repair serving line equipment and commercial dishwashers; garbage disposals and air curtains which are classified as real property.

C.5.8.9.1.4 Provide and maintain appropriate utility to the first valve or electrical connection from the food service equipment in DFAC's.

C.5.8.9.2. Inspect, Service and Clean Kitchen and Grill Hoods

Clean all kitchen and grill hoods identified in section C.7, TE-5.8.9.2 and to the frequency identified unless more frequent cleanings are necessary per fire inspections.

The Contractor shall do the following:

C.5.8.9.2.1. Thoroughly steam clean kitchen and grill vapor exhaust systems beginning with the portion of the hood or canopy above the filters and extending through to the final point of exhaust, including all items located in the exhaust air duct applicable to the various exhaust systems including contaminated area of roof walls and exhaust wells resulting from kitchen discharge, and areas contaminated as a result of leaking seams. The cleaning of the filters is the using agency's responsibility.

C.5.8.9.2.2. Remove all deposits of grease, dirt and any other foreign substance from the entire grease vapor exhaust system down to bare metal or surface.

C.5.8.9.2.3. Inspect all hoods after cleaning for defects and make required repairs.

C.5.8.9.2.4. Affix a small adhesive-backed tag or label on the outer hood, close to the wall identifying date of inspection and cleaning. Remove the previous tag prior to affixing a new one.

C.5.8.9.2.5. Return all electrical switches, detection devices and system components to an operable state using qualified personnel after cleaning is completed. Replace cover plates and dampers and diffusers shall be positioned for proper airflow.

C.5.8.9.2.6. Take precautions to ensure that cleaning substances do not come in contact with nor adversely affect fusible links, nozzles, or any other detection devices of the automatic extinguishing systems.

C.5.8.9.2.7. Keep all affected areas clean of debris, grease, and any residue generated by the cleaning process.

C.5.8.9.3. Special Services for Building 8000

Building 8000 has special components that require specific maintenance for occupant comfort.

The Contractor shall do the following:

C.5.8.9.3.1. Inspect, adjust and provide maintenance and repair on the electric window operating systems in the high bay areas.

C.5.8.9.3.2. Perform maintenance on the electric window operating systems each spring to ensure proper operation.

C.5.8.9.3.4. Inspect and clean industrial waste sand traps located in the tunnels under the maintenance bay's on a recurring basis.

C.5.8.9.4. Maintenance of Aboveground and Underground Storage Tanks (AST's & UST's)

Maintenance includes, but is not limited to: leak detection equipment, Above Ground Storage Tanks (ASTs) relocation, Automatic Tank Gauges (ATGs) on USTs, and repairs. These maintenance and repair activities apply to all tanks listed in current inventory and any tanks that may be installed in the future on Fort Carson. Reference section C.7, TE-5.8.9.4-A and TE-5.8.9.4-B.

The Contractor shall do the following:

C.5.8.9.4.1. Ensure that all ASTs and USTs are properly grounded.

C.5.8.9.4.2. Remove water and/or fuel from pits. If water is contaminated with fuel, place in the nearest oil/water separator inlet side. .

C.5.8.9.4.3. Repair and maintain AST's and associated components IAW Federal Regulation, FC Regulation 200-1 and the State of Colorado regulation, 7 C.C.R.1101-14. Removal of oil and sludge shall be done by others.

C.5.8.9.4.4. Ensure ASTs located in Contractor's work area including all fixed emergency generator sub-tanks, are inspected IAW FC 200-1 using the State of Colorado Aboveground Storage Tank Monthly Visual Inspection Form and ullage log. Forms are located in C.7, TE-5.8.9.4.4 and TE-5.8.9.4.6

C.5.8.9.4.5. Report any evidence of product/material releases to the environment to the COR immediately.

C.5.8.9.4.6. Provide copies of the inspection records and ullage logs NLT the 10th day of the month to the COR IAW C.8, report C.5.8.9.4.6 and C-5.8.9.4.6-A.

C.5.8.9.4.7. Ensure any tank relocations are properly approved through the DPW Environmental AST Program Monitor.

C.5.8.9.5. Gate Arm Operation

C.5.8.9.5.1. Ensure gate arms are serviced IAW manufacturer's recommendations. Ensure all hydraulic systems have no leaks, and reservoirs are full, and all emergency lighting is operational. Reference section C.7, TE-5.8.9.5.1 for a list of gate arm locations.

C.5.8.9.5.2. Perform winterization NLT 1 October. Activate the hydraulic heat tape system in accordance with seasonal weather conditions. Accomplish de-winterization no earlier than 1 May. Deactivate the hydraulic heat tape system in accordance with seasonal weather conditions.

C.5.8.9.6. Mechanical Rooms

Mechanical or boiler rooms do not belong to the facility for storage of material. They house real property systems for the respective building and shall be properly maintained. Failed parts shall be removed from the mechanical rooms after repairs are completed.

The Contractor shall do the following:

C.5.8.9.6.1. Ensure mechanical rooms are kept free of debris and broom swept upon completion of any service. Maintain all mechanical rooms free of: liquid leaks, damaged lagging, mechanical parts, inoperative lighting, stored items, etc.

C.5.8.9.7. Other Mechanical Services

This includes, but is not limited to; barracks washer and dryers, cranes, and gate arms.

The Contractor shall do the following:

C.5.8.9.7.1. Repair, replace and maintain lubrication and dispensing systems, and associated equipment and components IAW O&M manuals and manufacturers recommendations. These systems support vehicle and aircraft maintenance facilities.

C.5.8.9.7.2. Monitor cold storage installed equipment at bldg. 350 using EMCS, 24 hours a day, seven days a week. Maintain installed equipment IAW AR 420-1, the approved operation and maintenance plan and manufacturer's instructions.

C.5.8.9.7.3. Inspect, repair, certify and maintain cranes and floor lifts IAW manufacturer's recommendations and TB 43-0156. Inventory of cranes and vehicle lifts is identified in section C.7, TE-5.8.9.7.3.

C.5.8.9.7.3.1. Perform load tests annually or whenever a major component has been replaced and stencil next due date on the crane.

C.5.8.9.7.3.2. Perform magnetic particle or other suitable crack detection method annually on hooks.

C.5.8.9.7.3.3. Maintain records of inspection reports on all cranes and floor lifts for a period no less than five years. Allow review of records by the Government as required.

C.5.8.9.7.4. Repair laundry washers and dryers in barracks listed in section C.7, TE-5.8.9.7.4 as service orders are identified. Machines that are no longer economically functional or serviceable shall be replaced by DPW Furnishings Management.

C.5.8.9.7.5. Clean dryer vents from appliance to vent in facilities listed in section C.7, TE-5.8.9.7.4. per contractor maintenance plan

C.5.8.10. Energy Management Control System (EMCS)

Integrate facility HVAC controls into the EMCS system. This work will include integration into the TRIDIUM Headwork's, standardizing system control and alarm pages and assuring proper operation. Integration will be accomplished by a TRIDIUM integrator. Perform integration of new construction as high priority over scheduled.

The Contractor shall do the following:

C.5.8.10.1. Operate, maintain and repair the Fort Carson EMCS. There are approximately 600 facilities on EMCS. A list of EMCS facilities is located on the EMCS computer database. TE 5.8.10.1 identifies known UCS and Tridium metered systems.

C.5.8.10.2. Maintain and operate the EMCS on a 24 hour continuous basis. Utilize appropriate DMO or PMO for Repair and Preventive Maintenance work.

C.5.8.10.2.1. Open a priority DMO for any critical alarm identified in the EMCS system.

C.5.8.10.2.2. Ensure temperature set points in facilities identified in TE 5.1.7.1.1 can and will be adjusted 24/7 due to occupancy requirements within one hour of request. Respond to request as appropriate to validate no mechanical issue preventing appropriate temperature.

C.5.8.10.2.3. Support accreditation requirements set forth by IMCOM. This includes but is not limited to EMCS, eMASS, APMS, UCS and SPIDERS. This is an ongoing and continuous effort as part of the Army's energy management program and cyber security requirements.

C.5.8.10.3. System Integration:

C.5.8.10.3.1. Have a complete understanding of the EMCS system to collect and analyze data to determine building level energy saving measures and practices to contribute to Fort Carson's energy reduction goals. Opportunities to consider in maximizing energy savings while maintaining occupant comfort include, but are not limited to, matching building occupancy

schedules with current operations, ensuring weekend setbacks are in place for appropriate zones or entire buildings, establishing outdoor air resets for boiler systems, ensuring heat recovery systems are operational, setting up HVAC systems to allow for economizing when outside air temperatures support it, validating existing building sensors are functional/accurate and reviewing temperature set points to ensure they comply with Fort Carson policies and Army Regulation 200-1. Analyze a minimum of three buildings each month that are connected to the EMCS system. Use the EMCS system to ensure effective and efficient management of consumed energy. Energy can be electric, water, gas, HTHW, and from HVAC systems.

C.5.8.10.3.2. Continuously analyze facility energy data from the EMCS system. Determine if energy use is excessive and if peaks of energy use are unreasonable based on, facility usage, time of day, and season. Utilize installed meters to validate energy reduction from system adjustments, identify and resolve anomalies in energy consumption. Validate and correct erroneous meter data within 14 days of identification.

C.5.8.10.3.3. Provide a monthly report to the COR NLT the 5th business day of completed analysis identifying energy reduction recommendations as stated in C.5.8.10.3.1 and C.5.8.10.3.2. Identify by building what facility systems were reviewed/analyzed, what adjustments were performed, what the expected outcome is by the adjustments. Include any meter data issues which were identified or corrected. Provide suggestions on energy saving devices to reduce energy within a facility based on analysis that are outside contract scope for future project work IAW C.8, report C.5.8.10.3.3.

C.5.8.10.3.4. Perform audits of installed facility systems, i.e. lighting systems, HVAC controls, etc., and make appropriate adjustments to ensure proper operation for occupancy requirements. Maintain records of completed audits for Government review for the term of the contract. This is not the same as EMCS review but, requires a hands on and visual review of the installed system(s) at the building. Conduct these audits once in the contract term for all facilities that were constructed after 2004 and as requested by the Government.

C.5.8.11. Fire Training Tower

The training tower is building P3668 and is real property and is propane fired.

C.5.8.11.1. Ensure the training tower is properly maintained to the manufactures recommendations for fire fighter training. This includes the training tower, the propane firing system and all components to make the training tower properly operate. Propane fuel for the training tower is provided by another service contract.

C.5.8.12. Vertical Transportation Systems (VTS)

Provide full maintenance for all VTS (elevators) and take total repair and service responsibility for the equipment identified. The TE provided identifies all the VTS by location and type. Perform all work on VTS in accordance with the most current version of the American Society of Mechanical Engineers (ASME) section A17 and applicable sub sections, National Fire

Protection Association Standards (NFPA), Unified Facilities Criteria (UFC) 3-600-01, Fire Protection Engineering for Facilities, Occupational Safety and Health Administration (OSHA) Standards and the recommendations of the original equipment manufacturer (OEM).

The Contractor shall:

C.5.8.12.1. Perform quarterly Service Inspections (MSI), annual certification and five year speed and load testing on all VTS. Locations of VTS and data is identified in TE-5.8.12.1 Include in the MSI comprehensive operational inspections and adjustments to ensure the satisfactory functioning of elevator and elevator related machinery and controls and other services as required to maintain all systems in a safe and proper operating condition.

C.5.8.12.2. Ensure there is a log book in the VTS mechanical room and it is updated and current with all completed activity.

C.5.8.12.3. Develop and maintain a service folder for each VTC. Include in the folder all completed inspections and certification reports and repair data for the respective VTS. The folders shall be made available to the Government for review.

C.5.8.12.4. Perform inspections, testing, and certification on an annual and five year requirements in accordance with the appropriate ANSI/ASME Standards and manufacturer's specifications. Provide copies of completed testing to the COR monthly by the 5th business day IAW C.8, report 5.8.12.4.

C.5.8.12.5. Perform necessary inspections and tests as required under ASME A17.1 Appendix N, using a Qualified Elevator Inspector (QEI).

C.5.8.12.6. Respond to repair calls following requirements outlined in C.5.1.7. Inoperative elevators shall be identified as safety issues and prioritized accordingly for repairs. Any VTS repairs costing over \$2,000 shall be approved by the KO before being made.

C.5.8.12.7. Prepare and submit to the COR a written report by the 5th working monthly IAW C.8, report 5.8.12.4. detailing all repairs made to any and all VTS. Include in the report the date of the service call, the location of the VTS, description of repairs performed, and the name of the technician performing the repairs, reason for the service call and list of material and parts used

C.5.9. PLUMBING SYSTEMS

Plumbing systems include, but are not limited to, hot and cold potable water systems; fuel gas piping systems; waste, vent, and drain systems; swimming pools; spas; fountains; and HTHW and CW distribution systems.

C.5.9.1. Potable Water and Waste, Vent, and Drain Systems

Potable water systems consist of all hot and cold-water piping and components inside the facility and underground service piping from the building to the connection point on the main distribution pipe. Waste, vent, and drain systems consist of piping and components in the facility upstream of the sewer main. Potable water system includes domestic water heaters.

The Contractor shall do the following:

C.5.9.1.1. Repair, replace, maintain, and operate plumbing systems and associated components IAW UFC-3-420-01 *Plumbing Systems*, AR 420-1, International Plumbing Code (IPC), and manufacturer's specifications.

C.5.9.1.2. Perform seasonal inspections on the HTHW Distribution System 1 November through 30 November and submit an annual inspection report to the COR NLT 31 December of each year IAW C.8, report C.5.9.1.2. and maintain documentation of all inspections and repairs made for the duration of the contract and make available to the COR upon request.

C.5.9.1.3. Ensure piping, plumbing fixtures or pumps do not leak.

C.5.9.1.4 Hot and cold water shall flow on demand through the piping systems with no leakage through closed faucets or valves

C.5.9.1.5. Ensure drains are obstruction free, do not leak, operate as designed and sewage gases are not vented into facilities.

C.5.9.1.6. Fuel oil or gas shall not leak from the system or components.

C.5.9.1.7. Water heaters with forced draft burners shall perform in compliance *Fort Carson Title V Air Operating Permit, ASME Boiler and Pressure Vessel Code*, and manufacturer's specifications. Maintain and operate hot water temperatures within parameters specified in AR 420-1.

C.5.9.1.8. Properly disconnect utilities from facilities identified for demolition.

C.5.9.1.9. Provide utility support to other's with valve turning for isolation, opening vaults, pits or manholes or locating a utility line,

C.5.9.2. Hot Tubs, Swimming Pool Circulation, Heating, and Filtration Systems

Fort Carson has two outdoor swimming pools, three indoor swimming pools, a cold plunge, and five spas. These pools and spas are used for recreation and therapy for Fort Carson civilian personal, retirees and soldiers. Indoor Swimming Pools and hot tubs are open year round. The Outdoor Pool/Baby Pool and Fountain are normally open from 31 May to 30 September, but may change based on weather.

The Contractor shall do the following:

C.5.9.2.1. Repair, replace, and maintain swimming pool and hot tubs circulation, heating, and filtration systems and associated components IAW: 5 CCR 1003-5 Colorado Swimming pool and Mineral Bath Regulation, TM 5-662 Swimming Pool Operations and Maintenance, Technical Bulletin Medical TB MED 575 Occupational and Environmental Health Swimming Pools and Bathing Facilities, Regulations, and manufacturer's specifications. A list of equipment is in section C.7, TE-5.9.2.1.

C.5.9.2.2. Maintain the pool heating system at a temperature required by the individual pool manager and within design specifications.

C.5.9.2.3. Properly operate the pools various systems per the manufactures recommendations and ensure there are no leaks in the systems.

C.5.9.2.4. Ensure that personnel responsible for swimming pool and hot tub operations and maintenance have the proper certification through the National Swimming Pool Foundation (NSPF) as a Certified Pool Operator (CPO).

C.5.9.2.5. Operate, repair and maintain the water treatment filtration systems and disinfection equipment at the Indoor Swimming Pool, Building 1446 and 1995, the Outdoor Swimming Pool, Building 1231, Baby Pools, Building 1232, the Hot Tubs at Buildings 1160, 1856, 1925 and 2357, and the cold plunge at Building 1661.

C.5.9.2.6. Water quality shall be in strict accordance with the Swimming Pool O&M, Technical Bulletin – Medical TM 5-662, Occupational and Environmental Swimming Pool and Bath Facilities TB MED 575 Regulations.

C.5.9.2.7. Maintain swimming pool and hot tub operation and maintenance records for the duration of the contract make available to the Government upon request. Submit the daily swimming pool chlorine residual, bromine residual, combined chlorine and ph. results to the Government IAW C.8 report C.5.15.1.12.

C.5.9.2.8. Maintain spa, hot tub, and swimming pool pH levels in the 7.2 – 7.8 range.

C.5.9.2.9. Maintain a free-chlorine and bromine residuals in the 1.0-5.0 Milligrams per liter (Mg/l) range for all swimming pools. Maintain a bromine level in the 2.0-10.0 Mg/l for hot tubs and spa's IAW CDPHE 5CCR 1003-5.

C.5.9.2.10. Maintain a daily operation record for each swimming pool, hot tub, spa and the cold plunge recording water quality parameters, system pressures, flow rates, filter backwash times and water temperature.

C.5.9.2.11. Document how and what type of adjustments were made to meet the water quality parameters. A copy of the daily operation records shall be included in the monthly water report as required in report C.5.15.1.12.

C.5.9.2.12. Operate and maintain indoor pool solar collection system to assure optimum heat collection and transfer IAW manufacturers' specifications and approved maintenance plan. Maintain solar operation, maintenance checklists and records for the duration of the contract and make available to the Government upon request.

C.5.9.2.13. Prepare the outdoor pools for operation. This effort includes ensuring the boiler is operating correctly and the pool is filled with water and properly treated along with Dewinterize the facilities. Cleaning of the pool, pool deck, and facilities in preparation of opening is responsibility of others. At the end of the season, properly shut down the boiler and prepare the boiler and pool for off season lay-up and winterize the facilities.

C.5.9.3. Water-Free Urinals

As a water saving effort Fort Carson has numerous water free type urinals. These urinals must have their cartridges changed out periodically to keep sewer odors from entering the facilities and allow liquid to drain.

The Contractor shall do the following:

C.5.9.3.1. Repair, replace and maintain water free urinals per manufacturers requirements.

C.5.9.3.2. Respond as priority two service orders for, cartridge replacements, clogged drains and odor complaints.

C.5.9.4. Water Features and Fountains

Fort Carson has two water fountains and two water fall water features that enhances the landscaping for certain high usage areas. All of these features use recycled water and pumps to keep the feature functioning properly

The Contractor shall do the following:

C.5.9.4.1. Repair, operate and maintain the Civilian Memorial water fountain near Building 1500 and the water fountain behind Building 6215.

C.5.9.4.2. Repair, operate, and maintain the water feature located in the Resiliency Center Campus and the water feature west of Building 1046.

C.5.9.4.3. Maintain measurable chlorine residual in the water for each water fountain and water feature.

C.5.9.4.4. Winterize the water fountains and water feature as necessary to prevent damage to piping, pumps, and structures.

C.5.9.4.5. De-winterize the water fountains and water feature for warm weather operation. Normal warm weather operating months are May-September.

C.5.9.5. Fire Suppression Systems

Fire suppression systems include, but are not limited to, dry fire systems, wet fire systems, foam injection systems, standpipe systems, special gas systems and fire pumps. A list of fire suppression systems is in section C.7, TE-5.9.5.

C.5.9.5.1. Dry Fire Suppression Systems

Dry fire suppression systems include, but are not limited to, hose reel, deluge, compressors and reaction systems.

The Contractor shall do the following:

C.5.9.5.1.1. Test, inspect, repair, replace, maintain, and operate dry fire suppression systems and associated components IAW Unified Facilities Criteria (UFC) 3-600-02 and manufacturer's specifications.

C.5.9.5.1.2. Maintain system air and water pressures IAW individual system requirements.

C.5.9.5.1.3. Notify Fort Carson Fire and Emergency Services prior to taking fire suppression system out of service, and within one hour after system is restored to service. If the system is out of service for more than four continuous hours in a 24-hour period, notify the COR and facility occupants to provide a fire watch or evacuate the facility until the system is returned to service.

C.5.9.5.1.4. Maintain records by bldg. of each system. Document all work performed to the respective system in the record. These records shall be made available for review by the Government when requested.

C.5.9.5.1.5. Dry fire suppression systems shall be set and activated to alarm condition. Air and water pressures shall be set for the individual system requirements. Hose reels shall be connected and ready for immediate use.

C.5.9.5.2. Wet Fire Suppression Systems

Fort Carson has numerous wet pipe fire suppression systems to protect the facilities and building occupants. These systems are connected into the fire departments alarm system and automatically activate during a fire.

The Contractor shall do the following:

C.5.9.5.2.1. Test, inspect, repair, replace, maintain, and operate wet fire suppression systems and associated IAW, UFC 3-600-02, and manufacturer's specifications.

C.5.9.5.2.2. Notify Fort Carson Fire and Emergency Services prior to taking the system out of service, and within one hour after system is restored to service. If the system is out of service for more than four continuous hours in a 24-hour period, notify the COR and facility occupants to provide a fire watch or evacuate the facility until the system is returned to service.

C.5.9.5.2.3. Maintain records by bldg. of each system. Document all work performed to the respective system in the record. Copies of these records shall be made available to the Government when requested.

C.5.9.5.2.4. Inspect, adjust or repair room heating systems in sprinkler valve rooms to assure devices are operational and protected from freezing.

C.5.9.5.2.5. Activate the thermostat to verify that the electric heater is operational.

C.5.9.5.2.6. Notify the COR when installing temporary heat tape and ensure any temporary heat tape is properly installed and operating. .

C.5.9.5.2.7. Ensure wet fire suppression systems are set, charged, and activated to alarm condition. The system shall be charged to the water pressure required for the individual system.

C.5.9.5.3. Clean Agent Fire Suppression Systems

Clean Agent Fire Suppression Systems are the fire systems located in the Fort Carson communication node facilities. Locations and data about these systems can be found in C.7, TE-5.9.5.3.

The Contractor shall do the following:

C.5.9.5.3.1. Test, inspect, repair, replace, maintain, and operate clean agent fire suppression systems and associated components IAW UFC 3-600-02 and manufacturer's specifications.

C.5.9.5.3.2. Notify Fort Carson Fire and Emergency Services prior to taking the system out of service and within one hour after system service is restored. Notify the COR and facility owner if the system is out of service for more than four continuous hours in a 24-hour period.

C.5.9.5.3.3. Ensure maintenance and inspection records are kept on the clean agent fire suppression systems for the duration on the contract and be made available to the Government upon request.

C.5.9.6. Fire Pumps

Fort Carson has 14 facilities that house fire suppression pumps. These fire pumps were installed to maintain the required water pressure for both wet and deluge type automatic fire suppression systems.

The Contractor shall do the following:

C.5.9.6.1. Operate, repair and maintain the fire pumps at Buildings: 9613, 510, 515, 1210, 1435, 1447, 1500, 2330, 8002, 8009, 8010, 8143, 8930 and 20000. Fire pump data can be located in C.7, TE-5.9.6.1.

C.5.9.6.2. Maintain these fire pumps in a fully operable status. Notify the COR and the Fire Department immediately if for any reason the Contractor shuts these systems down

C.5.9.6.3. Maintain emergency parts or back up pumps to restore operation within 48 hours after shutdown.

C.5.9.6.4. Test run the fire pumps at Buildings: 510, 515, 1210, 1435, 1447, 1500, 2330, 8002, 8009, 8010, 8143, 8930, 9613 and 20000 monthly.

C.5.9.6.5. Maintain fire pump test data and maintenance records for each bldg. for the duration of this contract. The test data and maintenance records shall be made available to the government upon request.

C.5.9.7. Winterize/De-winterize Facilities

There are several facilities that are used seasonally and do not have heating systems. To protect the plumbing from damage these facilities must have their water and drain pipes isolated and drained of water before freezing temperatures. These facilities will also need to have the water turned back on for the warm weather season.

The Contractor shall do the following:

C.5.9.7.1. Perform winterization and de-winterization of all plumbing systems to prevent freeze damage. A list of facilities to be winterized and de-winterized is in section C.7, TE-5.9.7.1.

C.5.9.7.2. Coordinate winterization and dewinterization of IHP with the user 30 days prior to starting effort.

C.5.9.7.3. Ensure all facilities in TE 5.9.7.1 are winterized by 15 October of each year unless otherwise stated by the KO.

C.5.9.7.4. Winterize the Outdoor Pool within two weeks after the last pool event, usually mid-September, unless otherwise stated by the KO.

C.5.9.7.5. Winterize the Civilian memorial NLT 30 days after Labor Day.

C.5.9.7.6. Start de-winterization no earlier than 15 April and complete all effort within 30 days.

C.5.9.7.7. Dewinterize the outdoor pool and ensure the pool, to include office and changing bldg., and pool deck, is fully operational NLT one week before Memorial Day.

C.5.9.7.8. Ensure the Civilian Memorial is de-winterized and operational NLT one week before Memorial Day.

C.5.9.8. Propane Tanks

Propane tanks are provided to facilities that are not connected to a natural gas distribution system and are owned by the Government. Most of these are located down range. Filling of the tanks is not included in this PWS. A list of current active propane tanks and their locations are listed in section C.7, TE-5.9.8.

The Contractor shall do the following:

C.5.9.8.1. Repair, replace, maintain, and operate propane tanks and associated components IAW NFPA 31, NFPA 54, NFPA 58, NFPA 211, AR 420-1, TM 5-642, and manufacturer's specifications.

C.5.9.8.2. Inspect propane tanks annually in accordance with the NFPA 58, Liquefied Petroleum Gas Code. Results of the inspection shall be included in the annual gas leak survey IAW C.8, report C.5.10.7)

C.5.9.8.3. Relocate tanks to maximize use as identified by the COR. Anticipate up to four relocates each year.

C.5.9.9. Wash Racks

Military units use wash racks to wash their tactical vehicles. Wash racks are located in some of the vehicle maintenance facilities.

The Contractor shall do the following:

C.5.9.9.1. Repair, replace, and maintain wash racks and associated components IAW O&M manuals, and manufacturer's specifications.

C.5.9.9.2. Contact the Environmental Division-DPW COR for the Sludge contract if sludge removal is required. This effort is accomplished by other service contract.

C.5.10. NATURAL GAS DISTRIBUTION AND REGULATION SYSTEM

The maintenance boundary for the natural gas distribution system is located at and includes the low pressure valve to the gas meter in the control manhole next to the north wall of Evans Army Community Hospital (EACH). The maintenance boundary for natural gas distribution systems in Family Housing will be the point where the individual building gas service line(s)/valve(s) connects to the gas main.

Data about the gas distribution system can be located in C.7 in TE-5.10. Contractor shall provide updates to the COR NLT 31 January of each year IAW C.8, report C.5.10.

Fort Carson's natural gas system has two operating pressures maintained by a number of regulating stations. The high pressure distribution piping is regulated to operate at 50 PSI and the lower pressure distribution piping is regulated to operate at 25 PSI. Maintaining the correct pressure in the distributions systems is vital for proper operation. Over pressurization of the gas distribution systems can cause damage to the piping.

The Contractor shall do the following:

C.5.10.1. Maintain and service the natural gas distribution and regulation system IAW DOT (Department of Transportation) Pipeline Safety Regulations, 49-CFR 192, DOT Guidance Manual for Operators of Small Gas Distribution Systems, DPW Installation Design Guide and the Fort Carson Emergency Gas Plan, FC Regulation 420-24).

C.5.10.2. Perform specific tasks including, but are not limited to, adjusting pressure settings on regulating stations; repairing/replacing building service regulators excluding Family Housing buildings; turning system valves on/off; locating and marking gas lines; survey/maintenance of cathodic protection systems (See PWS Section C.5.18); performing gas leak surveys excluding the individual building service lines in Family Housing buildings; raising and color coding valve boxes.

C.5.10.3. Respond as a priority 1 and take action to safeguard life and property upon notification of a natural gas emergency. Ensure danger or risk is eliminated and notify the COR as to the status of the repair work if repairs cannot be complete.

C.5.10.4. Notify the COR immediately upon receiving an emergency natural gas call (such as, but not limited to: gas smell, line break) during duty hours and by 0730 the following duty day for non-duty hours emergency calls.

C.5.10.5. Notify the COR of the area, the approximate time and length of gas turn off for all scheduled outages. Also notify the Fire Department, and affected facility occupants of all outages and provide approximate times of outages.

C.5.10.6. Ensure personnel performing maintenance and repairs to the gas systems are qualified according to 49-CFR 192, Subpart N, sections 192.801-192.809.

C.5.10.7. Perform periodic gas leakage surveys (excluding the individual building service lines in Family Housing areas) in accordance with DOT Pipeline Safety Regulations, currently Part 192.723 and report the results to the COR NLT 30 September annually IAW C.8, report C.5.10.7. Requirements for the report are identified in C.7, TE-5.10.7

C.5.10.8. Respond to detection or reports of gas leaks. Gas leaks reported to the Contractor shall be responded to as Priority 1. The Contractor shall submit a gas leak report NLT 24 hours after response to the COR IAW C.8, report C.5.10.8. Classify gas leaks using the following criteria:

Grade 1: is a gas leak that represents an existing or probable hazard to life or property and requires immediate and continuous action until conditions are no longer hazardous. The immediate action in some instances may require one or more of the following: evacuating premises, blocking off an area, rerouting traffic, eliminating sources of ignition, venting the area, or stopping the flow of gas by closing valves or other means. Gas leaks that require immediate action and are classified as Grade 1 include but are not limited to any positive indication of gas in, under or around a building; any positive indication of gas in a confined space; any positive indication of gas under a wall-to-wall paved area within a building; escaping gas that has ignited or which can be seen, heard or felt at the ground surface; and any leak which, in the judgment of Fire Department personnel at the scene, is regarded as an immediate hazard. (Grade 1 gas leaks shall be reported to the Government within one hour of classification.)

Grade 2: is a leak that is recognized as being non-hazardous at the time of detection but justifies scheduled repair based on potential hazard. These leaks shall be monitored weekly and reported to the Government. Weekly monitoring shall include but not be limited to amount and migration of gas; proximity of gas to buildings and subsurface structures; extent of pavement; and soil type and conditions such as frost cap, moisture, and natural venting.

Grade 3: is a leak that is non-hazardous at the time of detection and can reasonably be expected to remain non-hazardous. These leaks shall be reevaluated at monthly intervals until the leak is regraded. Leaks requiring reevaluation at monthly intervals and are classified as Grade 3 include but are not limited to any positive indication of less than 4% gas in small associated substructures, such as small meter boxes or gas valve boxes, and any positive indications of leakage below 5% under a street where it is unlikely the gas could migrate to the outside wall of a building.

C.5.10.9. Maintain gas distribution valves and records in accordance with DOT Pipeline Safety Regulations, currently 49 CFR Part 192.

C.5.10.10. Maintain gas distribution regulation stations and gas valve maintenance records. These records shall be kept for the duration of the contract and be made available to the government upon request.

C.5.10.11. Provide updates to drawings when gas distribution changes are made by the Contractor or when drawings are found to be inaccurate in accordance with the procedures of PWS Sections C.5.1.15 and C.5.1.16.

C.5.11. ENVIRONMENTAL SERVICES

Environmental services include the Contractor's compliance with and the Contractor's assistance to Fort Carson's compliance with all environmental laws, regulations, and directives applicable or related to the performance of work under this contract. Specific tasks are stated both in this Section C.5.11, and elsewhere in this PWS.

C.5.11.1. National Environmental Policy Act (NEPA) Program Support

NEPA requires that Fort Carson consider the potential environmental consequences of any action it proposes to take. The formality of that consideration varies, based on the kind and extent of a proposed action and the possible environment effects it may involve.

The Contractor shall do the following:

C.5.11.1.1. Be familiar with NEPA requirements, which are outlined in AR 200-1 and FC Regulation 200-1.

C.5.11.1.2. Submit to DPW Environmental for a NEPA review all non-emergency work that requires disturbance of soil (e.g. digging or grading), pruning of trees and shrubs, removal of trees and shrubs, mowing of natural habitat, or repair/maintenance of historical structures or grounds within properties/areas listed in TE-5.2.2.5.

C.5.11.1.3. Notify DPW Environmental and the COR after emergency work is completed that would typically require a NEPA review as identified in C.5.11.1.2.

C.5.11.1.4. Comply with requirements identified under the NEPA review to perform the identified work.

C.5.11.1.5. Not require a NEPA review for repair work to non-historical facilities unless asbestos or lead based paint is encountered. Notify the COR before taking any action which has not had necessary NEPA review.

C.5.11.1.6. Respond to requests for information and questions raised in the course of a NEPA review by the DPW Environmental staff.

C.5.11.2. Hazardous Waste and Spill Response

Hazardous material (HM) is used in many processes that are essential for daily mission activities at Fort Carson, which may, in turn, create hazardous wastes (HW). Management of HM is governed by the requirements provided or referenced in appendices of FC Regulation 200-1.

Fort Carson must comply with all DoD, DA, Federal, State, and local regulations concerning the procurement, storage, transportation, handling, use, tracking, reporting and disposal of hazardous material.

The Contractor shall do the following:

C.5.11.2.1. Comply with all DoD, DA, Federal, State, and local regulations concerning the procurement, storage, transportation, handling, use, tracking, reporting and disposal of hazardous material.

C.5.11.2.2. Pursue all reasonable actions to avoid or reduce the use of HM; maintain a consolidated inventory of HM stored and used on Fort Carson; employ best management practices (BMPs) throughout the HM life cycle in order to minimize risk to public health or damage to the environment; reduce the generation of HW; and prevent or minimize pollution from releases of HM to the environment.

C.5.11.2.3. Provide cleanup of all hazardous spills IAW FC Regulation 200-1 in the Contractor's area of responsibility (AOR). Spills shall be contained immediately and collected and disposed of within 24 hours. Spills greater than 5 gallons, and spills that enter a storm water collection system, or ditch/creek shall immediately be reported to the FCFD and DPW-Environmental Division spill line (526-0973).

C.5.11.2.4. Report spills within one hour of occurrence to the COR and DPW Environmental Division with written report following within three work days.

C.5.11.2.5. Report actions on spills using the Spill Report Form to the COR NLT 72 hours after spill response IAW C.8, report C.5.11.2.5. Keep on file for the duration of the contract and make available for viewing by the Government and other interested parties. All files shall be turned over to the Government at the contract end. A sample Spill Report Form (FC 1200-E) is located in C.7, TE-5.11.2.5.

C.5.11.2.6. Obtain approval through the DPW Environmental Division for any new hazardous materials (e.g., chemicals, solvents, and paints) prior to purchasing and using them on Post. Safety Data Sheets must accompany the approval request.

C.5.11.2.7. Prepare an electronic hazardous material inventory of all materials purchased locally (i.e., not through the on-post Hazardous Materials Control Center) and provide it to the COR NLT the 10th of each month IAW C.8, report C.5.11.2.7.

C.5.11.3. Environmental Protection Officer (EPO)

Identify an EPO and Alternate, for the Contractor's work areas IAW FC Regulation 200-1. These persons shall be responsible for all contractor occupied facilities and work under control of the Contractor.

The Contractor shall do the following:

C.5.11.3.1. Ensure the EPO and alternate attend all initial and refresher training required under FC Regulation 200-1 and ensure that the Contractor is in total compliance with all federal, state and installation environmental codes, laws and regulations.

C.5.11.3.2. Ensure the EPO is responsible for preparing the Contractor for all environmental inspections, whether by Army, local, state or federal agencies, attending such inspections, and complying with the results from all such inspections.

C.5.11.3.3. Report to the COR the results of all inspections.

C.5.11.3.4. Ensure the EPO duties include compliance responsibilities of all Satellite Accumulation points operated by the Contractor.

C.5.11.4. Asbestos and Lead Based Paint (LBP)

Asbestos containing materials include all types of construction materials and may be encountered during maintenance and repair operations. Asbestos removal under this contract is limited; see Section C.5.1.16. Lead based paint (LBP) is typically found on trim work, window frames, door frames, baseboards, door panels and some wall board.

The Contractor shall do the following:

C.5.11.4.1. Obtain samples of suspected asbestos or lead containing material and have them analyzed by an accredited laboratory National Voluntary Laboratory Accreditation Program (NVLAP) for asbestos prior to disturbing the suspected asbestos or lead-containing material. Provide a copy of all sampling results to the DPW Environmental Division Asbestos Program Manager by the 5th of each month.

C.5.11.4.2. Sample, analyze, abate, encapsulate, clean up spills, and dispose of asbestos and asbestos containing materials, up to the limits described in Section C.5.1.16. Coordinate disposal through DPW Environmental Division.

C.5.11.4.3. Remove asbestos materials and identified LBP only when incidental to DMO's, miscellaneous requirements or PMO's performed under this contract and maintain abatement records as required by applicable CFR.

C.5.11.4.4. Ensure all abatement work is performed in compliance with 29 CFR 1926.1101, 40 CFR 61, subpart M Section 61.140 through 61.156; and 40 CFR 763 *Asbestos*, and Colorado Regulation #8 and #19.

C.5.11.4.5. Ensure all personnel performing abatement are licensed by the State of Colorado.

C.5.11.4.6. Ensure all monitoring and disposal records are maintained by the Contractor per regulations and be available for government review.

C.5.11.4.7. Ensure all abatement is monitored by DPW Environmental Division. Coordinate inspections directly with DPW Environmental Division.

C.5.12. INDUSTRIAL WASTEWATER TREATMENT PLANT (IWTP) AND CENTRAL VEHICLE WASH FACILITY (CVWF)

This facility treats the industrial wastewater from vehicle motor maintenance facilities and aircraft maintenance facilities. The discharge outlet of the Industrial Wastewater Plant goes into our sanitary sewer collection system.

C.5.12.1. Industrial Wastewater Treatment Plant (IWTP)

Data about industrial wastewater systems are located in C.7 in TE-5.10. The Contractor shall provide updates to the COR NLT 31 January annually IAW C.8, report C.5.10.

Equalization Basins: Due to the variation in flow and the large amounts of suspended solids in the industrial wastewater, equalization basins are provided at the treatment plant site for settling, solids, flow equalization, storm water storage, and oil removal.

Biological Ponds: Aerated biological ponds are provided at the IWTP site to remove Biochemical Oxygen Demand (BOD) prior to discharge to the STP.

FOS Facility: Rags, suspended solids and oil are removed prior to discharge to the Equalization Basins, under dry weather conditions.

The Contractor shall do the following:

C.5.12.1.1. Operate, maintain, repair, and replace IWTP systems and components IAW O&M manuals, manufacturer's specifications, and TM 5-814-3 *Domestic Wastewater Treatment Plant Laboratory Guidance*.

C.5.12.1.2. Ensure operation of the IWTP is under supervision of a certified operator IAW CDPHE Regulation 100, 5 CCR 1003-2 section 100.1.2. Perform continuous wastewater treatment plant operation IAW the most current edition of *Standard Methods for the Examination of Water and Wastewater*, O&M manuals and plans, and manufacturer's specifications.

C.5.12.1.3. Perform operational tours to evaluate plant operation and lift stations, collect and record data, perform adjustments necessary to control the quality of the operations, inspect for deficiencies and hazardous conditions to determine the plant maintenance and repair requirements. Explain missed operational tours in the operating log so that no spaces are left blank.

C.5.12.1.4. Perform routine cleaning of equipment subject to waste material accumulations, blockages, algae growth, vector attraction, and offensive odors. Dispose of grit, scum, and screening material removed from the facility waste streams.

C.5.12.1.5. Collect samples for analysis IAW the IWTP Operation and Maintenance Plan, and as requested by the COR. Prepare samples and submit to appropriate laboratory for analysis.

C.5.12.1.6. Perform maintenance cycles IAW checklists in the maintenance plan. Operations and maintenance includes, but is not limited to; flushing and unplugging lines, unplugging wash rack drains, operational checks of lift stations, sedimentation basin cleaning, repair of service water pumps, scheduled lubrication of equipment, coordination with the STP for transferring excess CVWF water to the industrial lagoons, and scheduled maintenance of equipment. Clean each sedimentation basin to keep sludge accumulation from solidifying.

C.5.12.1.7. Annually clean sludge from the IWTP free oil separator facilities and the transfer pump stations. Coordinate with the COR for the disposal of sludge by other service contract.

C.5.12.1.8. Maintain the entire collection system in a free flowing condition at all times. Upon notification of plugged lines, respond within two hours to unplug the system. Follow DPW SOP 61 (Fort Carson Wastewater Overflow Procedures) when an industrial sewer overflow condition is discovered.

C.5.12.1.9. Keep the IWTP in continuous operation and meet or exceed manufacturers O&M manual requirements. IWTP shall be capable of processing to its rated treatment capacity.

C.5.12.1.10. Notified the COR within one hour of initiating a system bypass, and again within one hour after system is returned to normal operating condition. Process waters, sludge, or lubricating oil shall not leak except as noted in O&M manuals, and manufacturer's specifications. Discrepancies identified during operational tours shall be resolved upon discovery.

C.5.12.1.11. Submit a monthly report of samples taken to the COR NLT the 5th business day IAW C.8, report C.5.13.1.5.

C.5.12.1.12. Perform and record all industrial wastewater. Wastewater monitoring equipment and procedures shall be in accordance with 40 CFR 136, Guidelines Establishing Test Procedures for the Analysis of Pollutants which is the standard used by Fort Carson for chemical analysis. Proper chain of custody procedures shall be followed. Monitoring results shall be recorded and filed daily.

C.5.12.1.13. Update utility drawings and provide to the Government IAW Sections C.5.1.15 and C.5.1.16 when the Contractor makes IWTP collection system additions or changes.

C.5.12.2. Central Vehicle Wash Facility

The major operating components of the CVWF are as follows:

Bird Bath: A 280,000 gallon, recessed, 4-lane concrete basin; inclined entrance and exit ramps allow easy access; and divided into a 2-lane multipurpose vehicle bath and a 2-lane tracked vehicle bath.

Post Wash Lanes: Consist of 20 wash stations, each comprised of a vehicle wash lane with tower, pressure wash distribution hose and nozzle and a freeze proof hydrant.

Wash Water Basins: Stilling Basin – closed loop conduit transports wastewater from the wash area to this 4.92 million gallon storage basin. Stilling basin water is filtered for use in the wash area cannons and hydrants. Storage for this wash water is the 4.68 million gallon Wash Water Basin.

Wash Water Pump Station: Located between the Stilling and Wash Water Basins, nine pumps of varying horsepower and volumes deliver water to the sand filter or to the wash area.

Sand Filter System: Four cells, each with a surface area of 11,000 square feet, filter stilling basin water for use as wash water. Filtered water is stored in the Wash Water Basin.

The Contractor shall do the following:

C.5.12.2.1. Properly operate and maintain the CVWF including all components or equipment associated with the CVWF.

C.5.12.2.2. Core hours of operation shall be 0900 – 1700, Monday to Friday.

C.5.12.2.3. Follow FC Reg. 700-5 for scheduling and use of the CVWF.

C.5.12.2.4. Perform scheduled periodic and day to day maintenance IAW the CVWF O&M manual and the maintenance plan.

C.5.12.2.5. Give priority for weekend usage to the National Guard or Reserves.

C.5.12.2.6. Log usage of the CVWF. Ensure log entries include ambient temperature, operational duties, basin levels, unusual conditions, accidents and complaints. Submit to the log to the COR NLT the 7th business day IAW C.8, report C.5.12.2.6.

C.5.12.2.7. Coordinate with the COR, for KO approval, any operational change or maintenance function that requires a disruption of CVWF operations. Provide no less than a 72-hour notice of a scheduled disruption.

C.5.12.2.8. Notify scheduled units of any weather closing no less than one hour before their scheduled use of the facility.

C.5.13. SANITARY WASTEWATER TREATMENT AND COLLECTION SYSTEM

The Fort Carson sanitary wastewater treatment and collection system collects and delivers sanitary wastewater to the Sewage Treatment Plant (STP) from Fort Carson and Cheyenne Mountain Air Force Base (CMAFB) by means of collection lines, sewer mains, and lift stations. Data about the sanitary wastewater and collection system can be located in C.7 in TE-5.10. ~~Provide updates to the COR NLT 31 January annually IAW C.8, report C.5.10.~~

Domestic sewage from Cheyenne Mountain Air Force Base (CMAFB), which is approximately 65,000 gallons per day (gpd): ~~The Contractor's maintenance responsibility for CMAFB begins at manhole 5534. The maintenance boundary for Evans Army Community Hospital (EACH), Building 7500, begins where the line exits the building. Responsibility in housing areas is limited to main lines and manholes.~~

The STP utilizes primary, secondary and tertiary treatment. It is designed for 97-99% removal of Biochemical Oxygen Demand (BOD) and 97% removal of total suspended solids (TSS). Currently, the average flow is 1.6 million gallons per day (mgd). The STP design is 4.02 mgd and is intended to satisfy requirements of the current National Pollutant Discharge Elimination System (NPDES), permits (Wastewater Treatment Permit number CO-0021181), (Landfill Number 5 Facility Permit number CO-0034771), and (Bio Solids Permit number COG652000). A brief description of the major operational units in the STP can be found in section C.7, TE-5.13.

Sanitary wastewater treatment and collection system components include, but are not limited to, electrical, electronic, and digital systems and controls; level controls, control panels, phone dialers, switches, metering devices, sewage basins, alarms, lift stations, wet wells, dry wells, pumps, motors, air compressors, filters, piping, indicator lights, mounting hardware, meters, manholes, lagoons, valves, fittings, and laboratory equipment and components.

The Contractor shall do the following:

C.5.13.1. ~~Operate, repair, and maintain sanitary wastewater treatment and collection system and associated components IAW AR 420-1, UFC 3-240-03N *Wastewater Treatment Systems Augmenting Handbook*, 5 CCR1003-2 *Water and Wastewater Facility Operator Certification Requirements*, TM 5-814-1 *Sanitary and Industrial Waste Water Collection—Gravity Sewers and Appurtenances*, TM 5-814-2 *Sanitary and Industrial Wastewater Collection—Pumping Stations and Force Mains*, O&M manuals, and manufacturer's specifications.~~

C.5.13.1.1. Operate and maintain the STP on a 24 hours per day, 365 days per year basis.

C.5.13.1.2. Operate and maintain the STP in accordance with established maintenance schedules, manufacturer's recommendations and in compliance with NPDES permit numbers CO-0021181, COG652000 and CO-0034771 section 1.3.1 and 1.3.2. Report any violation of the

NPDES permit standards to the COR and DPW-ED Environmental Compliance Branch within two hours of the incident with a detailed explanation of the noncompliance.

C.5.13.1.3. Be required to service portable latrines or pump sand traps on an emergency basis, at the direction of the KO.

C.5.13.1.4. Provide certified operators in compliance with the Colorado Department of Public Health and Environment Regulations (CDPHE) Water and Wastewater Facility Operators Certification Requirements Regulation 100, 5 CCR 1003-2.

C.5.13.1.5. Prepare and submit to the COR by the 7th business day of each month a copy of Facilities Engineering Operating Log (Sewage) IAW C.8, report C.5.13.1.5. Example of log is in C.7, TE-5.13.1.5.

C.5.13.1.6. Maintain a maintenance log and a daily operator's log notebook that includes routine operational duties, unusual conditions, accidents and complaints.

C.5.13.1.7. Submit the annual bio-solids and the drying bed percent solids hauling reports to the COR NLT 15 January annually IAW C.8, report C.5.13.1.7.

C.5.13.1.8. Prepare and submit to the COR test data required by the current NPDES permit in Section 1.7.2.1., Self-Monitoring Requirements the following reports as stated in C.8: NLT the 7th business day monthly report C.5.13.1.8; NLT the 1st working day after receiving results report C.5.13.1.8-A; and NLT 1st working day after receiving results report C.5.13.1.8-B

C.5.13.1.9. Provide the total daily sewage treatment plant effluent flows and total daily reclaimed water usage to Colorado Springs Utilities, Water Conveyance Section. Report this information by telephone no later than 0900 hours on a daily basis. The Water Conveyance Section phone number is 719-668-1934.

C.5.13.1.10. Operate, maintain and perform laboratory testing in the STP Laboratory, Bldg. 3900, in accordance with requirements established in NPDES permits. The STP laboratory must maintain state certification to perform microbiology testing.

C.5.13.2. Sewage Collection

C.5.13.2.1. Ensure sanitary wastewater treatment and collection system is free of leaks, free flowing, and clear of obstructions from the point of demarcation identified in the CSU IGSA inward (toward the facility). Repairs to buried sewer lines shall be inspected and approved by the Contractor's QC department prior to backfilling. Planned outages shall not exceed eight hours. Prepare and submit to the COR NLT the 5th business day monthly an unplanned outage report IAW C.8, C.5.13.2.1.

C.5.13.2.2. Maintain sewer mains, laterals and service lines in a free flowing condition. Sewage mains shall be flushed according to best management practices and guidance from the

California State University, Sacramento Manual; *Operation and Maintenance of Wastewater Collection Systems*. Follow DPW SOP 61 (Fort Carson Wastewater Overflow Procedures) when a sanitary sewer overflow condition is discovered.

C.5.13.2.3. — Ensure manholes are structurally sound, clean and free of debris. Ensure all manholes are covered with cast iron manhole lids.

C.5.13.2.4. — Update utility drawing when wastewater collection system additions or changes are made and provide copies to the Government according to the procedures of PWS Sections C.5.1.14 and C.5.1.15.

C.5.13.2.5. Pump sewage from the Wilderness Road Complex septic tanks and numerous down range latrine holding tanks as necessary.

C.5.13.3. Reclaimed Water

Operate and maintain the Fort Carson reclaimed water system that supplies wastewater treatment plant effluent to the 5.5 acre golf course pond for irrigation storage IAW CDPHE Reg. 84 for Category 2 type reclaimed water.

C.5.13.3.1. Ensure proper maintenance and operation of the primary pump station in building 3867 and the booster-pump station located on Sheridan Avenue.

C.5.13.3.2. Operate, maintain and repair all components of the reclaimed water system IAW CDPHE Regulation No.84 (Reclaimed Water Control Regulation), O&M manuals, manufacturer's specifications and best management practices.

C.5.13.3.3. Maintain the Golf Course pond at a minimum 75% capacity, with minimum of overflow. Use potable water to maintain pond level during high use periods. Maintain the golf course pond IAW PWS section C.5.13.5.

C.5.13.4. Grease Traps/Interceptors

Inspect, pump, clean and maintain grease traps/interceptors listed in section C.7, TE-5.13.4. Inspect all grease traps at least monthly.

C.5.13.4.1. Remove any grease that is floating on the liquid surface of grease traps holding less than 10,000 gallons of liquid monthly as a maintenance task and disposed of at the STP or other approved location.

C.5.13.4.2. Pump and clean all grease interceptors according to the following criteria, but no less than annually:

- When the floatable grease (FOG) layer exceeds six inches in depth as measured by a sludge core sampling device;

- When the settled solid layer exceeds eight inches in depth as measured by a sludge core sampling device;
- When the total volume of captured grease and solid material displaces more than twenty-five percent of the capacity of the interceptor as calculated using a sludge core sampling device; and
- When the interceptor is not retaining/capturing oils and greases.

C.5.13.4.3. Ensure built-up solids are scraped from walls, floors, baffles, and all pipe work interior of the interceptor when it is determined that a grease interceptor requires a complete pumping or cleaning

C.5.13.4.4. Maintain records for each grease interceptor which include the date and time of each inspection, FOG depth, sludge depth, details of any repairs, dates cleaned, dates pumped, amount of liquids and solids removed on each occasion and the service order number for repairs made and any other records pertaining to the interceptor. Make records available for review upon request by the DPW.

C.5.13.4.5. Obtain approval from DPW Environmental Division for any chemicals, enzymes, live bacteria or other grease cutters prior to use through the KO.

C.5.13.5. Sewage lagoons

Operate, repair and maintain wastewater treatment lagoons at PCMS, Range 143 and Range 145 according to best management practices and guidance from the California State University, Sacramento Manual, *Operation of Wastewater Treatment Plants*.

C.5.13.5.1. Keep weeds, trees and shrubs out of the lagoon area to prevent damage to the lagoon liner.

C.5.13.5.2. Maintain the lagoon dikes and take actions necessary to correct and prevent erosion of the dikes.

C.5.13.5.3. Maintain the reclaimed water storage pond aerators and take appropriate actions to control the algae growth in the pond water.

C.5.13.6. Sewage Drying Beds

The sewage drying beds are located at the Sewage Treatment Plant. The sludge produced from the sewer treatment process is placed on the drying beds to de-water the sludge. The water travels through drainage pipes and is pumped back into the sewage treatment process. The sludge is hauled off post in accordance with the treatments plants discharge permit.

The Contractor shall do the following:

C.5.13.6.1. Operate and maintain STP sludge drying beds to support generation rates and critical equipment failures. Keep pace with generation rates and Sewage Treatment Plant operational requirements and remove sludge from the sewage drying beds as necessary.

C.5.13.7. Oil Water Separators

Oil water separators are pre-treatment devices installed on certain facility wastewater service lines to prevent oil from entering the sanitary sewage collection system.

The Contractor shall do the following:

C.5.13.7.1 Inspect, pump, clean and maintain oil water separators listed in section C.7, TE-5.13.7.1.

C.5.13.7.2. Inspect all oil water separators semi-annually. Measure oil floating on top of the water inside the oil water separator; when there is two inches or more of oil in any chamber, the oil will be removed.

C.5.13.7.3. Clean the oil water separator when the sludge depth in the bottom of the separator exceeds eight inches in depth. Ensure maintenance and cleaning of the coalescing plates inside the oil water separator are IAW the manufactures' operation and maintenance manuals. Dispose of oil and sludge IAW local and state standards.

C.5.13.7.4. Maintain records for each oil water separator which include the date and time of each inspection, oil depth, sludge depth, details of any repairs and any other records pertaining to oil and sludge removal of the separator. Make records available for review upon request of the Government.

C.5.14. STORM SEWER COLLECTION SYSTEM

The Fort Carson storm sewer collection system collects and discharges storm sewer runoff to storm sewer outfalls and drainage. Storm sewer collection system and components include, but are not limited to, drainage ditches, catch basins, retention/detention ponds, manholes, storm vaults, culverts, bridge openings, collection lines, and sediment basins. Storm collection is detailed on the Basic information maps. Reference section C.7, TE-5.14.

The Contractor shall do the following:

C.5.14.1. Operate, monitor, maintain and repair the storm sewer collection system and associated components IAW TM 5-820-4 *Drainage for Areas Other Than Airfields*, TM -814-2, AR 200-1, and AR 420-1.

C.5.14.2. Ensure drainage ditches, culverts, catch basins and detention basins are clear of debris enabling free flow of runoff.

C.5.14.3. Inspect and maintain all storm-water systems and components annually.

C.5.14.4. Inspect culverts, drainage ditches and catch basins for proper flow of storm/surface water. This includes all storm ditches and subsurface storm collection piping in the cantonment area, BAAF, the main ditches in the housing areas and openings under all bridges.

C.5.14.5. Storm catch basins shall be cleaned if the storm water does not flow freely from the catch basin. The ditch on the input and output side, shall be cleaned to the next culvert or drain, if flow is obstructed or water stands in the ditch line.

C.5.14.6. Clean and grade all storm ditches if flow is restricted or water stands in any point of the storm ditch.

C.5.14.7. If debris, trees, and/or brush are restricting flow of the ditch the debris, trees, and/or brush may be removed to re-establish flow. Trees not restricting flow shall not be removed.

C.5.14.8. Ensure all ditch slopes that are reshaped or cleaned employ erosion protection to prevent slope erosion and excess material from entering the ditch and potentially obstructing the culvert.

C.5.14.9. Maintain the storm sewer collection and treatment system in continuous operation and free of obstructions. Maintain all ditches and culverts no less than 75% free flowing. Ensure ponding does not occur in culverts and drainage ditches.

C.5.14.10. Perform repairs to retention and detention ponds identified on TE 5.14. Receive inspection results monthly from DPW Environmental throughout the year on retention and detention pond condition requiring some level of repair effort.

C.5.14.11. Perform the repairs and maintenance based on results from inspections accomplished by DPW Environmental. Prioritize effort identified on inspection sheets as determined in C.5.1.7 and return inspection sheets with documentation identifying repairs have been completed with a copy of the closed service order to the COR NLT the 5th business day of each month IAW C.8, report C.5.14.11.

C.5.14.12. Perform functions to support health or environmental concerns that may arise, such as but not limited to, the elimination of standing water in response to a West Nile Virus outbreak in any area of Fort Carson or PCMS.

C.5.14.13. Inspect, after storm events, four areas on the perimeter fence that have drainage ditches running through the fence: two on the East boundary, one South of Gate 20, and one on the west side south of Gate 1 as these areas contain re-bar and collect a large amount of debris.

C.5.15. POTABLE WATER SYSTEMS AND DISTRIBUTION

Fort Carson's main potable water system is an extension of the City of Colorado Springs water system. The water utilized in the main water system is purchased from Colorado Springs Utilities. The distribution system handles an average daily flow of 3.2 million gallons per day (MGD). The water storage tanks are filled by a combination of altitude valves, pumping stations and solenoid controlled valves. During periods of demand, water is fed back by gravity from the tanks to the distribution system. The maintenance boundary for the distribution system to Evans Army Community Hospital, Building 7500, begins where the line exits the building. Responsibility in housing areas is limited to main lines and main line equipment.

Water systems and distribution data is located in section C.7, TE 5.10. ~~The Contractor shall provide updates to the COR NLT 31 January of each year IAW C.8, report C.5.10.~~

Fort Carson has 12 individual downrange water systems in this contract.

A description of downrange and other systems is in section C.7, TE-5.15.

C.5.15.1. Water Distribution and Quality

The Contractor shall do the following:

C.5.15.1.1. Operate, maintain and repair all components of Fort Carson's water systems not encompassed within the CSU IGSA, IAW AR 420-1, UFC 3-230-02 *Operation and Maintenance: Water Supply Systems*, Colorado Water Quality Control Commission Regulation 11, Colorado Primary Drinking Water Regulations (5 CCR 1002-11), DPW Installation Design Guide and manufacturer's specifications.

C.5.15.1.2. Maintain water quality throughout the distribution system to meet current standards of the CDPHE, federal and local standards. Water line disinfection is identified in these standards.

C.5.15.1.3. Respond to all water line breaks as a priority one and notify the COR, DPW-E, and Environmental Health within one hour of response. Submit a water line repair report to the COR NLT 24 hours after repair completion IAW C.8, report C.5.15.1.3. For State standards, the STP laboratory will be certified to conduct Total Coliform and E-coli analyses.

C.5.15.1.4. Operate and maintain potable water systems at a standard to provide adequate quantities of water at adequate pressures. Use the 2013USACE Water Study as a guide for pressure settings (available from the DPW Technical Monitor).

C.5.15.1.5. Conduct inspections daily during the work week of all water storage tanks and record levels in the daily log. Maintain 75% capacity or greater at all times in storage tanks for fire protection. Ensure the access doors to these tanks remain locked to prevent vandalism and contamination.

C.5.15.1.6. — Check the main Pressure Reducing Valves (PRVs) (#2, #18 and Gate #1) and the two altitude valves daily, during the work week, for proper operation and downstream pressure setting.

C.5.15.1.7. — Check remaining PRV's on a quarterly basis and recorded in the daily log. Ensure PRV pits are clean, free of debris, and locked. Ensure there are no leaks or water in the pits. Ensure piping and valves are kept painted with corrosion resistant paint.

C.5.15.1.8. — Perform water quality flushing annually IAW UFC 3-230-02 *Operation and Maintenance: Water Supply Systems*. This effort shall be included in the required maintenance plan for C.5.15. with an annual report submitted to the COR NLT 31 January IAW C.8, report C.5.15.1.8.

C.5.15.1.9. Provide to the Government utility drawings according to the procedures of PWS Section C.5.1.14 and C.5.1.15 when the Contractor makes changes.

C.5.15.1.10. Monitor water system operation using the Supervisory Control and Data Acquisition system (SCADA). Assure normal system parameters IAW manufacturer's operating manuals by reacting accordingly to alarms and abnormal data readings.

C.5.15.1.11. Provide certified operators in compliance with the CDPHE Regulation 100, 5 C.C.R. 1003-2 and maintain certification records.

C.5.15.1.12. Prepare daily operating logs per C.7, TE-5.15.1.12, and submit to the COR NLT 5th day after receiving result in February, May, August, and November IAW C.8, report C.5.15.1.12.

C.5.15.1.13. Submit monthly well water reports to the COR NLT 1st business day of each month IAW C.8, report C.5.15.1.13. An example well report can be found in C.7, TE-5.15.1.13.

C.5.15.1.14. Maintain daily water distribution meter readings and usage data along with gate, butterfly and pressure reducing valve operation records for the duration of the contract and make available to the government upon request.

C.5.15.1.15. Submit monthly CDPHE reporting form for bacteriological analysis on the Turkey Creek Recreational Area to the government as part of report C.5.15.1.13.

C.5.15.1.16. Prepare and submit test data required by the CDPHE Yearly monitoring schedule as changes occur from the state, provided by the COR. Submit to the COR the following reports IAW C.8: Quarterly test data report C.5.15.1.16 NLT 1st working day after receiving results in February, May, August, and November and 3 year testing, report C.5.15.1.16-A NLT 1st working day after receiving results. Next 3 year test is due between 1 June and 30 September 2019.

C.5.15.1.17. — Perform seasonal inspections on the HTHW Distribution System 1 November

through 30 November and submit an annual inspection report to the COR NLT 31 December of each year IAW C.8, report C.5.9.1.2.; maintain documentation of all inspections and repairs made for the duration of the contract and make available to the COR upon request.

C.5.15.1.18. Be familiar with the requirements in the U.S. Army Public Health Command *Water System Emergency Response Plan* for responding to potable water concerns

C.5.15.2. — Fire Hydrants

C.5.15.2.1. — Perform fire hydrant flow testing annually for all areas not included in the CSU IGSA IAW UFC 3-230-02 *Operation and Maintenance: Water Supply Systems* unless otherwise directed by the Government. Provide the completed report to the COR NLT 5 November annually IAW C.8, report C.5.15.2.1.

C.5.15.2.2. — Maintain maintenance records for the duration of the contract for each fire hydrant. Include fire hydrant location, manufacture, model, and install date. Make the records available to the Government upon request.

C.5.15.3. Backflows

C.5.15.3.1. Cross Connection/Backflow Preventer Program: manage the Fort Carson backflow prevention program.

C.5.15.3.2. Annually test backflow preventer devices including but not limited to the devices identified in section C.7, TE-5.15.3.2. Also, test new backflow devices for compliance immediately after they are installed or discovered IAW regulations specified by the UFC 3-230-02 and TB MED 576 Sanitary Control and Surveillance of Water Systems at Fixed Installations.

C.5.15.3.3. Ensure testing is performed by a person certified by the State of Colorado as a General Tester. Keep up to date records of tests, repairs, and inspections for the duration of contract performance. Submit a backflow test report to the COR NLT 5th business day monthly IAW C.8, report C.5.15.3.3.

C.5.15.3.4 Perform the Post Cross Connection Survey. A cross connection is a connection between potable water supply and a non-potable source, where a contaminant could potentially enter the drinking water supply. This survey must be performed on every facility on Fort Carson with the exception of Fort Carson Family housing, Evans Army Hospital, COARNG leases and any MEDCOM controlled facility. The excepted facility backflow devices shall be added to TE-5.15.3.2 and identified as “maintained by others”.

C.5.15.3.5 Conduct Annual Cross Connection Survey within CDPHE & EPA Guidelines, Submit to the COR by 1 Oct. report C.5.15.3.5

C.5.16. OPERATION OF THE CLEAN FILL SITE

The Clean Fill site is located near the intersection of Butts Road and Route 5A. No contaminated material will be placed in this site. Other contractor's may place material in or remove material from this area with approval of the KO.

The Contractor shall do the following:

C.5.16.1. Operate and maintain the Clean Fill Site per the State of Colorado Solid Waste Laws and Regulations and DPW Clean Materials Area Utilization Policy, Revision 1 dated 9 April 2012.

C.5.16.2. Control access to the Clean Fill Site at all times. Ensure no material is placed in the site without Contractor oversight.

C.5.16.3. Permit access during normal business hours only to authorized persons and activities.

C.5.16.4. The site shall be secured and access gates locked at times when facility is not in operation.

C.5.16.5. Erosion BMP's must be regularly maintained to control stock pile run off.

C.5.16.6. Utilize the rock screen as necessary to segregate fines and unwanted material/debris reuse of rock. Track usage of the rock screen and provide a report to the COR NLT the 5th business day monthly IAW C.8, report C.5.16.6.

C.5.17. WAREHOUSING AND SUPPLY

Manage and control of warehousing, storage, supply and material issue and ordering of supplies and materials for use within the scope of this PWS.

The Contractor shall do the following:

C.5.17.1. Establish a supply system that includes the following:

- Determination of stock levels and stock control of inventory;
- Receipt, notification of receipt to the end user;
- Storage locations, warehousing and security procedures. As part of the discussion of storage locations address the requirement for warehouse storage and the concept for the use of shop and bench/truck stock;
- Property accountability, records maintenance, inventory, issue, transportation, delivery, disposition, management of the supplies, parts and materials; and
- Best value by competitive bidding.

C.5.17.1.1. Purchase maintenance, repair or replacement items. Contractor shall obtain KO approval prior to purchase IAW FAR contract terms and conditions. Notify the KO in writing prior to any subcontracts (ref C.5.1.4) and any tool purchase with the extended cost in excess of \$2500 per individual item. The Contractor may be required to purchase replacement equipment items with KO approval and following applicable FAR requirements.

C.5.17.1.2. Maintain a stock of parts and materials in a warehouse to perform maintenance and repair to real property and systems described in this PWS.

C.5.17.1.3. Ensure all material is charged to appropriate DMO or PMO when issued.

C.5.17.1.4. Obtain appropriate user roles in DLA's Account Management and Provisioning System (AMPS) for ordering fuel in performance of this PWS. Order fuel for AST's located at 8200, 1860B tank farm, PCMS fuel oil boilers and equipment fuel. Provide notification to the COR when fuel is ordered and receipts of delivered fuel to the COR by end of day the fuel is delivered.

C.5.17.2. Facility Management Warehouse Supplies

IAW DPW facility managers program Soldiers are authorized to make minor repairs to their facilities utilizing the supplies in the facility manager warehouse.

The Contractor shall do the following:

C.5.17.2.1. Procure and provide materials in support of the DPW Facility Management program. A standard material inventory is identified in section C.7, TE-5.17.2.1. Stock and replenish the inventory so material items will not be out of stock. Issue materials to authorized personnel identified on a DA Form 1687, supported by an Assumption of Command Memorandum.

C.5.17.2.2. Reserved

C.5.17.2.3. Track these costs and supply the Government with quarterly reports.

C.5.17.2.4. Quantity of items that can be issued will be identified and provided to the Contractor. The KO shall approve changes or make exceptions to the quantities issued for troop support issue items.

C.5.17.2.5. Issue ice melt, snow shovels, and light bulbs to facility occupants without a DA Form 1687. However, issuance of these items shall be tracked as with other items.

C.5.17.2.6. Issue Weed control products to authorized Facility Managers as detailed in C.5.4.11.

C.5.18. CORROSION CONTROL AND CATHODIC PROTECTION:

The Contractor shall assure corrosion control and cathodic protection of Fort Carson and the Pinion Canyon Maneuver site assets. Work shall include but not limited to corrosion control of metallic natural gas lines, water lines, high temperature hot water lines (HTHW), well casing and steel water storage tanks. These systems can consist of galvanic or impressed current systems, rectifier's protective coatings, underground valves, fire hydrants and any underground metical utility they can also include remote monitoring equipment.

The Contractor shall do the following:

C.5.18.1. Establish a maintenance plan for the implementation of a corrosion control program IAW Public Works Technical Bulletin 420-49-29 *Operation and Maintenance of Cathodic-Protection (CP) Systems*.

C.5.18.1.1. Take prompt action to correct deficiencies in cathodic protected structures to obtain proper corrosion protection within 60 days of identifying inadequate corrosion protection. Notify the COR of any system were proper protection cannot be achieved.

C.5.18.1.2. Ensure that corrosion control and cathodic protection procedures, including those of operation, maintenance, troubleshooting and testing are performed under a person accredited by the National Association of Corrosion Engineers (NACE) with knowledge of CP theoretical concepts, applications and data interpretation.

C.5.18.1.3. Maintain detailed record for all cathodic protected structures, test stations, rectifiers and other applicable system components. Maintain the records in an electronic data base format. Include in the records all work completed on the cathodic protected structures to include dates and locations, type of work completed, findings and all test results. Also include structure ID/Asset names and GPS coordinates for all above ground structures and equipment and any anode bed locations.

C.5.18.2. Natural Gas System Corrosion Control

Ensure the corrosion control methods implemented for natural gas distribution meet requirements of the Department of Transportation (DOT) Pipeline Hazardous Materials Safety Regulations, Part 192.451 thru Part 192.491 Requirements for Corrosion Control.

C.5.18.2.1. Submit to the COR, a monthly report outlining cathodic protection work that has been completed on the natural gas system to the COR NLT 5th business day monthly IAW C.8, report C.5.18.2.1.

C.5.18.2.2. Perform an annual cathodic protection survey of the natural gas distribution system IAW DOT Part 192.451 thru Part 192.491. Provide a report of the annual survey to the COR NLT 15 January annually IAW C.8, report C.5.18.2.2.

C.5.18.2.3. Check gas system rectifiers monthly for proper operation, readings or changes and keep a record of the results and findings in the cathodic protection data base outlined in Section C.5.18.1.3.

C.5.18.3. Water Distribution Systems Corrosion Control

Implement and maintain corrosion control methods for water distribution, steel water tanks and wells IAW requirements of UFC-3-230-02.

C.5.18.3.1. Maintain records identifying cathodic work completed on the water systems and storage tanks for a period no less than five years and make the records available to the Government upon request.

C.5.18.3.2. Check water system rectifiers monthly for proper operation. Keep a record of results and findings in the cathodic protection data base outlined in Section C.5.18.1.3.

C.5.19. STRUCTURAL REQUIREMENTS, MAINTENANCE AND REPAIR

Structural requirements, maintenance and repairs include all Fort Carson and Pinion Canyon Maneuver Site facilities designated as real property. Repairs and maintenance shall include all structural aspects of the building. Locks and keys for barracks mailboxes are not part of the PWS. All structures and facilities identified as Real Property in section C.7, TE-5.2.2.2-A are applicable except those annotated in section C.7, TE-5.2.2.2-B as Non Real Property.

C.5.19.1. Facility Maintenance

A comprehensive facility maintenance program as described in TM 5-610 Preventive Maintenance for Facilities Engineering, Buildings and Structures shall be developed and maintained as part of the maintenance plan identified in C.5.19. This program shall be cyclical to prevent premature failure of facility components and systems. The expectation is facilities in the Fort Carson and PCMS Cantonment areas have this preventive effort performed. This effort shall also include other trades, i.e. plumbing, electrical, HVAC, etc. Facilities may be removed from or added to the program in coordination with the KO. Preventive maintenance of installed building equipment and systems is addressed in other sections of this PWS.

The Contractor shall do the following:

C.5.19.1.1. Incorporate the buildings identified in C.7, TE-5.19.1.1 into a regular facility maintenance schedule rotation as described in TM 5-610 Preventive Maintenance for Facilities Engineering, Buildings and Structures. Submit the initial schedule to the COR NLT 30 days after contract start and 10 days before change implementation IAW C.8, report C.5.19.1.1.

C.5.19.1.2. Coordinate with the facility manager(s) of the respective buildings to complete a walk through and identify the deficiencies that need to be addressed on a maintenance sheet. Coordination with the facility manager(s) will be critical during the facility maintenance process.

C.5.19.1.3. Ensure all tasks identified during the walk through are completed during the current cycle. Maintain records of work performed under this task for a period no less than 24 months. These records must be available for Government review upon request.

C.5.19.1.4. Ensure separate DMO's are opened for those items that are identified as needing repair during the walk-through.

C.5.19.1.5. Determine labor hour estimates for each identified deficiency on the list prior to starting effort and obtain Government approval of identified work no less than three work days prior to starting. Provide actual time spent on each deficiency to validate time spent in respective building. This will be used to determine the amount of time needed to complete the maintenance in the respective facility and cycle.

C.5.19.1.6. Use this same process in C.5.19.1.5 when performing facility maintenance on facilities that are being turned in from the unit. This effort shall be coined as "Turnover Maintenance". Anticipate approximately 30 facilities each year requiring this effort on any facility type. Receive the requirement from the KO when effort is needed.

C.5.19.1.7. Ensure all maintenance actions for a respective facility are captured on one DMO during the respective cycle.

C.5.19.2. Facility Repairs

Response to repairs will be in accordance with priority system identified in C.5.1.7. These repairs will consist of failures identified by facility occupants and called into the customer service area. Repairs may include any failure of real property and all necessary efforts after a water leak resulting from broken water pipes, overflowing bathtub, or other water sources. Barracks DMO's will be ahead of other competing facilities.

The Contractor shall do the following:

C.5.19.2.1. Accomplish authorized repair work on specified areas identified in Section C.5.1.7 of this PWS. Repairs to barracks rooms resulting from water damage will have emphasis above other DMO's.

C.5.19.2.2. Show repair status of affected water damaged barracks rooms, by building with percentage complete, estimated completion date, and potential issues in the weekly meeting.

C.5.19.2.3. General Interior Work for Repair and Maintenance

C.5.19.2.3.1. Repair or replace damaged or deteriorated flooring, to match existing material, style and color. All repairs shall be professionally completed.

C.5.19.2.3.2. Repair or replace damaged and deteriorated walls, ceilings, and related trim matching existing material, surface, texture, and color. Remove as necessary any wood trim and ceiling fixtures to provide access to the damaged area.

C.5.19.2.3.3. Reinstall fixtures and trim, ensure nails are set and filled, and items repainted or refinished to restore them to their original condition upon completion. When removing wall or ceiling coverings, inspect exposed structural components and notify the COR immediately of deficiencies requiring engineer support before proceeding.

C.5.19.2.3.4 Mitigate water as needed to prevent further damage, mold growth and facilitate repairs. This may involve but is not limited to, opening enclosed walls, removing cabinets, employing dehumidifiers and fans, removing cove base and drilling holes in drywall.

C.5.19.2.3.4. Maintain or repair all doors to operate smoothly without binding or sticking. Damaged, deteriorated, or missing doors and associated hardware shall be repaired or replaced as identified to match surrounding or original type, style, and finish. Install door hardware from the damaged door unless the hardware is not repairable. Fill and finish small holes or scars in door face to match surrounding door surface. Install doors to provide 1/8-in maximum clearance between door and jamb after painted or sealed and not scrape the floor. Make necessary adjustments to doors out of alignment with the doorframe so the door fits squarely in the frame and operates freely. Install weather stripping on exterior doors when replaced or as identified.

C.5.19.2.3.5. Repair stair treads, risers, stringers, handrails, brackets, and other components identified as failed. Ensure repaired components are refinished to match existing or surrounding like items. Remove trim items susceptible to damage during the repair activity and reinstall upon completion of the repair activity.

C.5.19.2.3.6. Repair or replace window coverings which include curtains, drapes, vertical and horizontal blinds. Ensure drapery rods and blinds do not sag when properly installed. Secure loose brackets, replace broken cords, and broken or missing drapery slides. Install additional support brackets to support sagging rods. Ensure blinds operate smoothly and block out light from the exterior when installed.

C.5.19.2.3.7. Repair or replace damaged or deteriorated cabinets, shelving, and countertops as required. Replace missing or inoperative hardware. Ensure replacement cabinets and countertops match existing in color, style, material and finish.

C.5.19.2.3.8. Repair or replace damaged, inoperative, or missing interior accessories including, but not limited to, paper holders, soap trays, dispensers, towel bars, shower curtain rods, medicine cabinets, mirrors, and doorstops. Ensure damaged or missing items are replaced with items matching the original. Hardware items requiring lubrication shall be lubricated and restored to an operable condition.

C.5.19.2.3.9. Repair or replace as needed all structural components of unit mailboxes located in barracks areas.

C.5.19.2.4. General Exterior Work for Repair and Maintenance

The Contractor shall do the following:

C.5.19.2.4.1. Repair damaged or deteriorated wall areas to restore to a serviceable, structurally sound, and watertight condition. This work includes, but is not limited to: replacing damaged masonry units, repairing failed expansion joints, tuck pointing loose or eroded mortar joints, and sealing penetrations in wall openings; replacing damaged or deteriorated structural members. Repair or replace, to match existing, damaged; siding, exterior trim, miscellaneous hardware items, and removal of graffiti or other defects as identified.

C.5.19.2.4.2. Repair exterior trim, including all exterior moldings, millwork, shutters, soffits, fascia, and cornices as required. Properly prepare surfaces of sealant and paint build-up as needed to ensure proper trim installation. Replace damaged or deteriorated insulation board, sheathing or underlayment with material of the same type, thickness, and quality. Ensure bird screens and soffit vents are intact and free of corrosion and missing pieces. Prime and paint all wood trim items prior to installation.

C.5.19.2.4.3. Repair roof leaks and identified roof defects to prevent interior damage. Respond to roof defects causing leaks onto electrical and computer as a priority 1 to prevent electrical hazards and damage to computers. Ensure roof defects are repaired urgently to prevent further damage. Make repairs to interior damage caused from roof leaks as incidental to the roof leak. If the Contractor identifies recurring roof leaks requiring roof replacement, they will submit applicable information to the COR so the replacement can be programmed in the DPW annual work plan.

C.5.19.2.4.4. Repair gutters and downspouts that are broken, damaged, misaligned, or leaking. Limit gutter replacement to affected area only and match original gauge, type of material and finish. Replace missing or broken gutter and downspout components, to include splash blocks. Ensure splash blocks are properly positioned to receive the impact of water, allowing water to drain away from building.

C.5.19.2.4.5. Repair damaged exterior concrete surfaces such as patios, sidewalks, and steps so they are structurally sound, at original alignment and grade, and are free of cracks that may cause a tripping hazard. Remove plant roots and backfill the area where they cause or contribute to concrete damage. Masonry fences, planters, and steps shall be repaired to replace missing or broken masonry units.

C.5.19.2.4.6. Repair or replace damaged or deteriorated wood and metal stairs and stairways, including treads, risers, and other stair components as identified.

C.5.19.2.4.7. Ensure doors (including storm doors), windows (including storm windows), and screens operate smoothly without binding or sticking in accordance with the manufacturer's design. Repair or replace damaged, deteriorated, or missing doors, windows, and screens, and associated components as identified. Use metal doors when replacing an exterior wood door unless matching an architectural feature or existing jamb will not support a metal door. Ensure caulking, glazing, and weather stripping is fully intact to maintain a completely weather tight seal. Ensure replacement glass is of the same size, type, color, and quality as the existing glass.

C.5.19.2.4.8. Ensure exterior doors are installed during the same workday as removal of original door. Fill and finish small holes or scars in door face to match surrounding door surface. Reinstall removed hardware from the damaged doors unless the hardware is un-repairable or incompatible. Replace cracked and broken glass in doors with the same quality (unless safety or security glass is required), type, and size. Install doors to provide 1/8-in maximum clearance between door and jamb after painted or sealed and will not scrape the floor when in operation. Make necessary adjustments to doors out of alignment with the doorframe so the door fits squarely in the frame and operates freely.

C.5.19.2.4.9. Adjust, repair, or replace damaged, inoperable, or missing hardware including, but not limited to; hinges, locks, striker plates, latches, keepers, window operating mechanisms, and door closures. Ensure replacement hardware matches existing in type, size, quality, and finish. Ensure hardware is installed in accordance with the manufacturer's recommendations and does not hinder proper operation.

C.5.19.2.4.10. Ensure overhead, hangar or rolling doors are properly operating. Perform cyclical maintenance, annually to ensure proper operation and preclude failure. Maintain an inventory of repair parts for existing overhead, hangar and rolling doors.

C.5.19.2.4.11. Perform inspections, maintenance, and repairs to playground equipment at CDC's and common use parks. Inspect playground equipment according to Certified Playground Safety Institute guidance with a frequency of monthly visual inspections and an annual in depth inspection. Maintain all records for COR review.

C.5.19.2.4.11.1. Provide results, by location, of visual inspection and annual inspection to the COR NLT 5th business day monthly IAW C.8, report C.5.19.2.4.11.1. Repairs to playground equipment must meet manufactures recommendations. Maintain documentation of all inspections, maintenance, and repair of equipment for a period not less than five years.

C.5.19.3. General Requirements for Painting

Painting is typically incidental to other work or spot repairs on both the interior and exterior of all types of surfaces on buildings and structures, as well as painting of other miscellaneous items such as signs, parking bumpers, and railings. Wholesale painting of building exterior, interior, to include individual offices and individual barracks rooms is not part of this PWS. Painting will include at a minimum; proper surface preparation, necessary caulking, and protection of items and areas not to be painted. Paint being used must match surrounding painted surfaces in color and sheen. Paint and paint preparation products used shall contain low VOC/HAP, non-asbestos, non-lead content.

The Contractor shall do the following:

C.5.19.3.1. Perform touch-up painting correcting minor defects in interior and exterior surfaces (i.e., filling and painting of nail holes, nicks, and scrapes; painting over graffiti, marks, and scuffs) to include walls, ceilings, and other surfaces.

C.5.19.3.2. Protect all furnishings, equipment, floor coverings, and other surfaces that are not to be painted by carefully moving, covering, or masking items or areas prior to surface preparation and painting. Return to its original position, all removed items, furnishings and other property. Ensure painted items such as windows, doors, and cabinets operate smoothly without binding.

C.5.19.4. General Requirements for Welding and Metal Working

This includes maintenance and repair of metal components, installed equipment such as exhaust fans, metal transmission pipe and repairing of metal components in support of other repair activities.

The Contractor shall do the following:

C.5.19.4.1. Perform metal working requirements to maintain, repair, or replace metal components, including but not limited to, air ducts, vehicle exhaust systems, structural members, pressure piping and signs.

C.5.19.4.2. Perform all types of welding and brazing in the accomplishment of maintenance and repair activity in the performance of this PWS. Perform welding on a variety of metals, in a variety of welding positions. Ensure welders are certified for the specific welding process in accordance with applicable American Society of Mechanical Engineers (ASME), ANSI, and American Welding Society (AWS) standards. Obtain a welding or burn permit before proceeding with welding, burning, or other open flame work.

C.5.19.4.3. Perform tasks such as drilling, tapping, boring, reaming, and grinding on a variety of materials such as steel, cast iron, stainless steel, aluminum, copper, brass, bearing bronze, and manganese. Install equipment requiring alignment of motors, pumps, blowers, and may be required to fabricate or modify brackets, frames or supports in support of other repair activity.

C.5.19.5. Locksmith Services

Locksmith services include, but are not limited to, maintaining accountability of master keys, providing technical assistance, and working on locks and vaults. Mailbox keys and cores are not included as part of any key effort within this document without prior approval from the KO.

Fort Carson uses a Master Key System database for key and lock interchangeable coring (IC). Stand-alone electronic locks may have one master combination assigned and maintained by DPW. High security areas may use an independent door key, coring, or code system. Only the Fire Department and MP's are authorized to possess a set of master keys. Door locks and keys

on all medical and dental clinic buildings are not included in this contract. All maintenance responsibilities for these doors will be under the Hospital Maintenance contract.

C.5.19.5.1. Master Keys, Duplicate keys and Locks

Master keys shall be to the building level only. Keying of interior doors shall meet physical security requirements as provided by Directorate of Emergency Services (DES). There are three lock systems used on Fort Carson, Best, Medco and ONITY. Best and Medco both use proprietary keyways for Fort Carson. Onity is a hotel style key card lock system used in about 12 barracks buildings with about 5500 doors. Customers submit requests for addition key duplication. Rekeying of buildings will not occur unless directed by the KO.

The Contractor shall do the following:

C.5.19.5.1.1. Accept key request letters directly from the customer and cut keys on a service order and keep in a file system as determined by the Contractor. Ensure the key request letter is electronically signed by the customers' Commander or First Sergeant, and accompanied by either a FLIPL or Statement of Charges. Key requests that exceed individual requests (i.e. whole facilities) must be approved by DPW. Commanders may request additional keys to augment basic issue for operational purposes.

C.5.19.5.1.2. Determine feasibility when the customer requests a priority and document the determination on the request letter. Refer to the COR any customer expressing dissatisfaction with the Contractor's determination.

C.5.19.5.1.3. Do not issue a mechanical room key, master key of any kind, or any other key for O&M Contractor/DPW use, unless approved in writing by the Government.

C.5.19.5.1.4. Have the customer sign for requested key(s) upon receipt of the key(s). The priority level for key requests will be the same as any other service order. Make the priority decision for key requests based on the customer's situation. E.g. the customer is deploying in five days.

C.5.19.5.1.5. Ensure master key and lock combination list are maintained on a stand-alone (non-networked) system that requires an individual to enter their password to gain access.

C.5.19.5.1.6. Maintain master keys and hard key codes in a secure Class 3 GSA approved container.

C.5.19.5.1.7. Not disclose keying codes to anyone unless approved by the Government.

C.5.19.5.1.8. Make appropriate lock adjustments when notified by the government security is compromised to a building or room. This may include re-coring a single door up to a whole bldg. and cutting necessary keys for the occupants. Provide a cost report to the COR for these requirements after work is completed.

C.5.19.5.1.9. Install or replace devices to increase security or safety requirements. Provide keys for new devices.

C.5.19.5.1.10 Maintain all Onity locks. Battery replacement is required annually. Assign new key cards between one and two years based on system operation limits. Coordinate with the barracks managers when this effort is required and plan at least 60 days in advance.

C.5.19.5.1.11 Keys shall not open more than one bedroom. Master key cards for barracks with Onity locks shall be to the floor level. Master keys shall only be issued to barracks managers.

C.5.19.5.1.12. Maintain the Onity computer database which stores the coding for each key card. Ensure this remains secured to prevent unauthorized key manufacturing. Do not write building or room number on the plastic key card when issuing new keys. Only use HiCo magnetic stripe cards for these locks.

C.5.19.5.2. Electronic and Computerized Locks

Electronic and computerized locks include, but are not limited to: keyless lock devices, secured access systems (Lenel) and their components, auditable combination locks, closed circuit systems, panic hardware, cipher locks, mortise locks, tubular locks, detention locks, and timed, auditable, and low voltage devices.

The Contractor shall do the following:

C.5.19.5.2.1. Repair, replace, maintain and operate electronic and computerized locks and components including, but not limited to, cores and user cards, batteries, circuit cards, cylinders, deadbolts, and latches IAW O&M manuals and manufacturer's specifications.

C.5.19.5.2.2. Locks shall function as designed by the manufacturer, be fully operational, and provide the intended level of security. External locks shall meet physical security requirement.

C.5.19.5.3. Vaults

Vault components include, but are not limited to, SCIF door locks, cores, mechanical combination locks, vault or container locking mechanisms, deadbolts, throw bolts, rim latches, panic hardware, doors, metal plates, security rings, and hinges.

The Contractor shall do the following:

C.5.19.5.3.1. Remove, repair, and replace vault components IAW AR 380-5 *Department of the Army Information Security Program*, O&M manuals, and manufacturer's specifications. Neutralize, repair, and set or change lock combinations on IAW Federal Standard 809B.

C.5.19.5.3.2. Locks shall function as designed by the manufacturer, be fully operational, and provide the intended level of security.

C.5.19.5.3.3. Provide training to units on changing combinations to arms room locks

C.5.19.5.4. Mechanical Locks

Mechanical locks include deadbolt, mortise, cabinet, pad, cipher, prison, arms room locks and others as directed by the Government.

The Contractor shall do the following:

C.5.19.5.4.1. Repair, replace, and maintain mechanical locks and associated components IAW O&M manuals and manufacturer's specifications.

C.5.19.5.4.2. Ensure locks function as designed by the manufacturer and are fully operational.

C.5.19.6. Signs

Signs include exterior facility identification, building number, directional, and regulatory (traffic) signs as stated in the Installation Design Guide (IDG). Interior signage is the facility occupant's responsibility.

C.5.19.6.1. Facility Signs

The Contractor shall do the following:

C.5.19.6.1.1 Accept requests for new signs and validate the request meets the IDG requirements. Take photos and provide power point renderings to reduce confusion on the sign requirement between the customer and installer.

C.5.19.6.1.2. Obtain clarification from the Government on any sign request that isn't clearly covered in the IDG. The average workload is expected to be 12 to 15 new sign requests per week. The actual quantity will vary.

C.5.19.6.1.3. Ensure new or replaced signs meet the design requirements for color, material, letter font and verbiage requirements of the IDG.

C.5.19.6.1.4. Re-lettering/re-wording of existing signs is classified as repairs and will follow requirements of C.5.1.7.

C.5.19.6.1.5. Continuously perform inspection of all installed signs identifying those that are failed for repair following requirements in C.5.1.7.

C.5.19.6.1.6. Replace each word of the month sign on the 1st work day of each month. The signs are located at Gates 1, 4, and 20. Coordinate the word list with 4th Infantry Division, Equal Opportunity Advisor by 15 December of each year for the following year.

C.5.19.6.1.7. Install/remove and update Combined Federal Campaign (CFC) signs annually. Actual date of installation/removal will be provided by the COR.

C.5.19.6.2. Regulatory, Downrange and Other Signage

The Contractor shall do the following:

C.5.19.6.2.1. Provide preventive maintenance and repairs to all culvert markers, regulatory and other required street markers and signs, to include street name signs in the cantonment area and downrange.

C.5.19.6.2.2. Conform all signage to Manual on Uniform Traffic Devices (MUTCD) and the Fort Carson Installation Design Guide. Restore all signs to original condition or replace with equivalent. Ensure culvert markers are securely anchored perpendicular to the ground and shall be five feet from the top of the marker to the ground.

C.5.19.6.2.3. Repair or replace signs within designated times from notification of deficiency or discovery on inspection: (regulatory signs) four hours; (warning signs) one day; (guide signs) seven days; (culvert markers) one month; (other downrange signs) seven days; and (street name signs) seven days.

C.5.19.6.2.4. Erect a temporary sign if there may be a life/safety issue, such as a “stop” sign, or school crossing sign.

C.5.19.6.2.5. Install replacement or new regulatory signs on the cantonment or downrange as identified.

**PERFORMANCE WORK STATEMENT
SECTION C.6
PUBLICATIONS INFORMATION**

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- C.6.1. Mandatory and Advisory Classification
- C.6.2. Provisions
- C.6.3. Reference Documents

C.6.1. Mandatory and Advisory Classification

The publications and forms listed in this Section shall be considered mandatory. The Contractor is obligated to comply with these publications and forms in their entirety.

C.6.2. Provisions

Cited references that are statutes include all regulations issued under their authority. Cited references that are federal statutes also include all state statutes and regulations issued under their authority.

The Contractor shall ensure that all publications are posted and up-to-date. Upon expiration or termination of the contract, the Contractor shall return to the Government all issued publications. The Contractor shall accomplish the tasks set forth in this contract IAW the reference documents listed herein.

Any task set forth in any such reference, which shall call for the exercise of discretionary Government authority that cannot be delegated, shall be subject to the final approval of the Government official having such authority.

All publications and forms shall be current issue.

The Contractor shall immediately implement changes to publications that result in a decrease to or no change in the contract cost. Prior to implementing any revision, supplement, or amendment that may result in an increase in costs, the Contractor shall submit a cost estimate to the KO and obtain approval. Said estimate shall be submitted within 30 days from the date the Contractor receives notice of such change, revision, supplement, or amendment, giving rise to the increase in cost of performance.

C.6.3. Reference Documents

Number	Title
7 USC 136-136y	Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA)
16 USC 470 et seq.	National Historic Preservation Act (NHPA)
16 USC 470aa-470mm	Archeological Resources Protection Act (ARPA)
16 USC 703-712	Migratory Bird Treaty Act (MBTA)
16 USC 1531-1544	Endangered Species Act of 1973 (ESA)
33 USC 1251-1387	Federal Water Pollution Control Act (FWPCA) a/k/a Clean Water Act (CWA)
42 USC 4321-4347	National Environmental Policy Act
42 USC 6901-6992k	Resource Conservation and Recovery Act (RCRA)

Number	Title
42 USC 7401-7671	Clean Air Act (CAA)
29 CFR 1910	Occupational Safety and Health Standards (OSHA)
29 CFR 1926	Safety and Health Regulations for Construction (OSHA)
	Manual on Uniform Traffic Devices (MUTCD) CO State Department of Transportation
Federal Register	Environmental Protection Agency, Part II
5CCR 1002-11	Colorado Primary Drinking Water Regulation
5CCR 1003-5	Colorado Swimming Pool and Mineral Bath Regulations
5CCR 1003-2	Water and Wastewater Facility operator Certification Requirements
	Colorado Water Quality Control Commission Regulations
5 CCR 1002-84	Colorado Reclaimed Water Control Regulation
EO 13423	Strengthening Federal Environmental, Energy, and Transportation Management
NFPA 10	Standard for Portable Fire Extinguishers
NFPA 11	Standard for Low and Medium - High Expansion Foam
NFPA 11A	Standard for Medium and High Expansion Foam Systems
NFPA 54	National Fuel Gas Code
NFPA 58	Liquefied Petroleum Gas Code
NFPA 80	Standard for Fire Doors and Other Opening Protectives
NFPA 90A	Standard for the Installation of Air-Conditioning and Ventilating Systems
NFPA 90B	Standard for the Installation of Warm Air Heating and Air-Conditioning Systems
NFPA 96	Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations
NFPA 101	Life Safety Code®
NFPA 110	Standards for Emergency and Standby Power Systems
NFPA 111	Standard on Stored Electrical Energy for Emergency and Standby Power Systems
AR 25-1	Army Information Technology
AR 25-2	Information Assurance
AR 40-5	Preventive Medicine
AR 70-13	Management and Oversight of Service Acquisitions
AR 95-2	Airspace, Airfields/Heliports, Flight Activities, Air Traffic Control, and Navigational Aids
AR 190-51	Security of Unclassified Army Property (Sensitive and Non-Sensitive)
AR 200-1	Environmental Protection and Enhancement
AR 380-5	Department of the Army Information Security Program
AR 380-67	Personnel Security Program

Number	Title
AR 385-10	Army Safety Program
AR 420-1	Army Facilities Management
AR 530-1	Operations Security
AR 710-2	Supply Policy Below the National Level
AR 735-5	Property Accountability Policies
DoD Instruction 4150.07	DOD Pest Management Program
DODM 4500.36	Acquisition, Management, and Use of DOD Non-Tactical Vehicles
DODM 5200.1	DoD Information Security Program
DOD 5200.2-R	Personnel Security Program
DODI 6055.4	Instruction Traffic Safety Program
DODI 8500.01	Cybersecurity
DOD 5500.07-R	Joint Ethics Regulation (JER)
DOD 8570.01-M	Information Assurance Workforce Improvement Program
MIL STD 188-124B (Notice 4)	Grounding Bonding and Shielding for Common Long Haul/Tactical Communication Systems Including Ground Based Communications-Electronics Facilities and Equipment
	National Board inspection Code (NBIC) parts 1, 2 and 3
	National Board of Boiler and Pressure Vessel Inspectors (NBBI)
Unified Facilities Criteria (UFC) 3-230-02	Operations and Maintenance: Water Supply Systems
Unified Facilities Criteria (UFC) 3-240-03N	Wastewater Treatment Systems Augmenting Handbook
Unified Facilities Criteria (UFC) 3-260-04	Airfield and Heliport Marking
Unified Facilities Criteria (UFC) 3-420-01	Plumbing Systems
Unified Facilities Criteria (UFC) 3-430-07	Operations and Maintenance: Inspections and Certification of Boilers and Unfired Pressure Vessels
Unified Facilities Criteria (UFC)3-600-01	Fire Protection Engineering for Facilities
Unified Facilities Criteria (UFC)3-600-02	Operations and Maintenance: Inspection, Testing, And Maintenance Of Fire Protection Systems
Unified Facilities Criteria (UFC)4-860-03	Railroad Track Maintenance and Safety Standards

Number	Title
DFAS- IN Manual 37-100	The Army Management Structure (AMS)
FAR subpart 45.5	Support Government Property Administration
FAR 52.204-2	Security Requirements
49 CFR-192 thru 195	Pipeline Safety Regulation (Natural Gas)
DoD 4525.6-M	Postal Manual 15 August 2002
FED-STD-809A	Federal Standard Neutralization and Repair of GSA Approved Containers
TB MED 530	Tri-Service Food Code
TB MED 575	Swimming Pools and Bathing Facilities
TB MED 576	Sanitary Control and Surveillance of Water Supplies at Fixed Installations
	Section 106 Cultural Programmatic Agreements with the State of Colorado
	Sacramento State Water Distribution System Operations and Maintenance
	Sacramento State Operation of Wastewater Treatment Plants Volume I
	Sacramento State Operation of Wastewater Treatment Plants Volume II
	Sacramento State Operations and Maintenance of Wastewater Collection Systems Volume I
	Sacramento State Operations and Maintenance of Wastewater Collection Systems Volume II

In Technical Library	
	National Pollutant Discharge Elimination System (NPDES) Permit, STP ,CO-0021181
	Title V Air Pollution Control Operating Permit, renewed July 1, 2007
	MS4 Stormwater Permit, 28 July 2014
	CGP Stormwater Permit, September 2010
	Fort Carson Stormwater program Dewatering Requirements and Guidance
	Best management Practices Operations and Maintenance Plan for Stormwater Management Structures, July 29,2014
IPC	International Plumbing Code 2006
ASME	ASME Boiler and Pressure Vessel Code Sections IV, V, VI, VIII,
ASME CSD-1-2009	Controls and Safety Devices for Automatically Fired Boilers
ANSI A300	American National Standard for Tree Care Operations, Standard Practices

AWWA M14	Recommended Practice for Backflow and Cross Connection Control
In Technical Library	
AWWA M17	Installation, Field Testing and Maintenance of Fire Hydrants
AWWA M23	PVC Pipe-Design and Installation
AWWA M55	PE Pipe-Design and Installation
AWWA C651-92	Standard for Disinfecting Water Mains
	State of Colorado Backflow regulations 5000 Series
PWE CB AIR 003	Fort Carson Fugitive Dust Management Plan (30 March 2012)
NFPA 31	Standard for Installation of Oil Burning Equipment
NFPA 54	National Fuel Gas Code
NFPA 70	National Electric Code
NFPA 70B	Recommended Practice for Electrical Equipment Maintenance
NFPA 70E	Arc Flash Electrical Safety
NFPA 72	National Fire Alarm and Signaling Code
NFPA 211	Standards for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances
NFPA 291	Flow Testing Hydrants
IEEE C2	National Electric Safety Code
	Installation Pest Management Plan, Fort Carson Mountain Post (IPMP) 21 September 2001
DPW SOP #60	Fort Carson Potable Water Hauling
DPW SOP #61	Wastewater Overflow Procedures
DPW SOP #65	Water Damage Protocol for Fort Carson Facilities
	Fort Carson Cross-Connection Plan
	DPW Clean Materials Area Utilization Policy
MAP	Fort Carson Bridge Map and Description
	U.S. Army Public Health Command <i>Water System Emergency Response Plan</i> , August 2009
	Flow Chart Responding to Water Contamination Threat
	RS Means Facilities Maintenance and Repairs Cost Data 2015
	Standard Methods for the Examination of Water and Wastewater
	Fort Carson Installation Design Guide, Spring 2013
DA Pam 420-6	Directorate of Public Works Resource Management System
DA Pam 420-11	Project Definition and Work Classification
DA Pam 710-2-1	Using Unit Supply System (Manual Procedures)
FC Regulation 200-1	Environmental Management and Protection
FC Regulation 200-6	Wildlife Management and Recreation
FC Regulation 210-18	Prohibited & Regulated Conduct
FC Regulation 420-6	Ice Control and Snow Removal Procedures
FC Regulation 420-	Assignment and Utilization of Buildings, Structures and

14	Furnishings
In Technical Library	
FC Regulation 420-17	Repairs and Utilities Procedures and Implementation of Self-Help Program
FC Regulation 420-19	Fort Carson Scheduled Utility Outage Requests (SUOR)
FC Regulation 420-20	Facilities Engineering Fort Carson Dig Permit (FCDP)
FC Regulation 420-22	Administration and Enforcement of Construction Warranties
FC Regulation 420-24	Fort Carson Emergency Gas Plan,
FC Regulation 700-5	Central Vehicle Wash Facility (CVWF)
TM 5-650	Central Boiler Plants
TM 5-662	Swimming Pool Operations and Maintenance
TM 5-814-1	Sanitary and Industrial Wastewater Collection – Gravity Sewers and Appurtenances
TM 5-814-2	Sanitary and Industrial Wastewater Collection – Pumping Stations and Force Mains
TM 5-814-3	Domestic Waste Water Treatment

**PERFORMANCE WORK STATEMENT
SECTION C.7
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PERFORMANCE WORK STATEMENT
Section C.8
Contract Reporting Requirements

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.1.2.2-B	Motor Vehicle Licensing Requirements	Document operating qualification training and provide report to the contracting office before equipment operation by contract employees	15 days prior to Contract start Within 15 days of new hire	KO and COR	Required from all employees: License issue and expiration date, employee name, employee number, CDL type, CDL medical card expiration, Department, Section, GFE licensed to operate and remarks.
C.1.3.1	Phase In Plan	Requirements for smooth transition	NLT 15 days after award	KO and COR	Min requirements listing in PWS
C.1.4.1	Designated Project Manager and Alternate Project Manager	Designate Primary and Alternate Project Managers	5 days prior to Phase in Within 5 business days of change	KO and COR	N/A
C.1.4.1.1	Meeting Minutes	After every meeting between Contractor and Government	Within 3 days of meeting	KO and COR	Date, Location, Subject, Attendees, Discussion points, Decisions, Follow-up
C.1.4.11.1	Organization Chart	Show accurate depiction of organizational structure, personnel assigned and line of supervision	5 days prior to Contract Start Annually by 1 Oct When requested	KO and COR	Show; Management, Lines of supervision, Work areas, Personnel assigned to each area
C.1.4.11.2	Automated Systems Access Roster	Identify personnel needing access to Government network and databases	10 days prior to phase-in Within 5 days of new hire	KO and COR	DPW security manager will assist with CAC application
C.1.5.4	QC Evaluation Meetings	Provide advance copy of slides for QC meeting	7 days prior to quarterly meeting	KO and COR	Government to review and provide comment
C.1.6.1.1.1	Planned Visits	Provide notification of planned visit as stated in C.1.6.1.1.1	Within four hours of notification	KO and COR	Include the name(s), agency(s) of the visitor, reason for visit, and any remarks
C.1.6.1.1.3	Visit Report	Visit Planned or otherwise by regulatory agent or other federal, state, or local official	COB day following visit	KO and COR	Visitor name, agency, reason

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.1.6.2.5	Regulatory Agency Information Request	Information requested by Environmental regulatory agency	Notify 1 day after receipt of request. Provide info within 5 days of request	KO and COR	N/A
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.6.3.4	Report of Findings	Information regarding OSHA citations Unannounced OSHA visit/inspection	Verbal notification within 1 hour Written report NLT COB day following visit	KO and COR	N/A
C.1.6.3.6	Hazardous Condition	Provide report of hazardous condition identified by Contractor or Federal or State agency	Provide written report within 72 hours of identification	KO and COR	N/A
C.1.6.3.8.1	Accident Report	Complete an Abbreviated Ground Accident Report (AGAR)	Verbal notification with 1 hour AGAR report within 3 days	KO and COR	N/A
C.1.6.3.8.2	Third Party Accident Claim	Claims filed by a third party against the contractor.	Report within 24 hours after claim is received	KO and COR	Incident description, contributing factors, Contractor employees involved, management actions
C.1.6.4.1	Hazardous Materials Issued	Inventory of hazardous materials Issued	5 th business day of each Month	KO and COR	DMO/PMO number, Material item, item description, Issued date and qty.
C.1.8.4.1	Estimated Contract Value	Provide an excel spread sheet with applicable information and back-up support	NLT three months prior to end of option period	KO, COR and others provided by COR	See TE-1.8.4.1 for ECV requirements and example
C.1.8.4.2	Outstanding Obligations	Identifies all obligations to date in the current FY. Submit separate report for previous FY	Twice monthly October –June Weekly July – September	KO, COR and others provided by COR	See TE-1.8.4.2 for report requirements and example
C.1.8.4.3	Subcontract Report	Identifies all subcontracts issued for the period	NLT 5 th business day monthly	KO, COR and others provided by COR	Include at a minimum, Subcontractor's name, DMO number and dollar value for KO approval. See TE-1.8.4.3
C.1.8.4.4	Quarterly BPA Report	Report shows used BPS for the period	NLT 5 th business day following end of quarter	KO, COR and others provided by COR	Include BPA holder, effective date, maximum dollar amount and amount used to date

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.1.8.4.5	Burn Rate Report	Develop burn rate based on funding provided	Bi-weekly October –June Weekly July – September	KO, COR and others provided by COR	See TE-1.8.4.5 for report requirements and example.
C.1.8.6.1	Travel Request	Request for travel to PCMS or other TDY locations for training etc.	NLT14 business prior to travel	KO and COR	Include travel mode, travel cost if not Government vehicle. location per diem rate for lodging and meals
C.3.1.5	GFP Master Report	Provide updates/changes to GFP and CAP	NLT 1 st business day of January, April, July, and October	KO, PA and COR	Must be in MS Office excel spreadsheet. No abbreviations; use one tab for GFP and one tab for CAP
C.5.1.4.8.1	Priority 2 and 3 DMO’s affected by Construction wage rate	Provide detailed cost estimate of labor, material and equipment with explanation of work to be performed Work determined by the contractor to fall within this statute	Prior to proceeding with repair work falling into the construction wage determination statute	KO and COR	See PWS paragraph
C.5.1.4.8.1A	Priority 1 DMO’s affected by Construction Wage Rate	Documentation of emergency utility repairs on DMO’s that exceeded 32 hours	1200 business day after utility restored	KO and COR	See PWS paragraph
C.5.1.4.8.2	KO Approved Wage Rate Requirement (construction)	Progress of approved work subject to the construction statute	0900 Monday each week	KO and COR	See PWs paragraph
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.5.1.15	Facility as-built update	Provide redline drawing with detail of incorrect facility or utility drawing/mapping	5 days after identified	COR and others provided by COR	N/A
C.5.1.16	ACM and LEAD sample testing results	Copies of all laboratory material and abetment clearance results and sampling protocol documentation	72 hours after receipt of results	COR and DPW Environmental	N/A
C.5.2.1.7	Non-Duty Trouble Calls	Describe calls received after normal business hours	Monday – Friday by 0730	COR and others provided by the COR	DMO, Priority, facility, description, action taken, and follow-up needed.

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.5.3.1.1	Miscellaneous Reports	Validation reports	2 business days of request	COR and others provided when requesting report	N/A
C.5.3.4.4	Utility Meter Reading	Past month and current month reading of reimbursable customers on master meter spreadsheet	NLT 17 th day of each month	COR and others provided by the COR	Contractor to develop spreadsheet
C.5.4.1.1.1	Seasonal mowing Schedule	Develop seasonal schedule for all mowing requirements	NLT 1 May annually 3 days prior to change	COR and others provided by the COR	N/A
C.5.4.10.9	Tree City USA Report	Provide information on planting, pruning watering and removal of trees	NLT 1 December Annually	COR and others provided by the COR	Requirements listed in PWS C.5.4.10.9
C.5.4.11.4	Employee Certification of Pesticide Application	Copies of employees certification/licenses allowing application of pesticides	1 February Annually 10 business days of new hire	COR and others provided by the COR	N/A
C.5.4.11.8	Pesticide Use Record	Provide information of applied pesticides for the month	NLT 5 th business day of each month	COR and others provided by the COR	using form in C.7, TE 5.4.11.8
C.5.4.11.8-A	Stored Pesticides Report	Inventory of chemicals stored to include Common name, Form, Active Ingredient, Pack, Percent, and Quantity	NLT 5 th business day of each month	COR and others provided by the COR	N/A
C.5.4.11.8-B	Daily Pesticide Report	Report of pesticide product utilized each day	NLT 0800 Monday – Friday for previous days efforts	COR and others provided by the COR	Use form in C.7, TE-5.4.11.8-B
C.5.4.11.10	Out of Cycle Pesticide Request	Request use of pesticide not currently approved	Annually NLT 1 Oct	COR and others provided by the COR	Use Form in C.7, TE-5.4.11.10-A
C.5.5.1.4	Traffic Paint Usage	Report usage of traffic paint for the month. Identify paint brand, type, color and amount used	NLT 10 th business day of each month	COR and others provided by the COR	N/A
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.5.7.1.2.2	Electrical Outage Report	Power outage affecting entire bldg. or large area affecting multiple buildings	Verbal notification within 30 minutes Written report within 12 hours	COR and others provided by the COR	Report must identify; DMO #, facilities affected, circuits,

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
					duration, description, cause, repairs made
C.5.7.5	Generator Hour Report	Report monthly generator run hours	NLT 7 th business day of each month	COR and others provided by the COR	Use Form in C.7, TE-5.7.5.
C.5.7.9.3	Resistive Testing	Report of static ground point and lighting protection System resistance testing	Annually within 5 business days of testing completion	COR and others provided by the COR	Locations and testing standard identified in C.7, TE-5.7.11
C.5.8.1.1.9	Boiler Run Time Report	Report boiler run time on boilers identified in PWS C.5.8.1.1.9.	NLT 5 th business day of each month	COR and others provided by the COR	Example form in C.7, TE-5.8.1.1.9
C.5.8.1.1.11	Boiler Operating Log	Report boiler operation on boilers identified in PW C.5.8.1.1.11.	NLT 5 th business day of each month	COR and others provided by the COR	Example form in C.7, TE-5.8.1.1.11
C.5.8.1.3.5	Opacity Observation Report	Perform Opacity Testing on boilers IAW Operating Permits, for all restarts, and/or 24 hour continuous runs on fuel oil	24 hours after Opacity Test completion	COR and others provided by the COR	N/A
C.5.8.1.5.3	Boiler Inspection Report	Provide copy of completed boiler inspection reports	NLT 5 business days after inspection is completed	COR and others provided by the COR	N/A
C.5.8.2.3.6	Chiller Plant Operating Log	Chiller plant daily operational requirement log	NLT 5 th business day of each month	COR and others provided by the COR	Example form in C.7, TE-5.8.2.3.6
C.5.8.9.4.6	AST Inspection report	Monthly visual inspection of each AST in Contractor's work area	NLT 10 th day of each month	COR and others provided by the COR	Form is located in C.7, TE-5.8.9.4.4
C.5.8.9.4.6-A	AST Ullage log	Stick readings of AST's as required by State of Colorado regulation, 7 C.C.R.1101-14	NLT 10 th day of each month	COR and others provided by the COR	Form is located in C.7, TE-5.8.9.4.6
C.5.8.10.3.3	Energy Reduction Analysis	Identifying energy reduction recommendations to reduce energy consumption peaks which will save energy. Include suggestions on energy saving devices to reduce energy within a facility	NLT 5 th business day of each month	COR and others provided by the COR	N/A

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.5.8.12.4	VTS Inspection , Testing and Maintenance Report	Identify maintenance and repairs performed on VTS's, Include copies of completed testing and certification results for annual and five year requirement.	NLT 5 th business day of each month	COR and others provided by the COR	See PWS section C.5.8.12.4 and C.5.8.12.7.
C.5.9.1.2	HTHW Distribution System Inspection	Perform inspection of all HTHW pits, valves, and check for leaks during the month of November	NLT 31 December of each year	COR and others provided by the COR	N/A
C.5.10	Utility Data Update	Provide updates to items listed in TE 5.10	NLT 31 January Annually	COR and others provided by the COR	N/A
C.5.10.7	Gas Leak Survey	Inspect gas distribution system for leaks as required by regulation	NLT 30 September of each year	COR and others provided by the COR	Reference TE 5.10.7 for survey requirements
C.5.10.8	Gas Leak Report	Provide a report of findings on gas leaks reported.	NLT 24 hours after response	COR and others provided by the COR	Example form is in C.7, TE-5.10.8
C.5.11.2.5	Hazardous Spill Report	Report of hazardous spill incident	NLT 72 hours after spill response	COR and others provided by the COR	Spill report log is in C.7, C.5.11.2.6
C.5.11.2.7	Hazardous Material Issued	Report all hazardous material issued to perform work described in the PWS	NLT 10 th of each month	COR and others provided by the COR	Include: item #, DMO/PMO #, Item description, date issued, and qty. issued.
C.5.12.2.6	CVWF Usage Report	Report usage of CVWF as stated in the PWS	NLT 7 th business day of each month	COR and others provided by the COR	Include unit and number and type of vehicles using CVWF
C.5.13.1.5	Monthly STP Operations Log	Report of STP Operations as required by PWS and operations permit	NLT 7 th business day of each month	COR and others provided by the COR	Operations log is in C.7, TE-5.13.1.5. This is a five sheet log
C.5.13.1.7	Annual Bio-Solids Report	Data on hauling sludge from drying beds	NLT 15 January of each year	COR and others provided by the COR	Include: Qty., percent solids, laboratory analytical report

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.5.13.1.8	Waste Water Test Data	Waste water data required by the NPDES with week and daily	NLT 7 th business day of each month	COR and others provided by the COR	N/A
C.5.13.1.8-A	Monthly Waste Water Test Data	Waste water data required by the NPDES with a frequency of monthly	NLT 1 st working day after results received from laboratory	COR and others provided by the COR	N/A
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.5.13.1.8-B	Quarterly Waste Water Test Data	Waste water data required by the NPDES with a frequency of quarterly.	NLT 1 st working day after results received from laboratory	COR and others provided by the COR	N/A
C.5.13.2.1	Waste Water Disruptions	Number of unplanned disruptions of wastewater service lasting more than two hours	NLT 5 th business day of each month	COR and others provided by the COR	Include: DMO #, date, facilities affected, cause, and action taken.
C.5.14.11	Detention/Retention Pond Repairs	Report repairs made to ponds based on inspection results by DPW Environmental	NLT 5 th business day of each month	COR and others provided by the COR	Include inspection sheets with DMO identifying action to correct failure
C.5.15.1.3	Water Line Repair Report	Submit repair report after each repair	NLT 24 hours after complete repair	COR and others provided by the COR	Include: DMO #, Date reported and completed, location, pipe size and type, Bac-t results, cause of failure
C.5.15.1.8	Water Quality Flushing Schedule	Water quality flushing results	NLT 31 January annually	COR and others provided by the COR	N/A
C.5.15.1.12	Water Report	Water operating log report	NLT 5 th business day of each month	COR and others provided by the COR	Report Example is in TE-5.15.1.12
C.5.15.1.13	Well Water Report	Meter reading of water wells identified on TE-5.15.1.13	NLT 1 th business day of each month	COR and others provided by the COR	Report is in TE-5.15.1.13
C.5.15.1.16	Quarterly Test Data Report	Required CDPH testing for water quality, Disinfection Byproducts. Locations are Bldg. 3900, 2059, 7800, and 9633.	NLT 1 st working day after results received from laboratory in February, May, August and November	COR and others provided by the COR	Test for THHMs and HAA5s

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.5.15.1.16-A	Three year test requirement	Required CDPH testing for water quality, Lead and Copper. Thirty, locations to be provided by the Government.	NLT 1 st working day after results received from laboratory. Next due between 1 June and 30 September 2019	COR and others provided by the COR	Test for lead and copper
C.5.15.2.1	Fire Hydrant Flow Tests	Perform hydrant flow testing IAW UFC 3-230-02 and provide report of effort	NLT 5 November annually	COR and others provided by the COR	Include number of hydrants, results by hydrant number, maintenance requirement due, color of hydrant cap
C.5.15.3.3	Backflow Device Inspection Report	Report all backflow devices inspected with next month's inspection schedule	NLT 5 th business day of each month	COR and others provided by the COR	Include: test date, facility #, device number, pass/fail, repair actions, re-inspect date and result
C.5.16.6	Rock Screen Use Report	Report hours of use by date in the month	NLT 5th business day of each month	COR and others provided by the COR	N/A
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.5.18.2.1	Gas Cathodic Protection Report	Cathodic work performed on gas distribution system	NLT 5th business day of each month	COR and others provided by the COR	Include: test station tested and results, rectifiers tested and results, anodes tested and results, all maintenance /repair work completed and scheduled
C.5.18.2.2	Gas Cathodic Protection Survey	Results of annual survey on gas distribution system IAW DOT Part 192.451 thru Part 192.491	NLT 15 January annually	COR and others provided by the COR	Include: test station tested and results, rectifiers tested and results, anodes tested and results, anode beds replaced, all maintenance /repair work completed and scheduled, areas of concern, areas of low potential, areas requiring closer investigation, suggestions for upgrades/replacements
C.5.19.1.1	Facility Maintenance Schedule	Schedule of cyclical facility maintenance IAW TM 5-610	NLT 30 Days after contract start NLT 10 days before change implementation	COR and others provided by the COR	Identified repairs to be completed on separate DMO's

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction
C.1.2.2-A	Affirmative Procurement of Bio-based Products	Ensure maximum use of bio-based products that are United States Dept. of Agriculture (USDA) designated items	NLT 31 October annually	KO	Include product type, dollar value purchased from previously fiscal year
C.5.19.2.4.11.1	Playground Inspections and Repairs	Report results of inspection and repairs by location	NLT 5 th business day of each month	COR and others provided by the COR	All repairs to meet manufactures recommendations

PERFORMANCE WORK STATEMENT
Section C.8.1
Contract plan submittal requirements

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction	Government Response Time
C.1.3.1	Phase In Plan	Describe how the contractor shall prepare to assume full responsibility of contract operations IAW terms and conditions	NLT 15 days after contract award	KO and COR	N/A	10 Business Days
C.1.4.11.4	Event Contingency Plan	Identify capability and ability to support the Government during sudden or unusual events	15 days prior to contract start When requested	KO and COR	ID emergency personnel with contact info, asset list, emergency operations, how to maintain critical functions, storm response	10 Business Days
C.1.4.11.4.4	Strike Contingency Plan	Identify ability and capability to support requirements in event of a labor strike	NLT contract start date Update as necessary	KO and COR	Government to review and provide comment	10 Business Days
C.1.5.2	Quality Control (QC) Plan	Identify ability to control quality of product supporting this PWS	30 Days after Contract Start 30 Days after option period start 10 days prior to change	KO and COR	Government to review and provide comment	10 Business Days
C.1.6.2.2	Environmental Protection Program	How does the program comply with applicable Army Regulations and Ft. Carson Plans	15 days prior to contract start 10 days prior to change	KO and COR	Minimize pollutant discharge, haz waste handling and disposal.	10 business Days
C.1.6.3.3	Safety Plan	How will contractor adhere to and enforce applicable OSHA requirements for employee safety	15 Days prior to contract start 10 days prior to change	KO and COR	N/A	10 Business days
C.1.6.3.7.1	Fire Prevention and Protection Plan	Complement the Ft. Carson Fire Prevention and Protection Plan for all Contractor operated facility	15 days prior to contract start 10 days prior to change	KO and COR	N/A	10 Business Days
C.1.6.4.2	Hazardous Material Spill Plan	Prevent and address spills in all contractor facilities	15 days prior to contract start	KO and COR	N/A	10 Business Days

Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction	Government Response Time
C.1.3.1	Phase In Plan	Describe how the contractor shall prepare to assume full responsibility of contract operations IAW terms and conditions	NLT 15 days after contract award	KO and COR	N/A	10 Business Days
			10 days prior to change			
Report Number	Report Title	Requirements	Required Date	Electronic Distribution	Special Instruction	Government Response Time
C.1.8.1	Voucher System	Proposed voucher system for meeting financial requirements	30 days prior to contract start	KO and COR		5 Business Days
C.3.1.4	Property Control Plan	Management of GFP	10 days after contract start 10 days prior to change	KO, PA and COR	N/A	10 Business Days
C.5.1.13	Traffic Control Plan	Describe traffic lane closure and/or diversion meeting MUTCD	14 days prior to each closure/diversion event	KO and COR and others provided by COR	Provide traffic plan with FC Form 420-E	3 Business Days
C.5.3.2.1	Operation and Maintenance Plans	How will Operations and maintenance be performed for identified functional areas	NLT 30 days after contract start Annually NLT 31 January	KO, MICC QA, COR	See TE 5.3.2 for list of functional areas and PWS for more information	15 Business Days
C.5.4.11.2	Pest Management Plan	Provide a Pest Management Plan for effort in C.5.4.11	10 Days prior to Contract Start	KO and COR	Follow IPMP and DODI 4150.7	10 Business Days
C.5.5.4.2	Snow and Ice Control Plan	Provide written plan addressing contractor operation to meet PWS requirements of snow plowing and ice control	NLT 1 September annually Annual briefing NLT 30 September	COR and others provided by the COR	Requirements identified in C.5.5.4.	10 Business Days